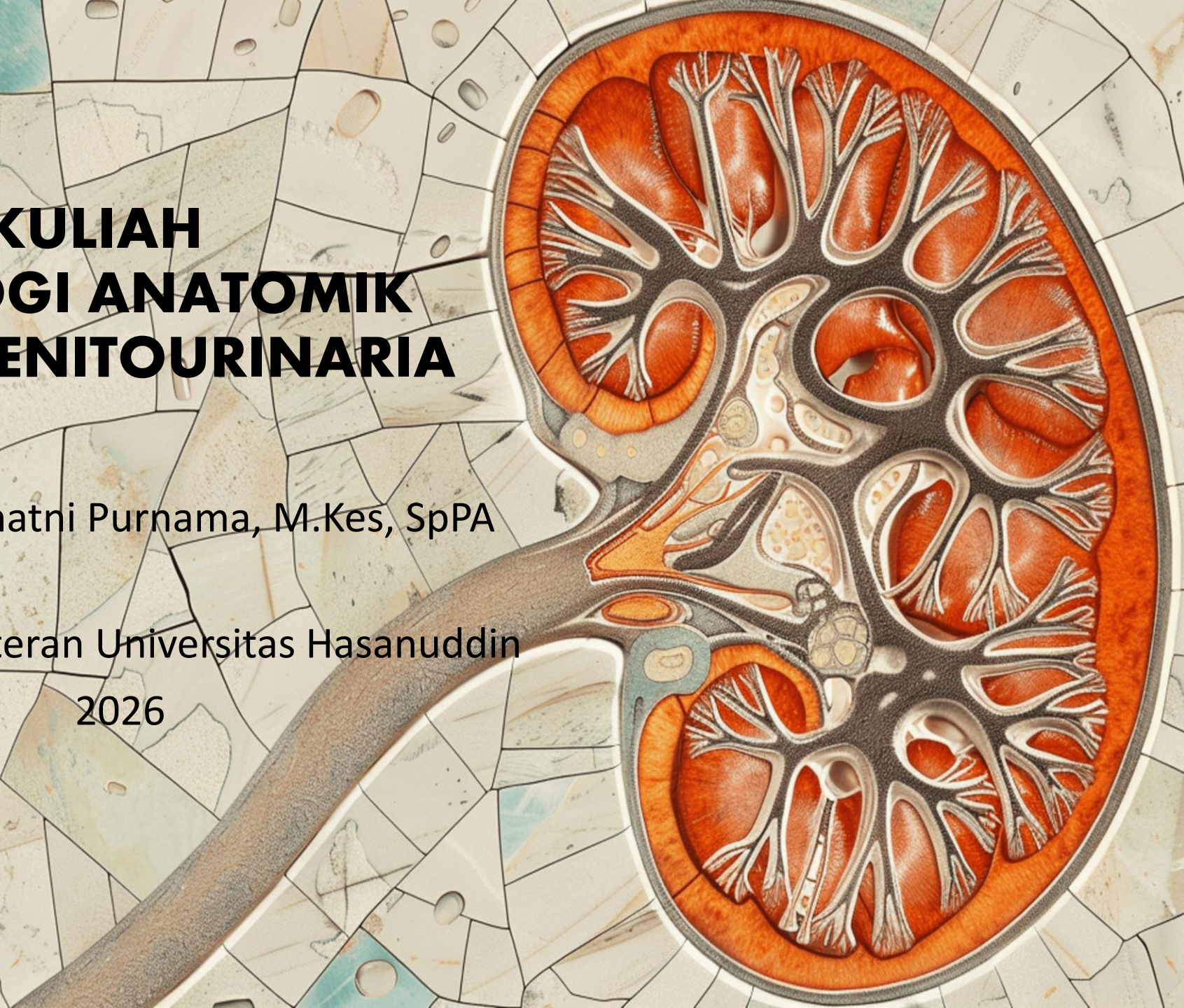


KULIAH PATOLOGI ANATOMIK SISTEM GENITOURINARIA

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2026



Tujuan pembelajaran

Setelah mengikuti kuliah ini, mahasiswa diharapkan mampu:

1. Menjelaskan **struktur dan fungsi dasar** organ sistem genitourinaria.
2. Menjelaskan **mekanisme dasar terjadinya penyakit** pada sistem genitourinaria.
3. Mengenali gambaran **makroskopis dan mikroskopis dasar** penyakit genitourinaria.
4. Membedakan karakteristik dasar **lesi non-neoplastik dan neoplastik**.
5. Menjelaskan prinsip diagnosis patologis melalui **histopatologi**.
6. Menghubungkan temuan patologis dengan **manifestasi klinis dan prognosis** pasien.

Pendahuluan

- Sistem genitourinaria merupakan salah satu sistem organ penting yang berperan dalam **homeostasis cairan dan elektrolit, ekskresi produk metabolisme, regulasi tekanan darah, fungsi reproduksi, serta fungsi seksual**. Sistem ini terdiri atas organ urinaria, yaitu **ginjal, ureter, vesika urinaria, dan uretra**, serta organ genital laki-laki dan perempuan.
- Dalam bidang Patologi Anatomi, pemahaman mengenai sistem genitourinaria menjadi penting karena berbagai penyakit pada organ-organ tersebut dapat menimbulkan gangguan fungsi yang signifikan dan memiliki manifestasi klinis yang beragam. Kelainan dapat terjadi mulai dari tingkat **seluler dan molekuler hingga perubahan struktur makroskopis dan mikroskopis jaringan**.

Pendahuluan

- Secara umum, penyakit pada sistem genitourinaria dapat dikelompokkan menjadi beberapa kategori, yaitu **kelainan kongenital dan perkembangan, penyakit inflamasi dan infeksi, gangguan vaskular, penyakit degeneratif dan metabolik, kelainan obstruktif, serta neoplasma jinak dan ganas.**
- Pada ginjal, misalnya, kerusakan dapat mengenai **glomerulus, tubulus, interstitium, maupun pembuluh darah.** Pola kerusakan tersebut menjadi dasar dalam memahami berbagai penyakit ginjal seperti glomerulonefritis, nefritis interstisial, dan penyakit ginjal kronik. Pemeriksaan histopatologi, imunofluoresensi, dan pada kondisi tertentu pemeriksaan mikroskop elektron dapat digunakan untuk menentukan diagnosis.

FUNGSI UTAMA

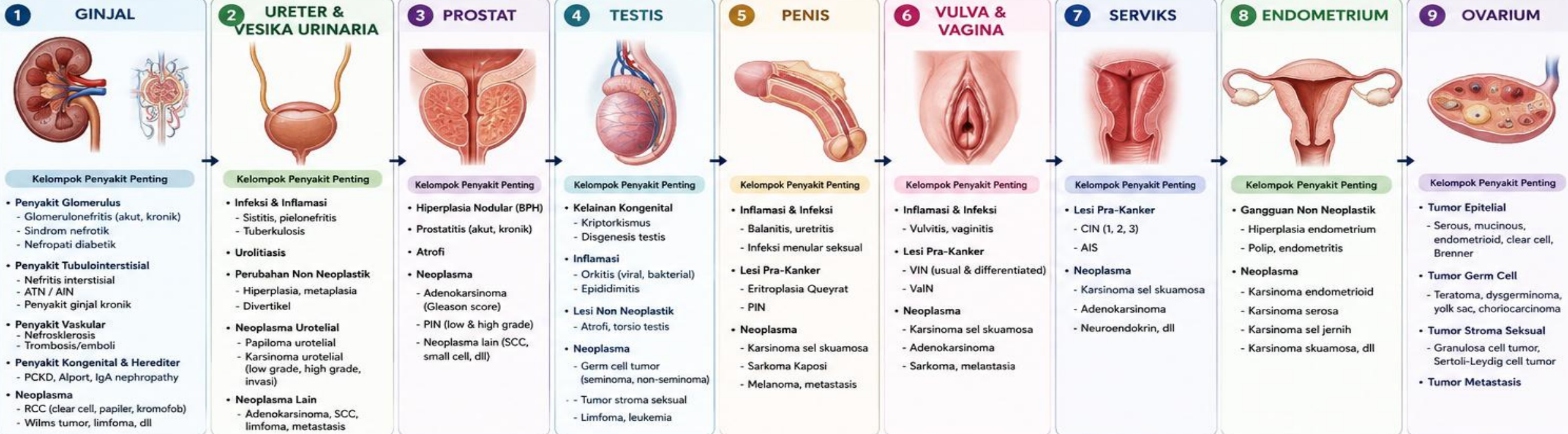
- Ekskresi & homeostasis cairan, elektrolit, asam basa
- Regulasi tekanan darah (RAAS)
- Produksi hormon (EPO, renin, kalsitriol)
- Reproduksi & fungsi seksual

PETA BESAR PATOLOGI SISTEM GENITOURINARIA

Pendekatan Patologi Anatomik: Struktur Normal → Mekanisme Penyakit → Perubahan Morfologi → Diagnosis → Prognosis & Terapi

PRINSIP KUNCI PATOLOGI

- Kelainan dapat mengenai sel, jaringan, organ
- Pola reaksi umum: adaptasi, cedera, inflamasi, perbaikan, neoplasia
- Diagnosis berdasar morfologi + IHC + molekuler



PROSES PATOLOGI UMUM PADA SISTEM GENITOURINARIA



PENDEKATAN DIAGNOSIS PATOLOGI ANATOMIK

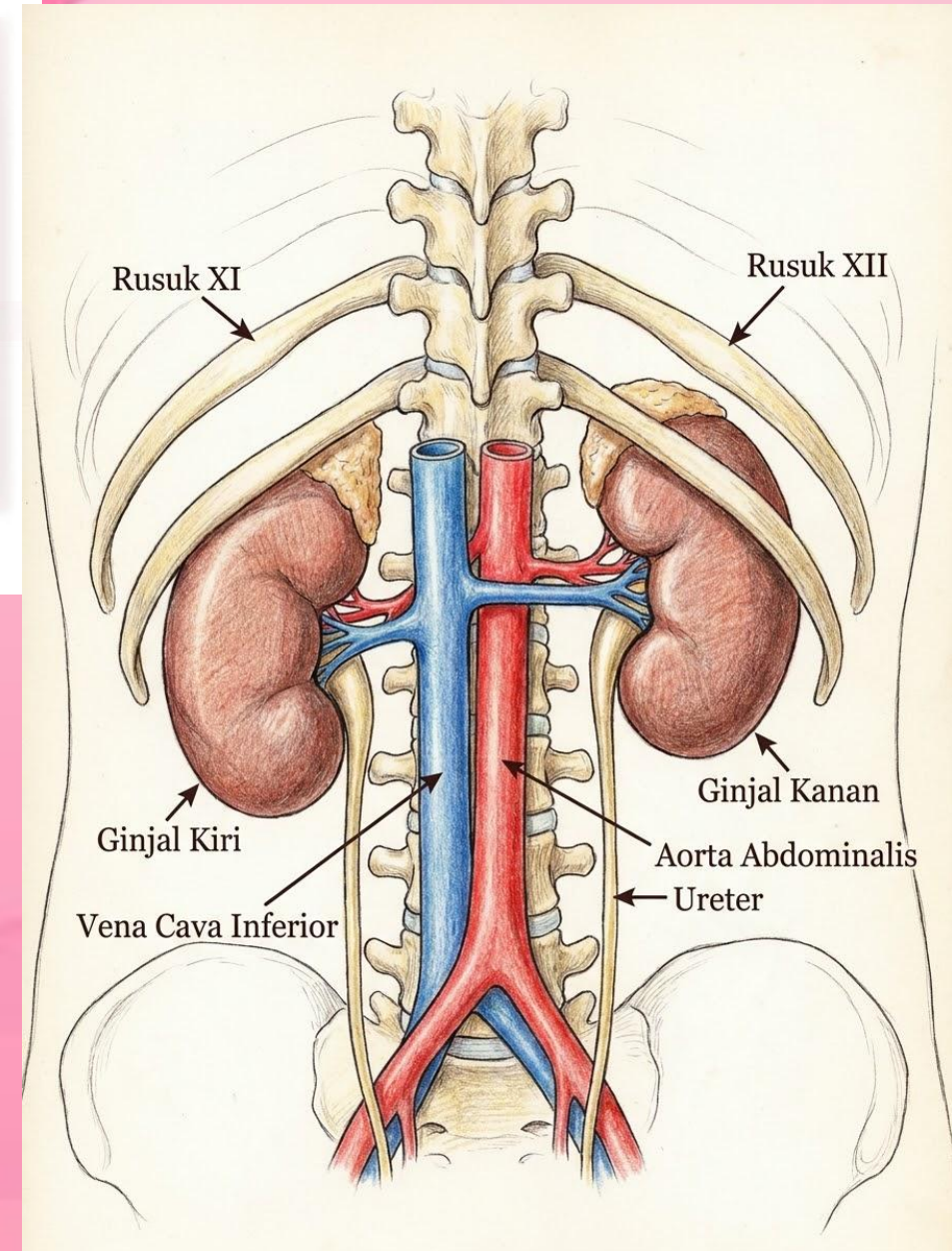
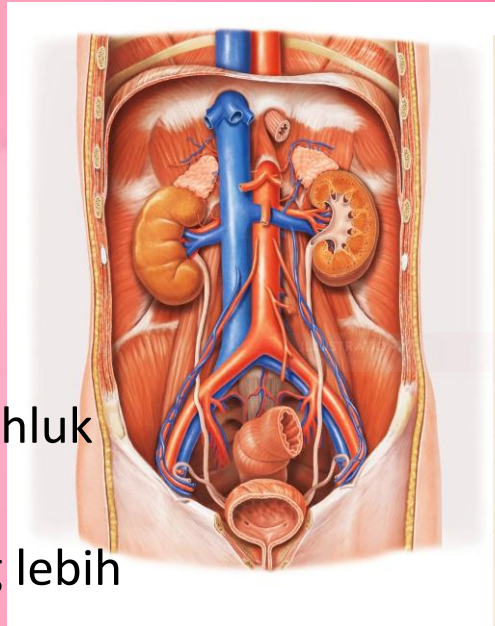


Inti: Memahami hubungan struktur normal – kelainan – gambaran morfologi – diagnosis – prognosis – terapi untuk setiap organ sistem genitourinaria.

GINJAL

1. Anatomi dan Posisi Ginjal

- Bentuk dan Berat: Ginjal merupakan organ berpasangan yang pada hampir semua makhluk hidup berbentuk mirip kacang.
- Setiap ginjal manusia memiliki berat kurang lebih 125 gram.
- Posisi Anatomis: Ginjal terletak di bagian dorsal kavum abdomen, di sebelah lateral vertebra torakalis bawah (beberapa sentimeter di kanan dan kiri garis tengah tubuh)
- Struktur Pelindung: Organ ini terbungkus oleh jaringan ikat tipis bernama kapsula renis
- Bagian anterior ginjal dipisahkan dari rongga abdomen oleh lapisan peritoneum, sedangkan bagian posteriornya dilindungi oleh dinding toraks bawah

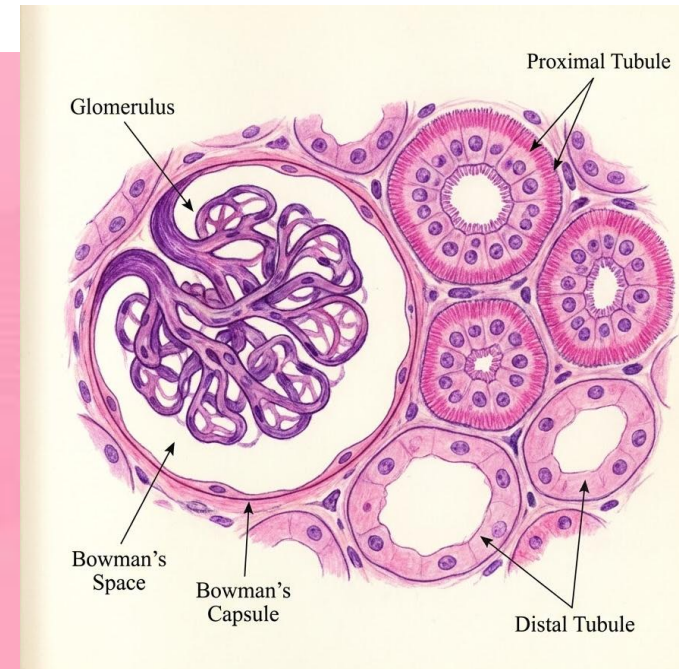
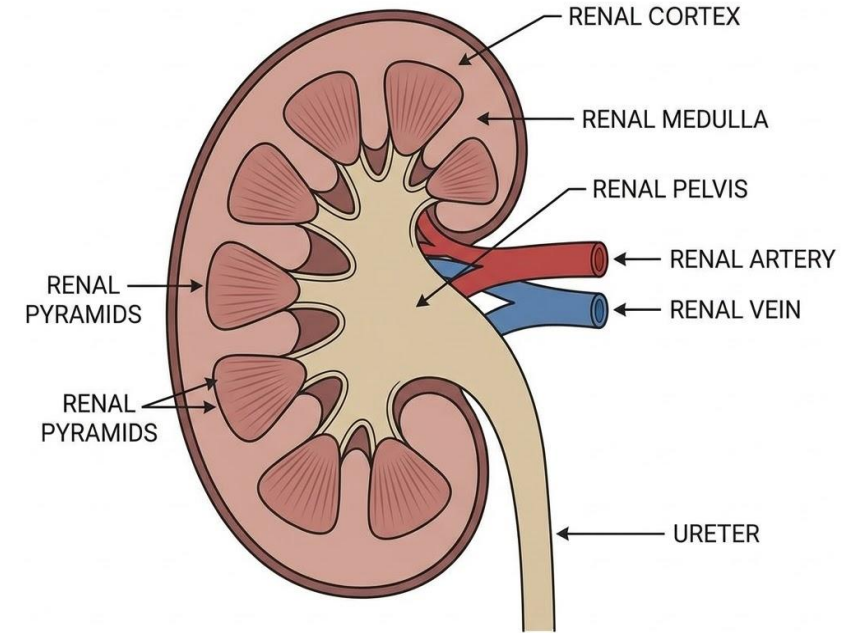


GINJAL

2. HISTOLOGI

Setiap ginjal memiliki lebih dari satu juta unit penyaringan mikroskopis yang disebut nefron. Nefron inilah yang bertanggung jawab penuh dalam menjalankan fungsi penyaringan darah. Setiap nefron terdiri atas komponen-komponen berikut:

- **Glomerulus & Kapsula Bowman:** Glomerulus merupakan kumpulan pembuluh darah halus tempat penyaringan plasma darah terjadi, yang dikelilingi oleh organ mirip cangkir bernama kapsula Bowman.
- **Tubulus Kontortus Proksimal (TKP):** Bagian saluran setelah glomerulus yang dibatasi oleh sel epitel selapis kubus dengan banyak mikrovili pembentuk brush border. Sel epitel di bagian ini sangat peka terhadap kekurangan oksigen (anoksia) serta rentan mengalami kerusakan/nekrosis jika terjadi kontak dengan zat-zat beracun atau obat-obatan yang diekskresikan lewat ginjal.
- **Lengkung Henle (Ansa Henle):** Saluran yang menghubungkan tubulus proksimal dan distal, bersifat permeabel terhadap urea di mana sebagian urea akan direabsorpsi secara aktif.
- **Tubulus Kontortus Distal (TKD):** Saluran yang tersusun atas epitel selapis kubus namun tidak memiliki brush border.
- **Duktus Pengumpul (Tubulus Urinarius):** Saluran akhir nefron yang menampung urine hasil penyaringan untuk dialirkan menuju pelvis ginjal
- Pelvis ginjal kemudian membentuk ureter, yaitu pipa berotot polos yang menyalurkan urine ke kandung kemih.



Renal Grossing/ Gambaran Makroskopik

- 1. Radical nephrectomy

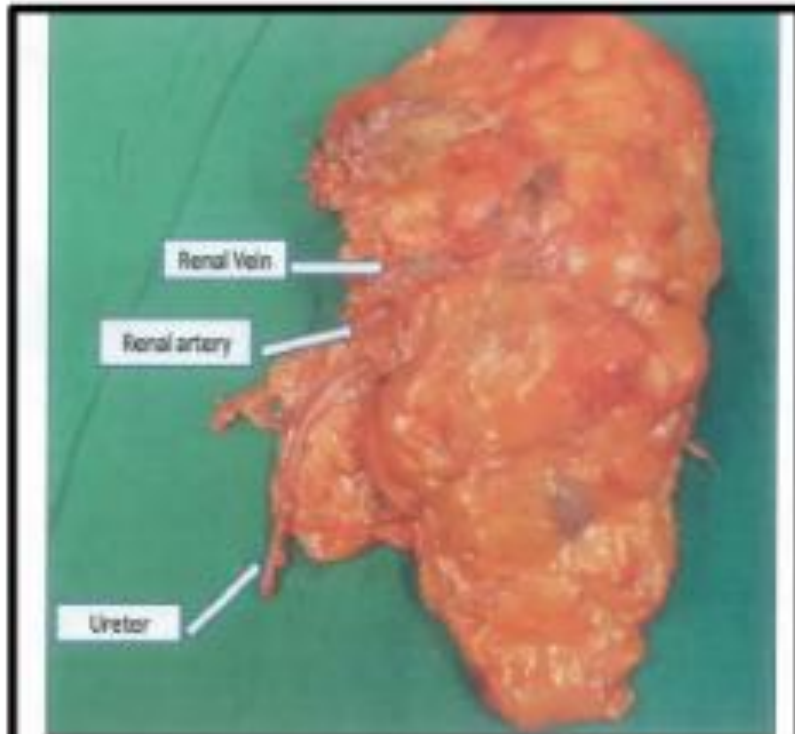
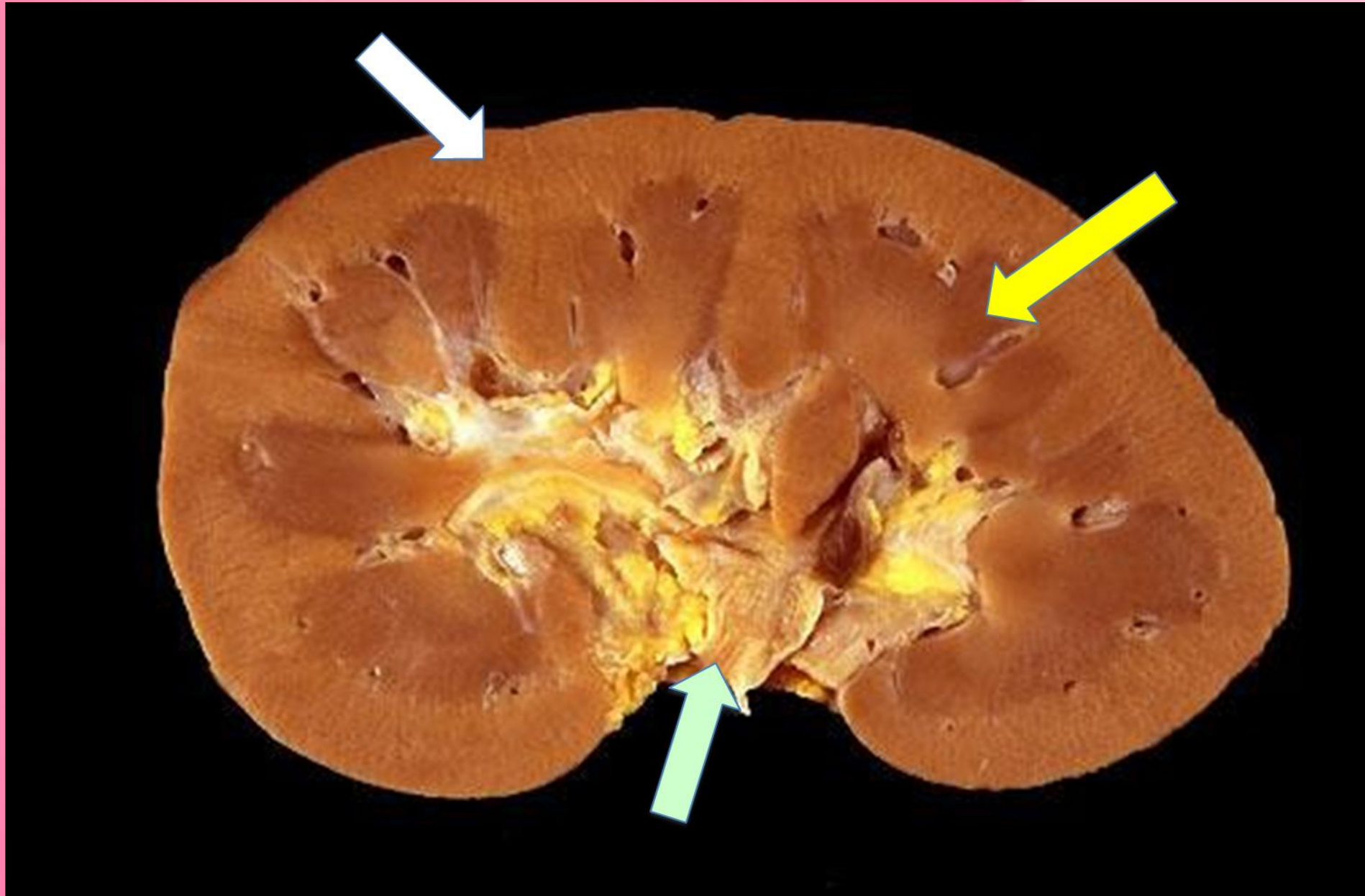


Figure 1: Left Kidney invested by perinephric fat and Gerota's fascia with vein, artery and pelvis/ureter related anterior to posterior,

- 2. Partial nephrectomy



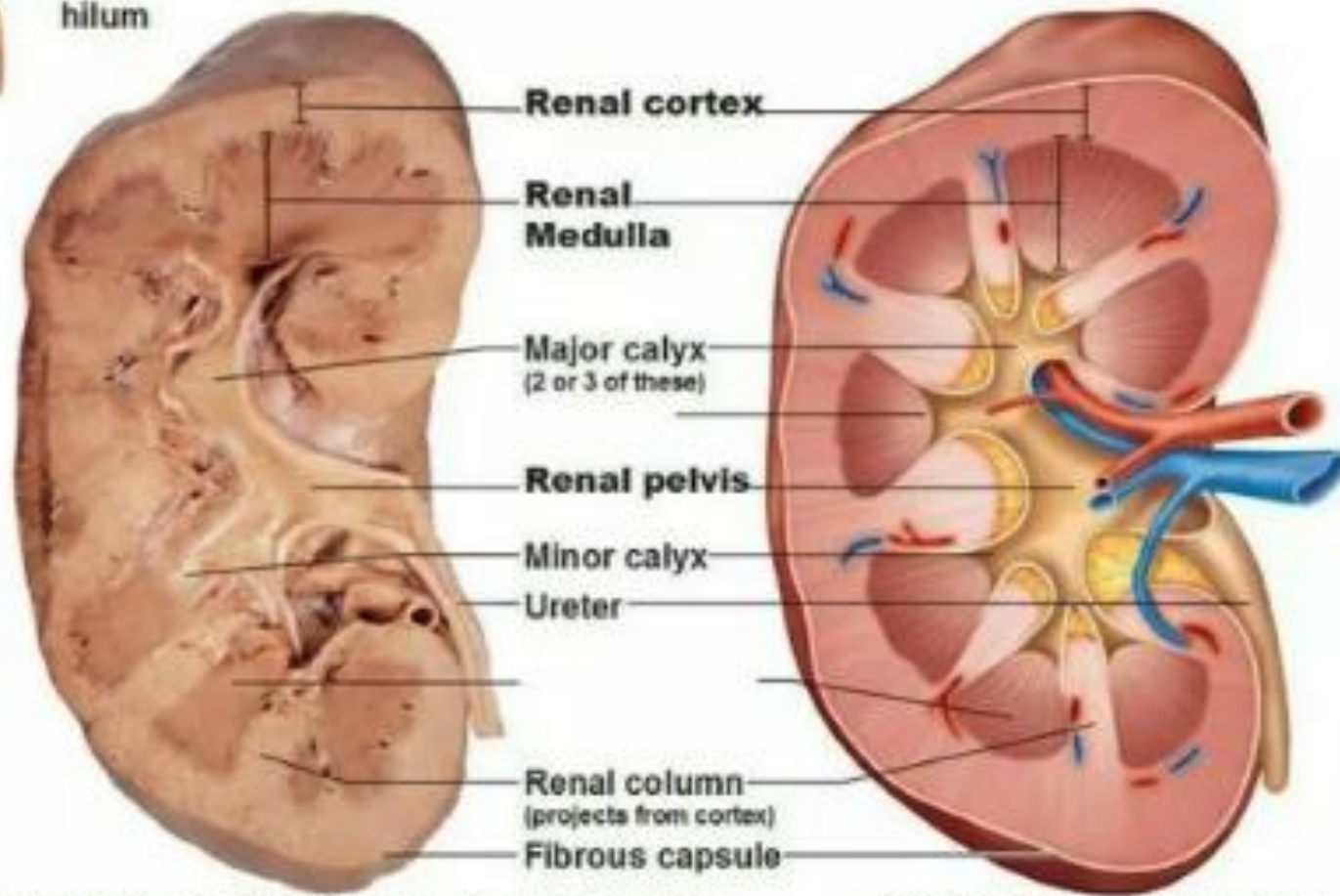


In cross section, this normal adult kidney demonstrates the lighter outer cortex and darker medulla with central pelvis.

Internal Gross Anatomy of the Kidneys



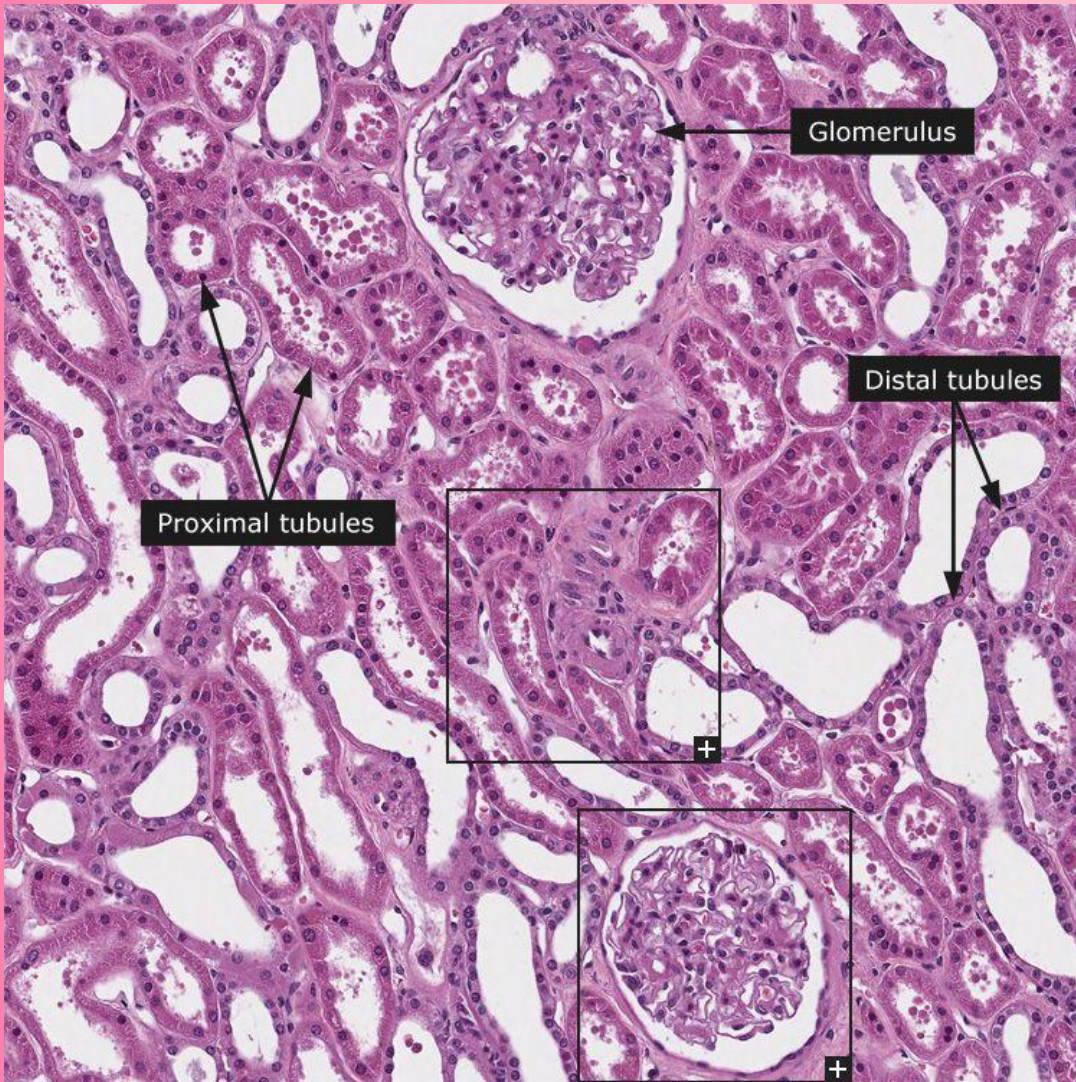
Renal
hilum



(a) Photograph of right kidney, frontal section

(b) Diagrammatic view

Gambaran Mikroskopis Renal



- Glomerulus: Bulatan padat di tengah; berfungsi sebagai saringan utama darah.
- Tubulus Proksimal (Proximal tubules): Saluran dengan sel berwarna pink gelap dan lumen agak 'kotor' (karena ada brush border); berfungsi reabsorpsi nutrisi/air.
- Tubulus Distal (Distal tubules): Saluran dengan lumen lebih bersih, jernih, dan bulat; berfungsi mengatur kadar ion/garam.

Gambaran Mikroskopis

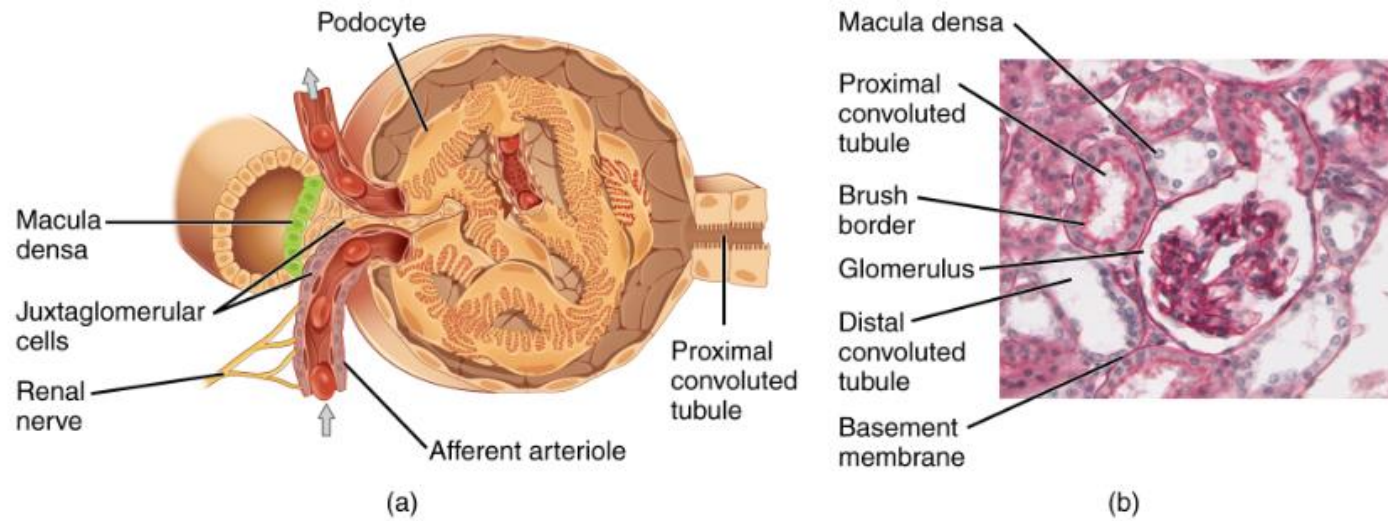


Figure 25.13 Juxtaglomerular Apparatus and Glomerulus (a) The JGA allows specialized cells to monitor the composition of the fluid in the DCT and adjust the glomerular filtration rate. (b) This micrograph shows the glomerulus and surrounding structures. LM $\times 1540$. (Micrograph provided by the Regents of University of Michigan Medical School $\copyright$ 2012)

Glomerulus & Podosit: Rumpun kapiler darah tempat terjadinya filtrasi pertama urin, dibungkus oleh sel podosit (foot processes).

- Aparat Jukstaglomerulus (JGA): Area pertemuan antara arteriol ginjal dan tubulus distal, terdiri dari:
 - Sel Jukstaglomerulus (Juxtaglomerular cells): Sel otot polos pada arteriol aferen yang menghasilkan hormon renin.
 - Macula Densa: Sel-sel rapat pada dinding tubulus distal yang bertindak sebagai sensor kandungan cairan/garam (Na^+).
- Tubulus Kontortus Proksimal (PCT): Saluran penampung filtrat awal yang memiliki Brush Border (mikrovili) untuk reabsorpsi.

LESI NON NEOPLASTIK PADA GINJAL

1. Kelainan kongenital dan Lesi kistik ginjal
2. Glomerulonefritis Progresif Cepat (GNPC / RPGN - Rapidly Progressive Glomerulonephritis)
3. Glomerulonefritis Akut (GNA / GNAPS)
4. Nekrosis Tubular Akut (NTA / ATN - *Acute Tubular Necrosis*) Nefrotoksik
5. Sindrom Nefrotik (*Nephrotic Syndrome*)
6. Gagal Ginjal Akut (AKI) dan Penyakit Ginjal Kronik (CKD)
7. Batu Ginjal (Nefrolitiasis / *Renal Stones*) dan penyakit obstruksi-vaskular
8. Interstitial Disease
9. Infeksi

KELAINAN KONGENITAL

1. Agenesis Ginjal (Renal Agenesis)
2. Hipoplasia Ginjal (Renal Hypoplasia) dan Displasia Ginjal (Renal Dysplasia) / Multicystic Dysplastic Kidney (MCDK)
3. Ginjal Tapal Kuda (Horseshoe Kidney)
4. Lesi kistik ginjal

1. Agenesis Ginjal (Renal Agenesis)

Agenesis ginjal adalah cacat lahir bawaan yang ditandai dengan tidak terbentuknya satu atau kedua ginjal karena kegagalan interaksi perkembangan awal saat masa janin, atau akibat involusi total dari ginjal displastik.

Gambaran Makroskopik:

- Ketiadaan fisik total dari satu ginjal (unilateral) atau kedua ginjal (bilateral) di dalam rongga retroperitoneal
- Agenesis Unilateral: Hanya terdapat satu ginjal tunggal yang berfungsi. Ginjal tunggal ini biasanya mengalami hipertrofi kompensasi (ukurannya membesar secara abnormal, melebihi dua standar deviasi ukuran normal) untuk mengambil alih fungsi ginjal yang hilang
- Agenesis Bilateral: Kedua ginjal sama sekali tidak ada. Bayi baru lahir akan menunjukkan Sindrom Potter (akibat oligohidramnion atau ketiadaan air ketuban), yang ditandai dengan hipoplasia paru berat (paru-paru tidak berkembang) dan kelainan fisik luar, sehingga tidak kompatibel dengan kehidupan luar rahim.

Gambaran Mikroskopik:

- Pada sisi agenesis, sama sekali tidak ditemukan adanya sediaan parenkim ginjal, glomerulus, maupun tubulus karena organogenesis tidak pernah terjadi
- Pada kasus unilateral, jika ginjal kontralateral (sisi lainnya) diperiksa di bawah mikroskop, nefron-nefronnya akan menunjukkan gambaran glomerulus dan tubulus yang mengalami hipertrofi (pembesaran ukuran sel epitel) akibat beban kerja kompensasi yang meningkat.

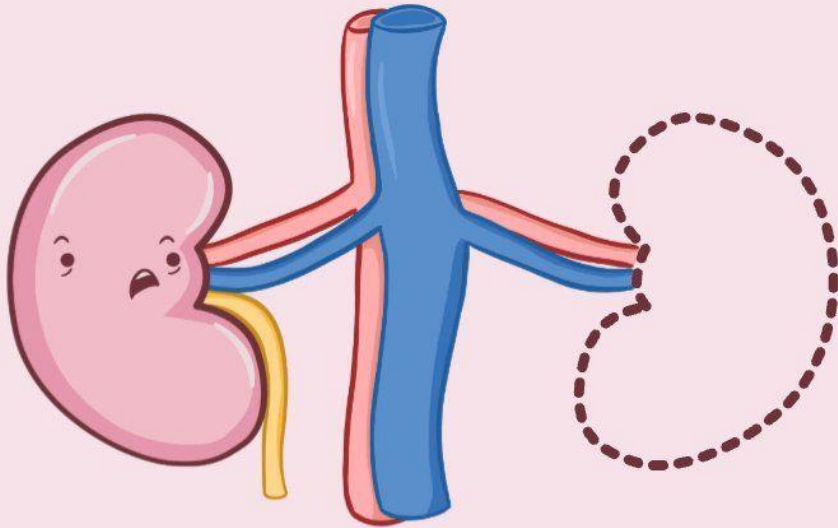
2. Hipoplasia Ginjal (Renal Hypoplasia) dan Displasia Ginjal (Renal Dysplasia) / Multicystic Dysplastic Kidney (MCDK)

- Hipoplasia Ginjal (Renal Hypoplasia): Secara makroskopis ukuran ginjal abnormal kecil dengan arsitektur jaringan yang normal, namun jumlah nefron berkurang secara signifikan (hanya sekitar 20–25% dari jumlah normal).
 - Glomerulus dan tubulus yang tersisa mengalami hipertrofi untuk mengompensasi fungsinya.
-
- Displasia Ginjal (Renal Dysplasia) / Multicystic Dysplastic Kidney (MCDK): Malformasi histologis di mana parenkim ginjal digantikan oleh kista-kista epitelial non-komunikasi yang tersebar di antara jaringan ikat stroma primitif (sering ditemukan kartilago metaplastik).
 - MCDK merupakan salah satu bentuk dysplasia terberat.
 - Kondisi ini sering dikaitkan dengan mutasi pada gen supresor/proto-onkogen seperti PAX2, HNF1B, dan SALL1.

RENAL AGENESIS

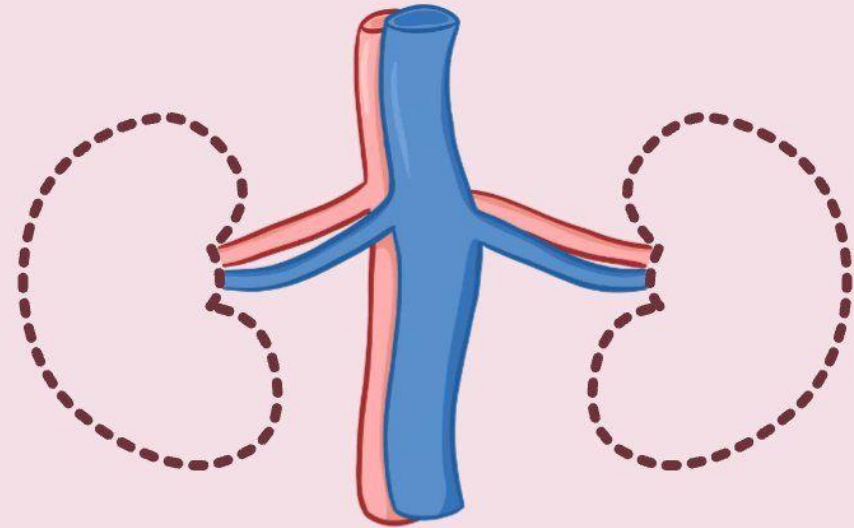
KIDNEYS DON'T FORM

UNILATERAL RENAL AGENESIS

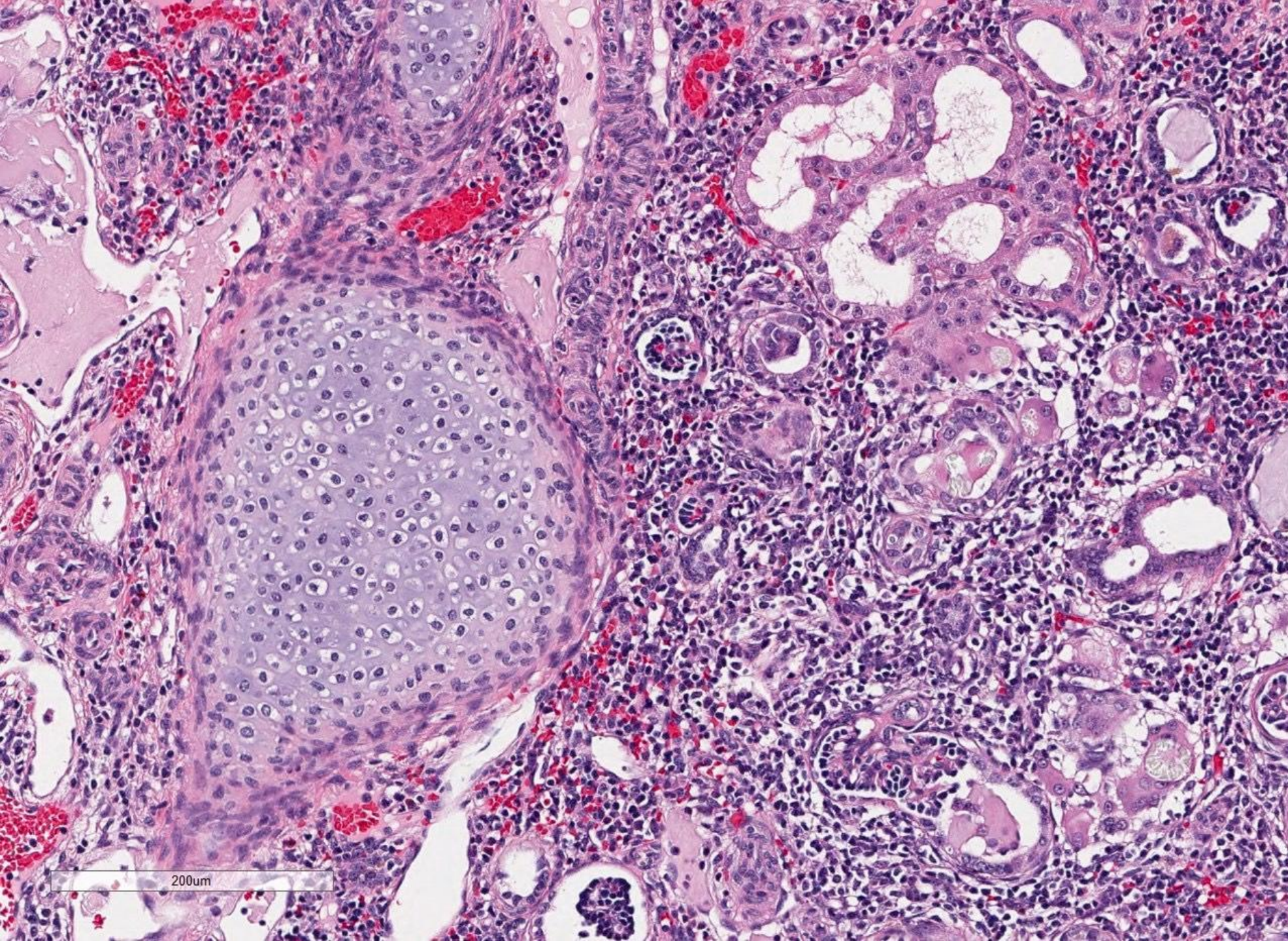


URA

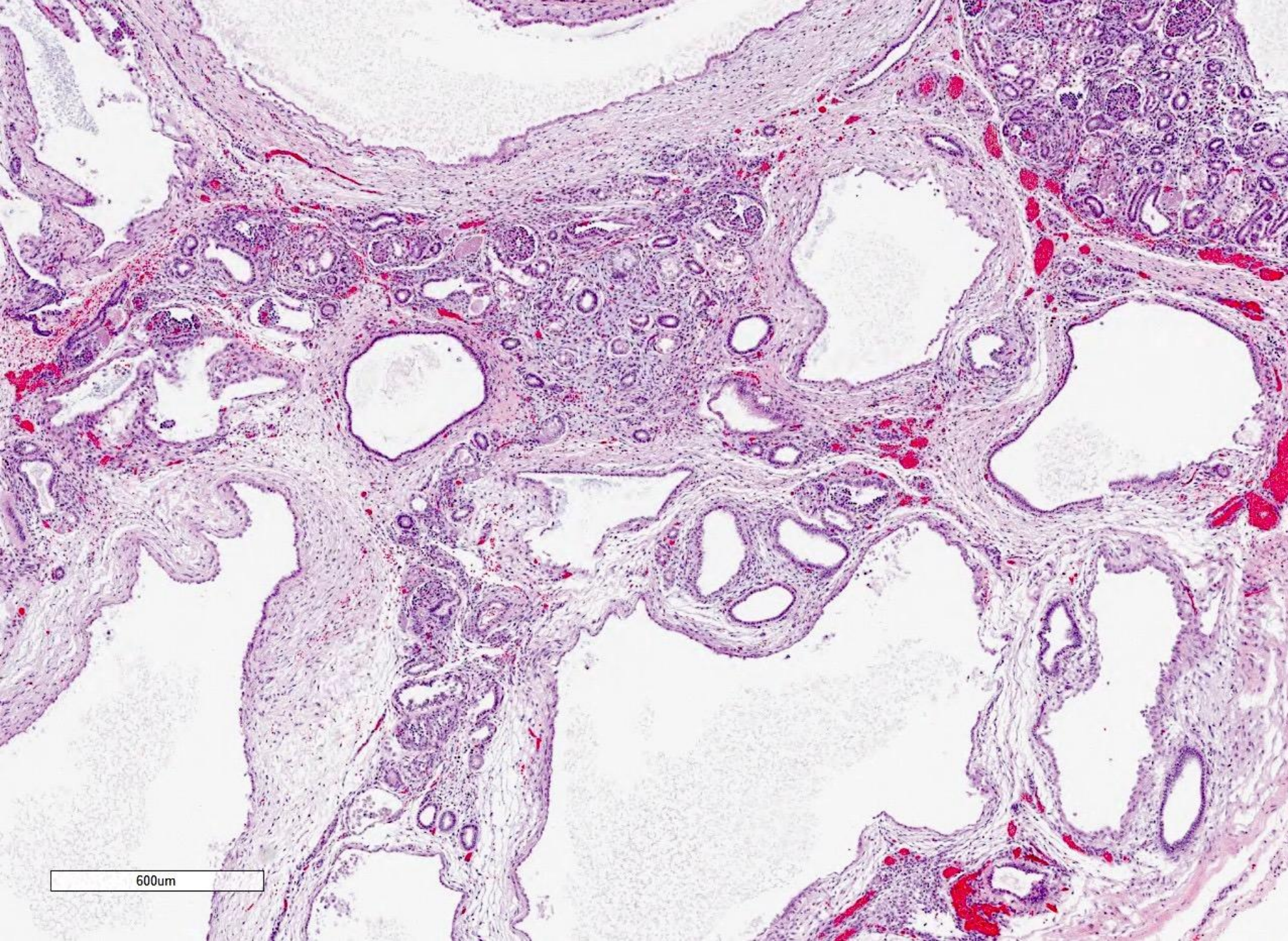
BILATERAL RENAL AGENESIS



BRA



- Gambar histopatologi ini menunjukkan gambaran khas Displasia Ginjal (Renal Dysplasia):
- Pulau Tulang Rawan Metaplastik (Kiri): Tampak nodul kartilago hialin (hyaline cartilage island) yang matang/primitif di dalam parenkim ginjal. Keberadaan jaringan tulang rawan di dalam ginjal merupakan ciri histopatologi terpenting (patognomonik) untuk displasia ginjal.
 - Duktus/Tubulus Primitif & Disorganisasi (Kanan): Tampak saluran tubulus abnormal yang belum berkembang sempurna, dikelilingi stroma mesenkim dan infiltrasi padat sel radang mononuklear.
 - Konteks Klinis: Merupakan kelainan bawaan akibat gangguan diferensiasi struktur nefron dan duktus pengumpul selama perkembangan janin (sering dijumpai pada kasus Multicystic Dysplastic Kidney / MCDK). Kadang elemen kartilago heterolog ini juga dapat ditemukan pada tumor ginjal pediatrik seperti Wilms tumor.



Gambar histopatologi ini menunjukkan gambaran klasik dari Ginjal Displastik Multikistik (Multicystic Dysplastic Kidney / MCDK):

- Struktur Kista Multilokular Besar: Tampak rongga-rongga kista berukuran besar dengan dinding tipis fibrosa yang menggantikan parenkim normal ginjal.
- Duktus Primitif / Imatur: Di dalam septa fibrosa di antara kista, terdapat tubulus/duktus primitif kecil yang dikelilingi stroma mesenkim longgar (fibromuskular imatur).
- Parenkim Ginjal Normal Terdesak (Kanan Atas): Masih terlihat sisa tubulus ginjal yang padat di sudut kanan atas, namun sebagian besar arsitektur korteks dan medula ginjal telah hilang digantikan oleh kista dan stroma disorganisasi.
- Konteks Klinis: MCDK adalah anomali kongenital non-hereditas yang paling sering menyebabkan massa kistik pada ginjal neonatus/bayi, umumnya bersifat unilateral dan ginjal yang terdampak tidak memiliki fungsi filtrasi yang normal.

3. Ginjal Tapal Kuda (Horseshoe Kidney)

Horseshoe kidney adalah kelainan fusi (penyatuan) ginjal bawaan yang paling sering dijumpai. Kelainan ini terbentuk pada masa embrio antara minggu ke-4 hingga ke-9 kehamilan, sebelum ginjal bermigrasi naik dari panggul ke rongga retroperitoneal atas.

Gambaran Makroskopik:

- Tampak kedua ginjal menyatu di garis tengah tubuh, paling sering terjadi pada bagian kutub bawah (lower pole) masing-masing ginjal, sehingga membentuk struktur melengkung seperti tapal kuda.
- Kelainan fusi ini mengubah bentuk, rotasi, dan pembuluh darah ginjal.
- Ginjal biasanya terletak lebih rendah di dalam rongga perut dibandingkan posisi normal.
- Ureter yang keluar dari masing-masing hilus ginjal akan berjalan lurus ke bawah menuju kandung kemih tanpa saling menyilang (uncrossed ureters).

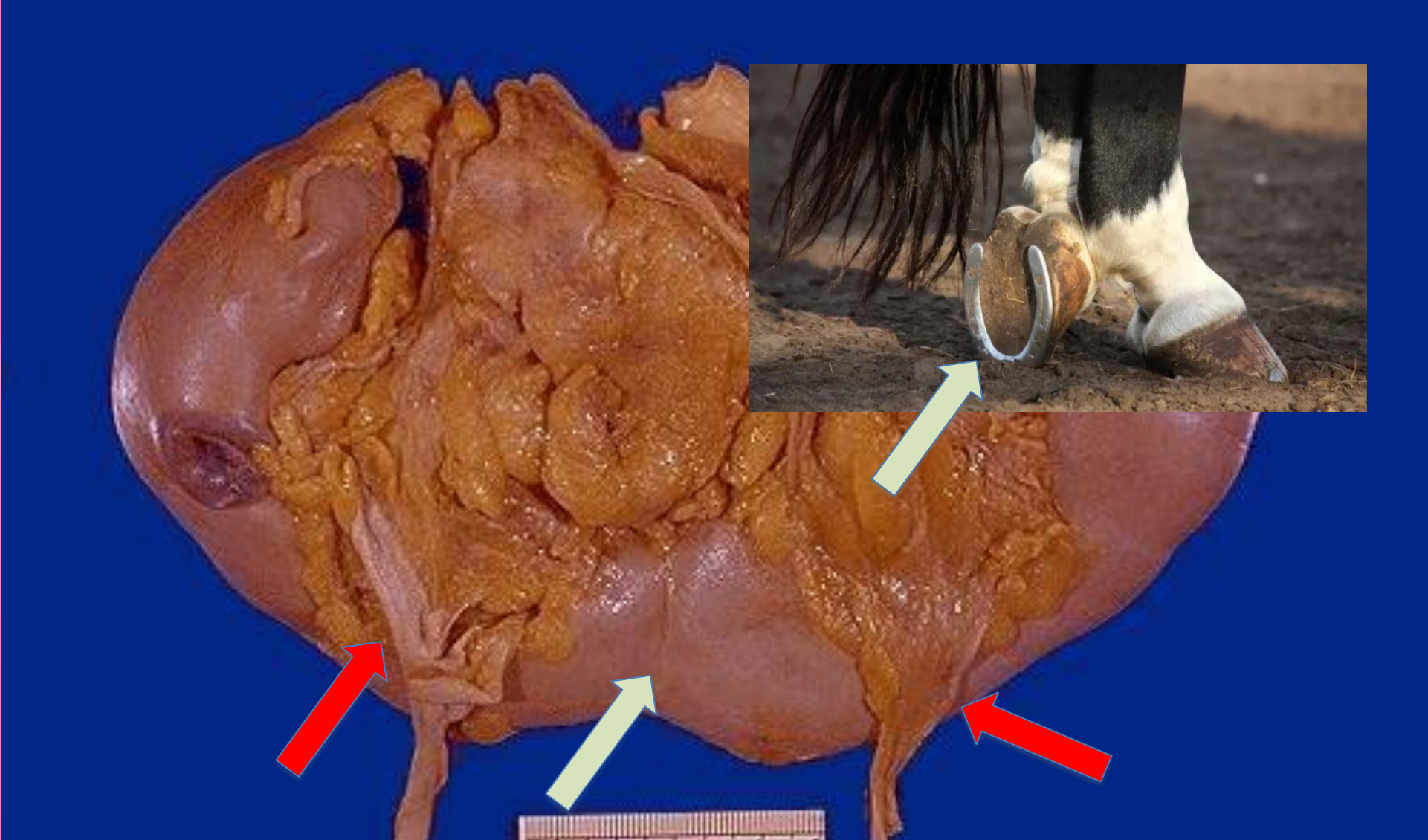
Gambaran Mikroskopik:

- Area penyatuan (isthmus) dan lobus ginjal terdiri atas massa jaringan ginjal fungsional yang normal (memiliki glomerulus, tubulus proksimal, dan tubulus distal yang sehat secara histologis).
- Struktur mikroskopis ini tampak normal, kecuali jika terjadi komplikasi sekunder. Sekitar 80% penderita mengalami hidronefrosis (pelebaran pelvis ginjal) akibat refluks urine atau sumbatan sambungan ureteropelvik (UPJO), yang secara mikroskopis dapat memicu atrofi tubulus atau jaringan parut (scarring) pada korteks seiring berjalannya waktu.

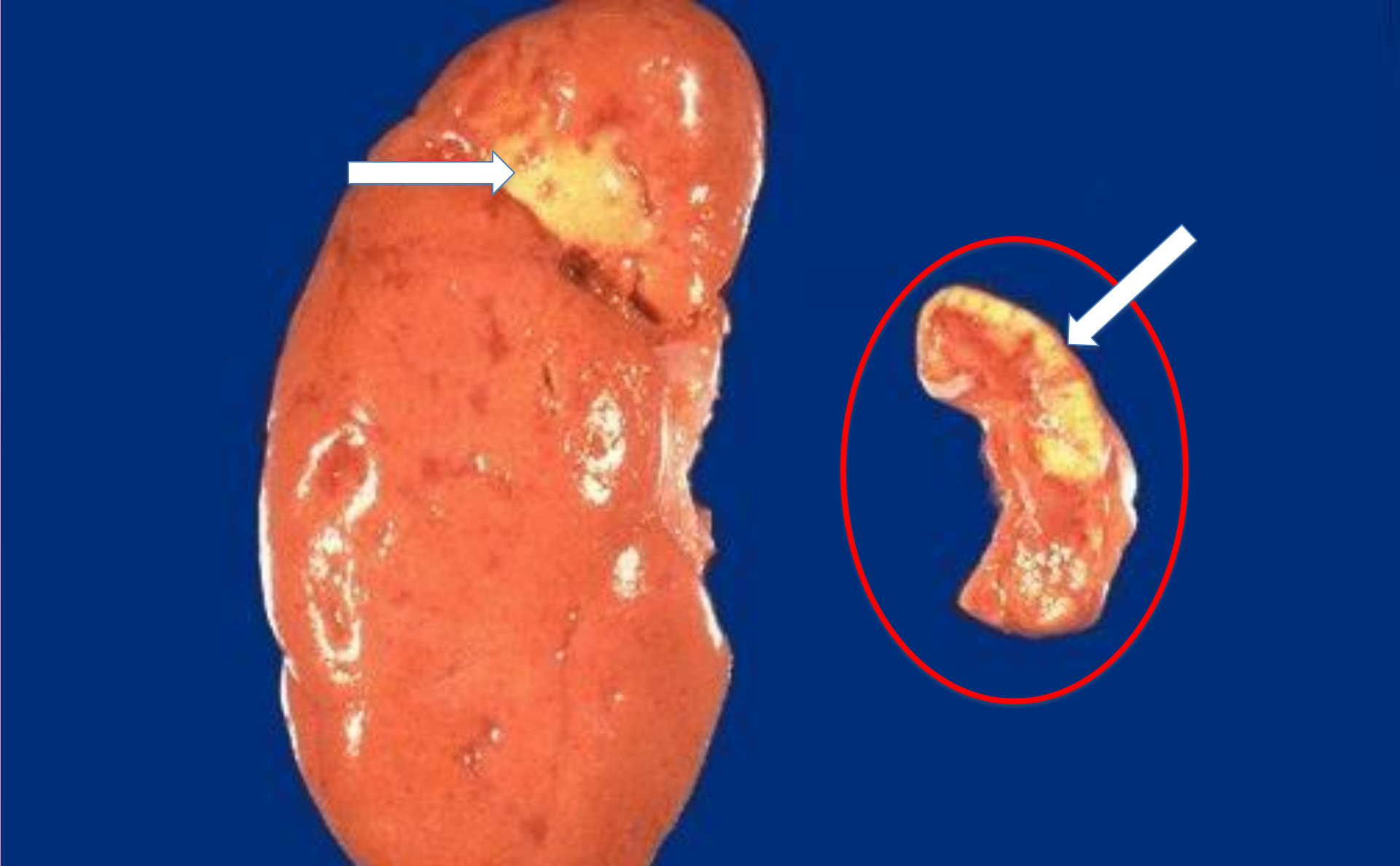
Horseshoe kidney

- Fusion of upper or lower poles of kidneys
- Mostly asymptomatic
- May be associated with obstruction, infection or stone.



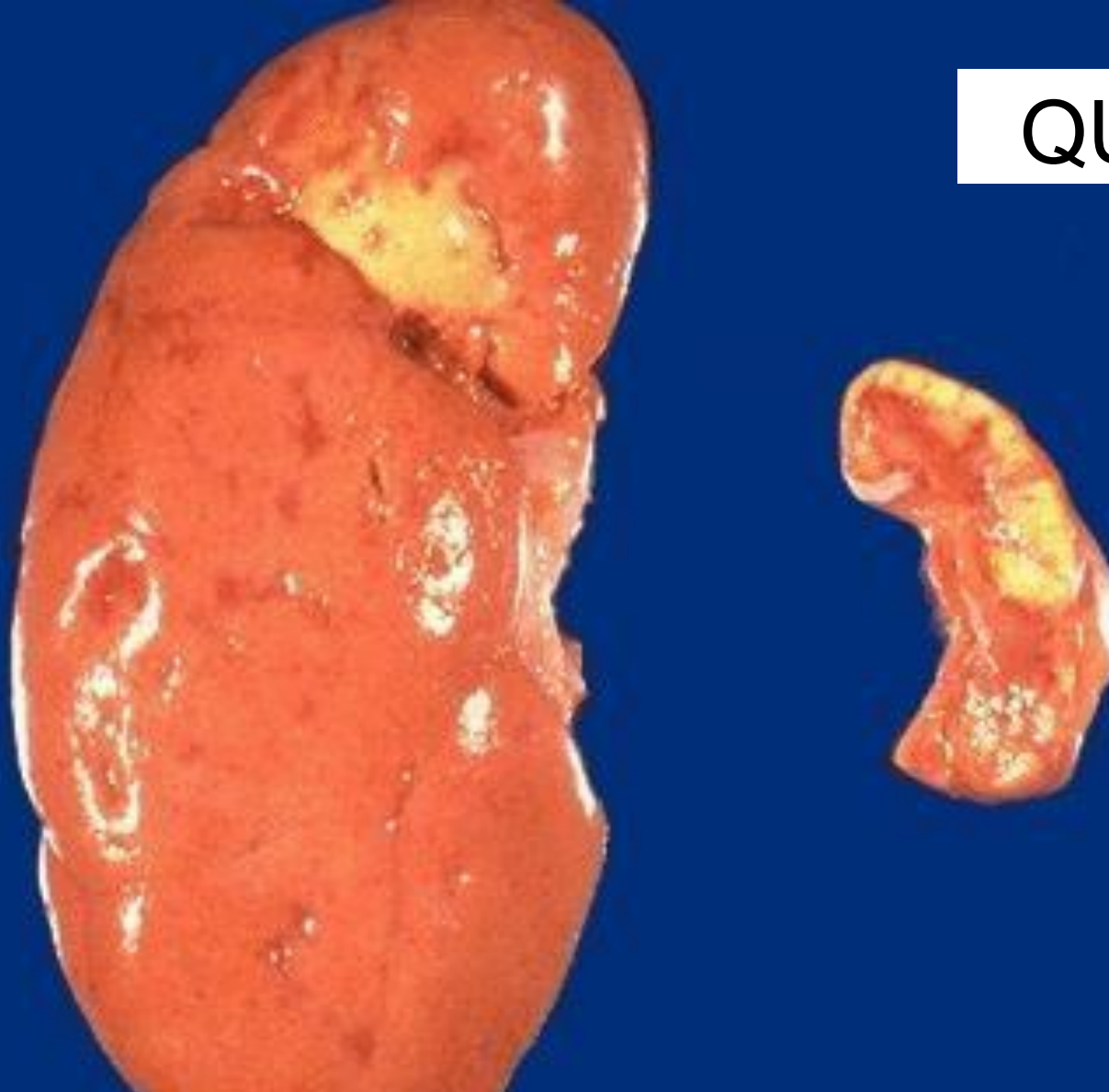


Here is a "**HORSESHOE**" **KIDNEY**. This is a congenital anomaly that most often occurs in association with other anomalies or syndromes with specific genetic defects such as **trisomy 18**. However, it can also occur as an isolated anomaly. The possible problem here is that **the ureters take an abnormal course across the "bridge" of renal tissue** and this can lead to **partial obstruction with hydronephrosis**.



There is a relatively normal kidney at the right with only a few scattered, shallow **cortical scars** and **one fairly large pale tan-yellow scar** in the upper pole. The left kidney is **atrophic because of renal arterial occlusion**. Such a situation can lead to hypertension (**GOLDBLATT KIDNEY**).

QUIZ

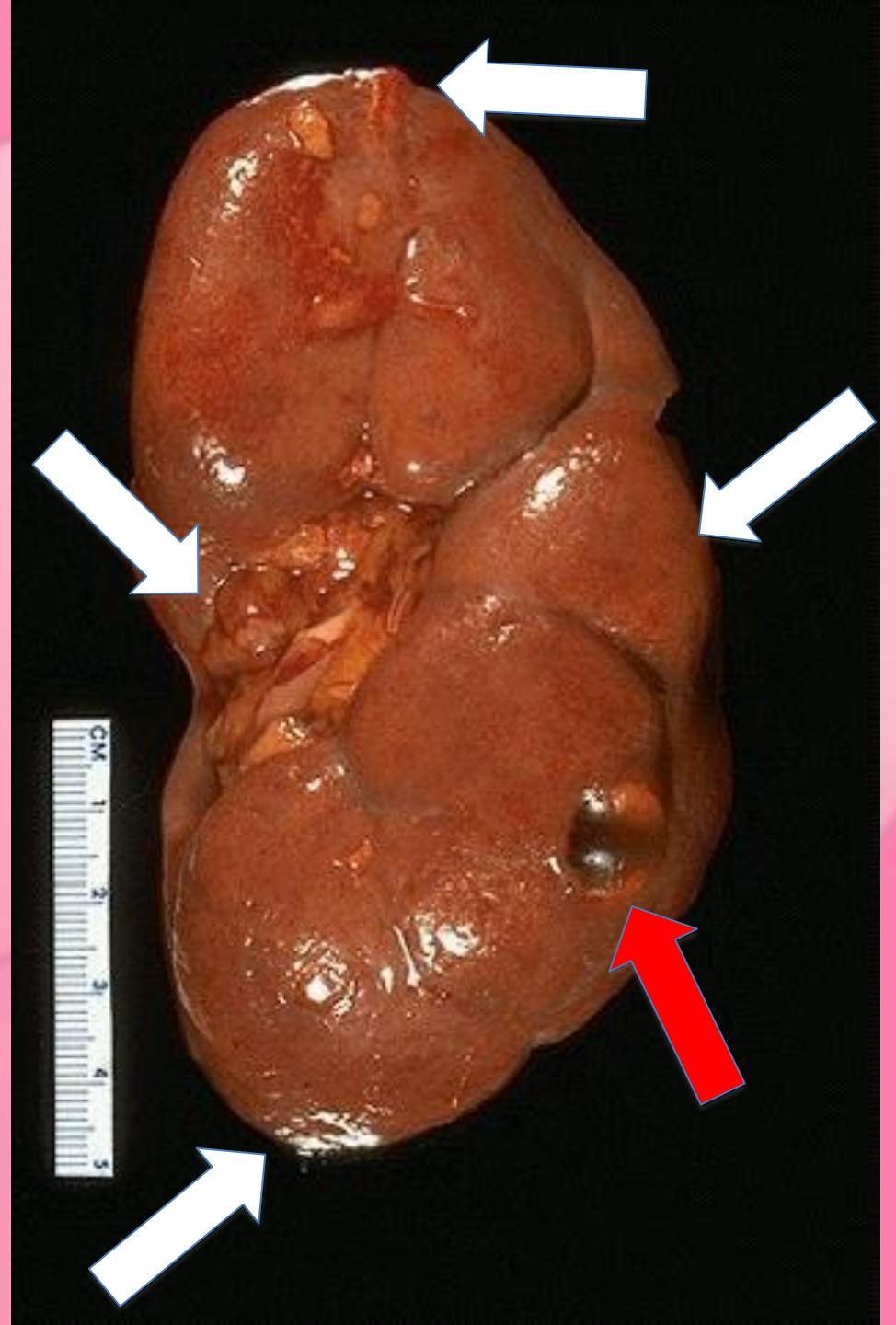


What is the plasma renin level follow with the finding shown here most likely to be ?

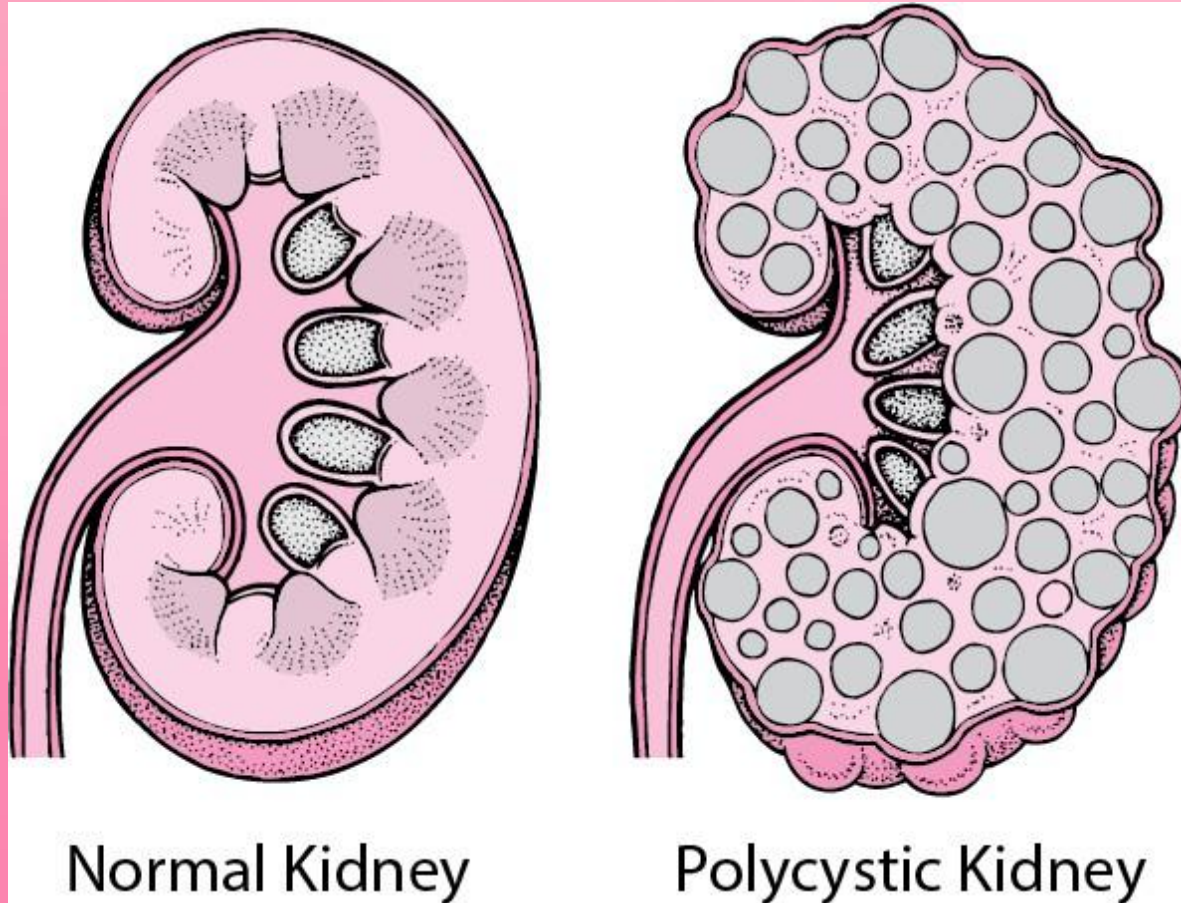
- A. Low**
- B. High**

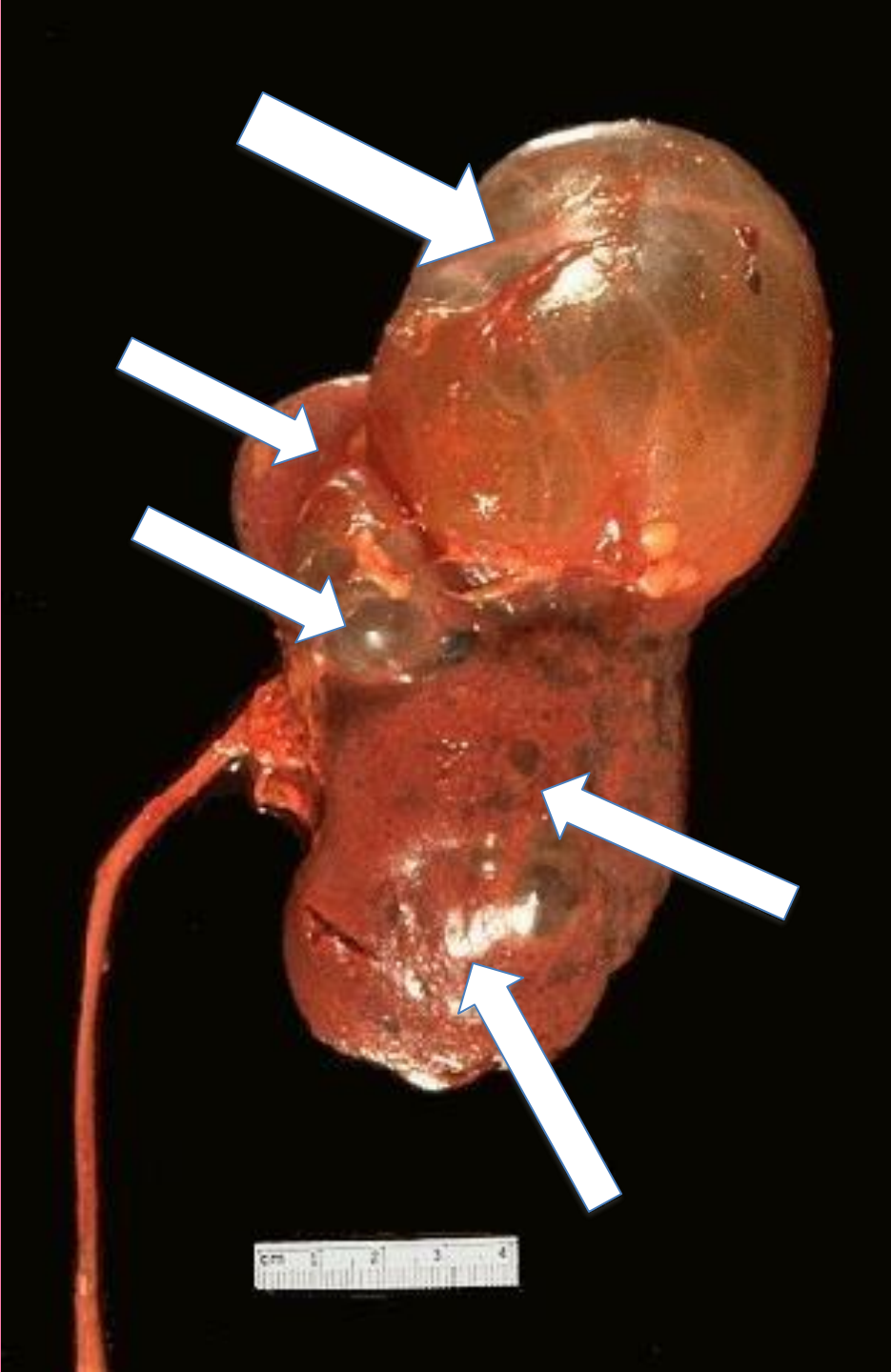
4. Lesi kistik ginjal

Here is a normal adult kidney. The capsule has been removed and a pattern of fetal lobulations still persists, as it sometimes does. The hilum at the mid left contains some adipose tissue. At the lower right is a smooth-surfaced, small, clear fluid-filled simple renal cyst. Such cysts occur either singly or scattered around the renal parenchyma and are not uncommon in adults. [



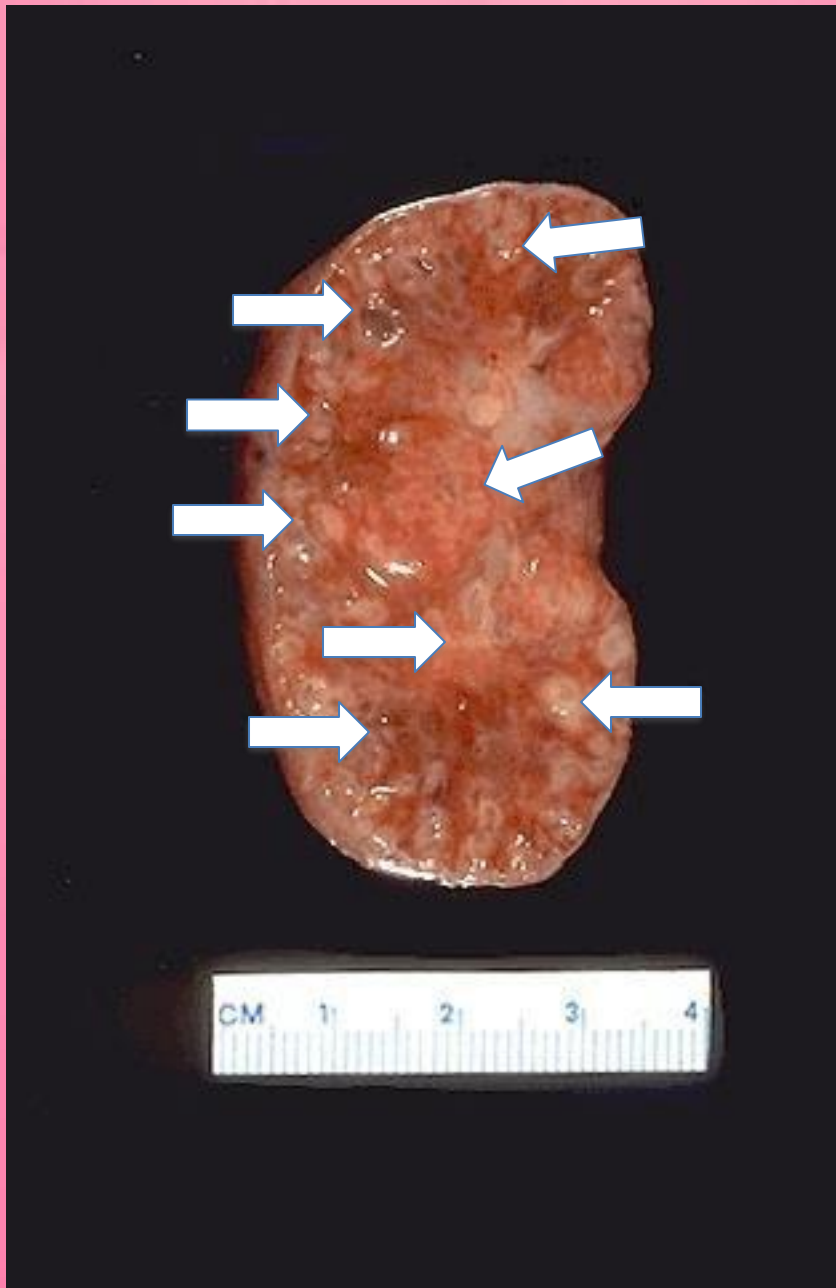
KELAINAN KONGENITAL GINJAL





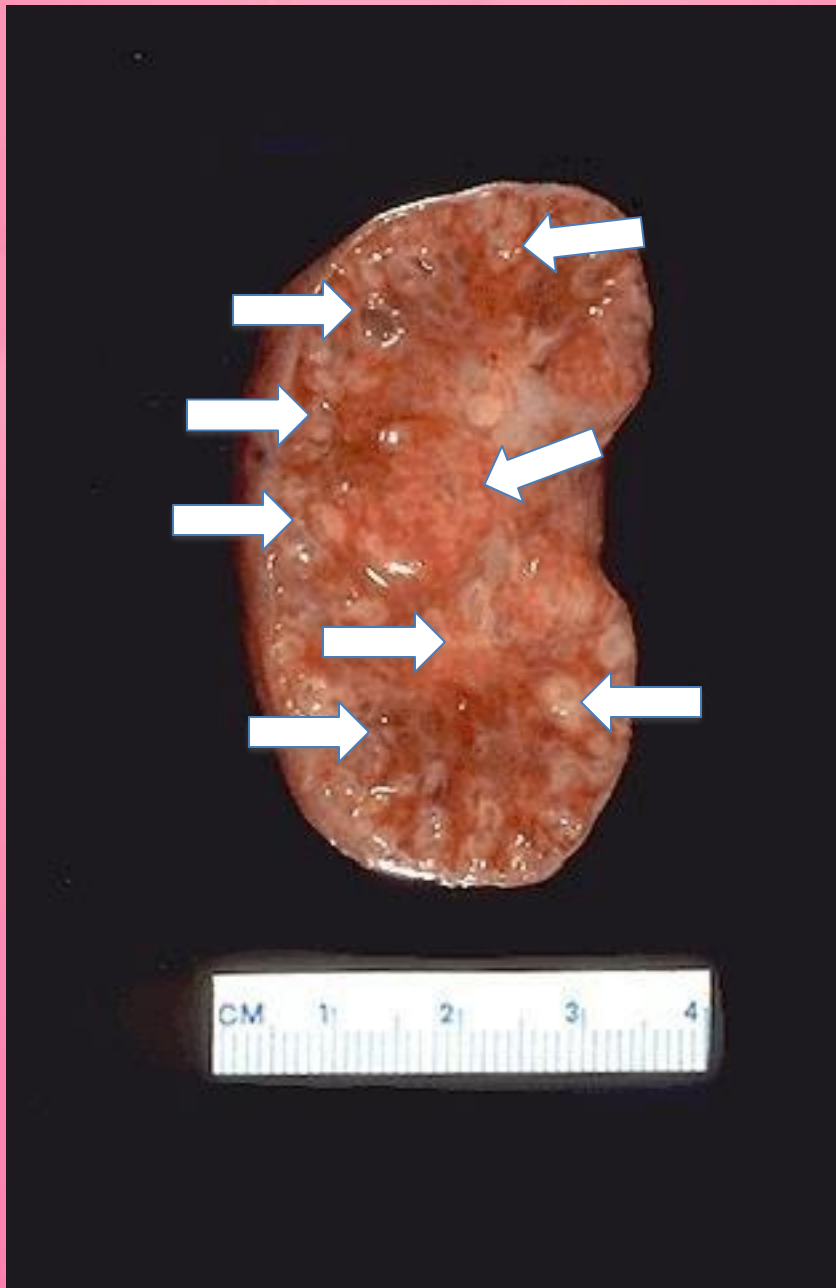
SIMPLE RENAL CYST

Here is a much larger simple renal cyst of the upper pole. Other smaller cysts are also scattered around the kidney. The ureter exits south on the left. Such a large renal cyst would be seen on a radiographic imaging procedure, but could probably be distinguished from a neoplasm by its fluid density.



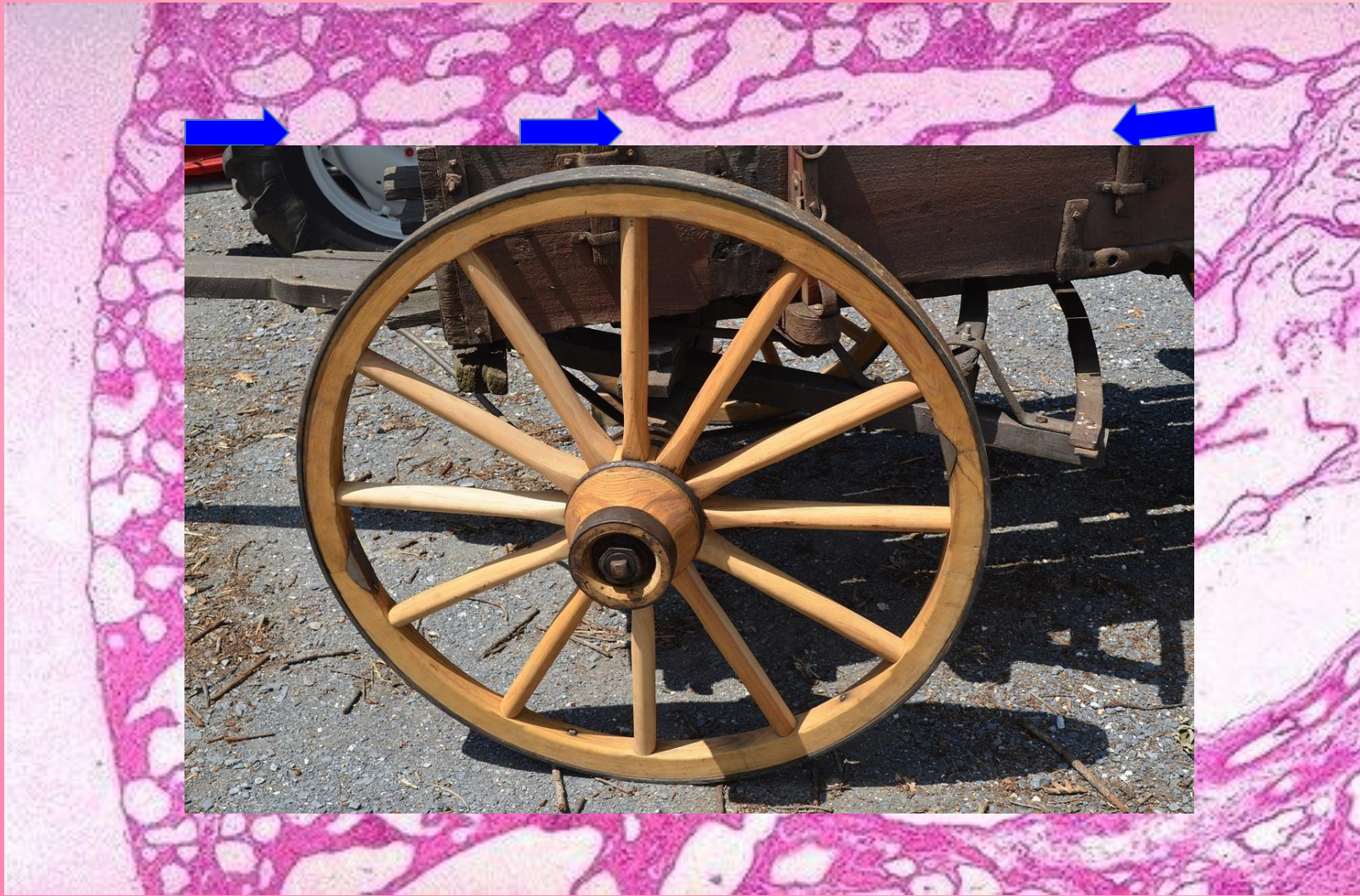
AUTOSOMAL RECESSIVE POLYCYSTIC KIDNEY DISEASE (ARPKD)

Here is a cut section of a kidney with autosomal recessive polycystic kidney disease (ARPKD). Note that the cysts are fairly small but uniformly distributed throughout the parenchyma so that the disease is usually symmetrical in appearance, with both kidneys markedly enlarged. The recurrence risk for this disease is, of course, 25% because of the autosomal recessive inheritance pattern. Affected babies usually do not survive long.

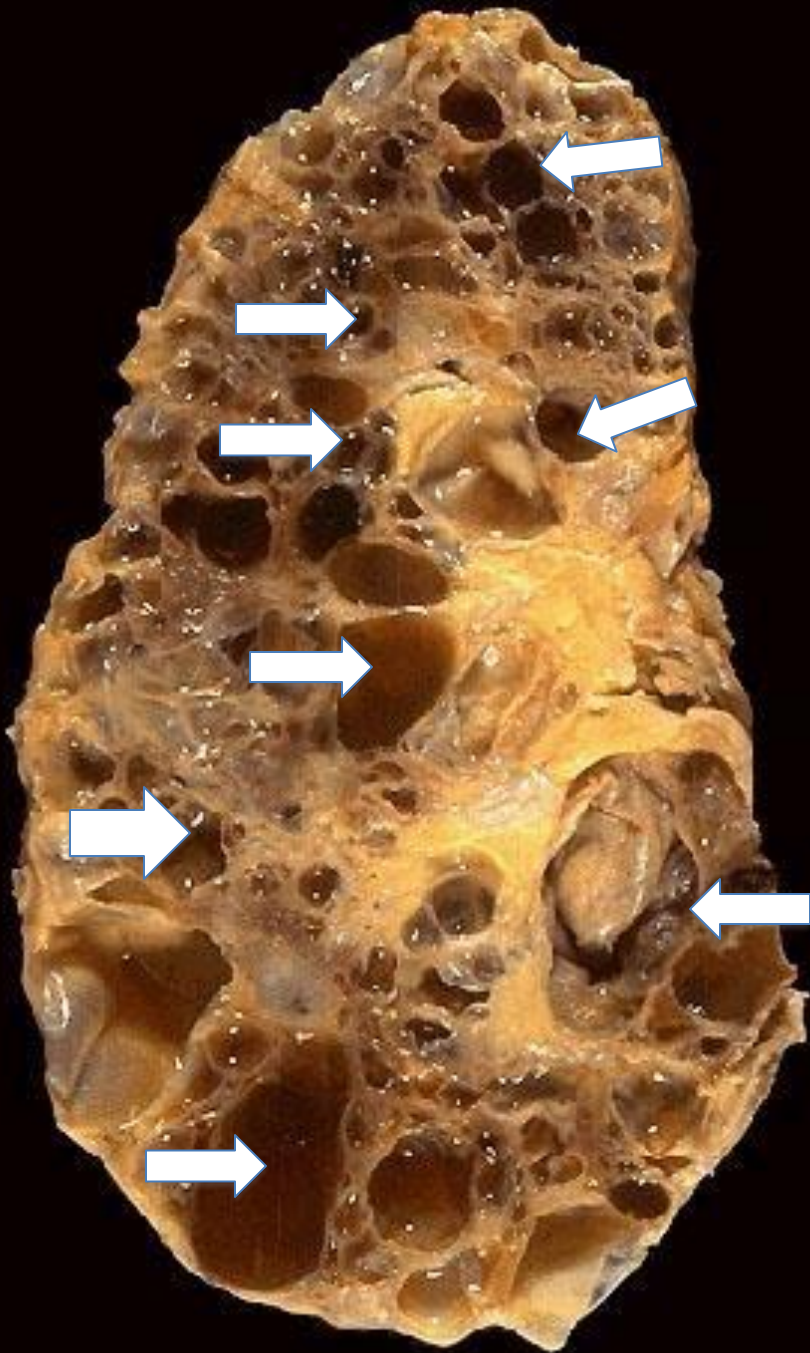


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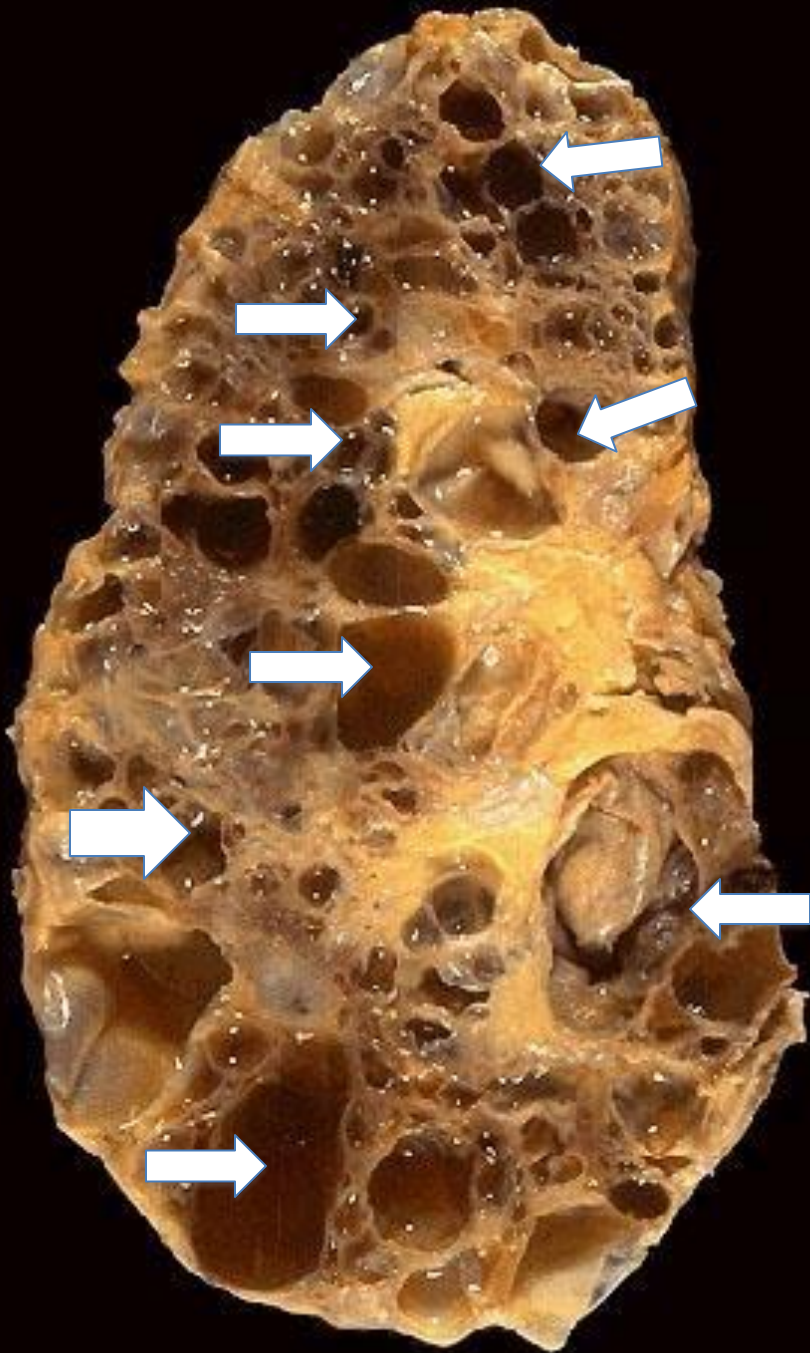


Autosomal recessive polycystic kidney disease (ARPKD) is characterized microscopically by elongated cysts that fill most of the parenchyma, and it is hard to find glomeruli. Many of the cysts are elongated and radially arranged from the center of the kidney on the right, much like spokes on a wagon wheel.



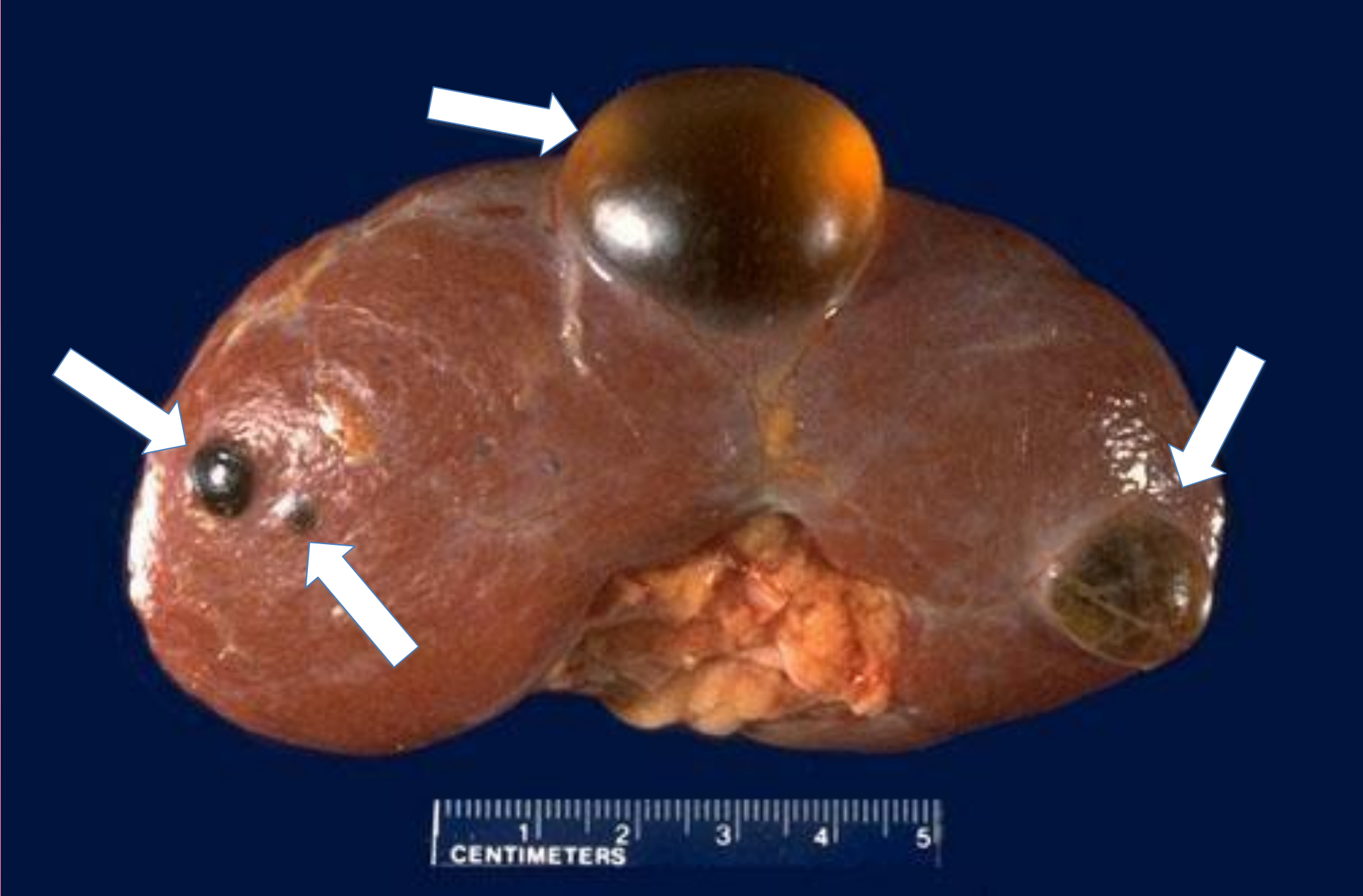
AUTOSOMAL DOMINANT POLYCYSTIC KIDNEY DISEASE (ADPKD)

This kidney in a patient with ADPKD weighed 3 kilograms! This disease is inherited with an autosomal dominant pattern, so the recurrence risk in the family is 50%. The cysts are not usually present at birth, but develop slowly over time, so the onset of renal failure occurs in middle age to later adult life.

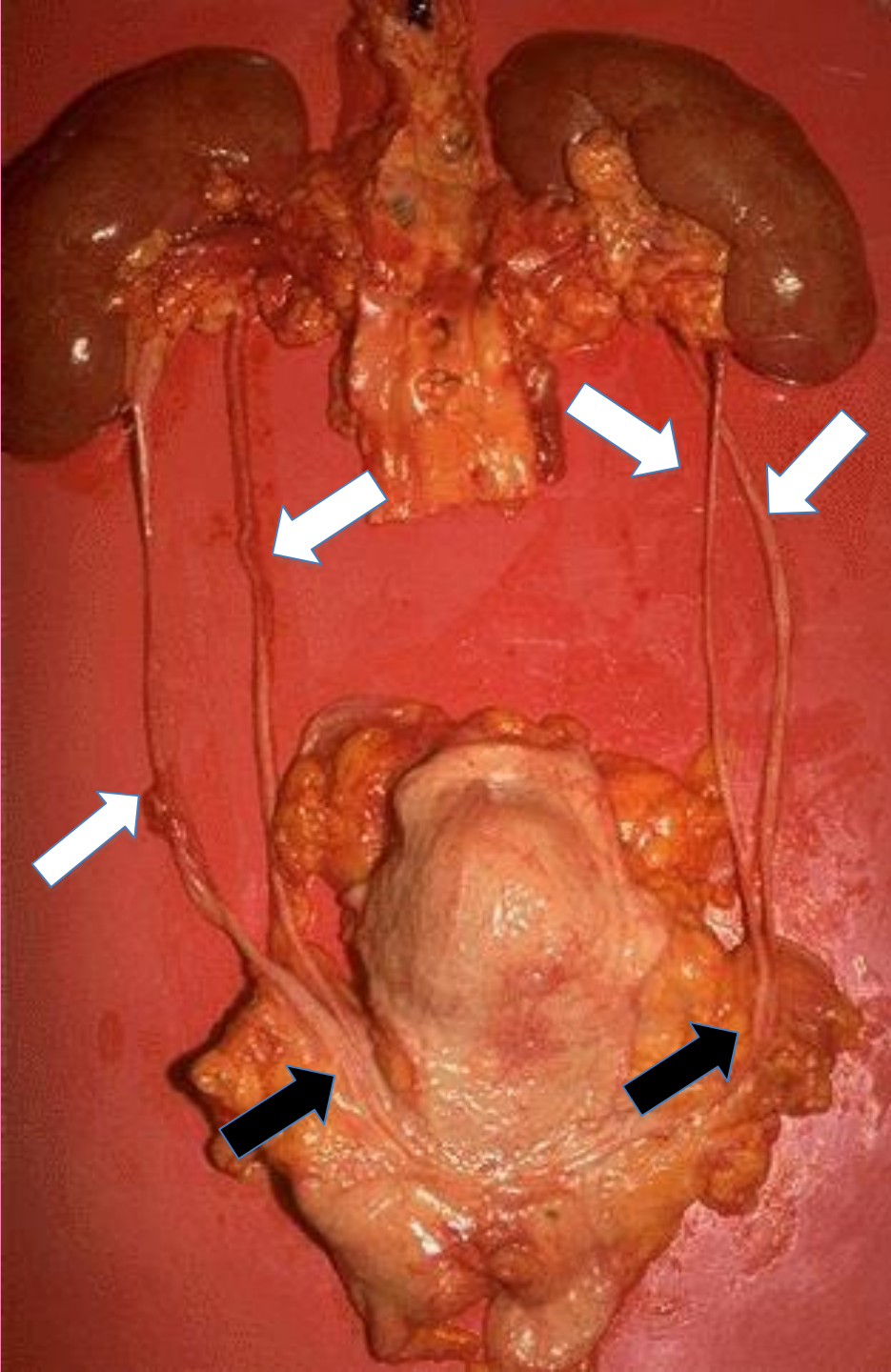


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SIMPLE RENAL CYSTS, as seen here, can also be multiple, but they are never as numerous as with polycystic change.



DOUBLE URETERS

Double ureters are seen exiting from each kidney and extending to the bladder that has been opened. A small segment of aorta is seen between the normal, smooth-surfaced kidneys. A partial or complete duplication of one or both ureters occurs in about 1 in 150 persons. There is a potential for obstructive problems due to the abnormal flow of urine and the entrance of two ureters into the bladder in close proximity, but most of the time this is an incidental finding (except to a urologist).

Glomerulonefritis Progresif Cepat (GNPC / RPGN - Rapidly Progressive Glomerulonephritis)

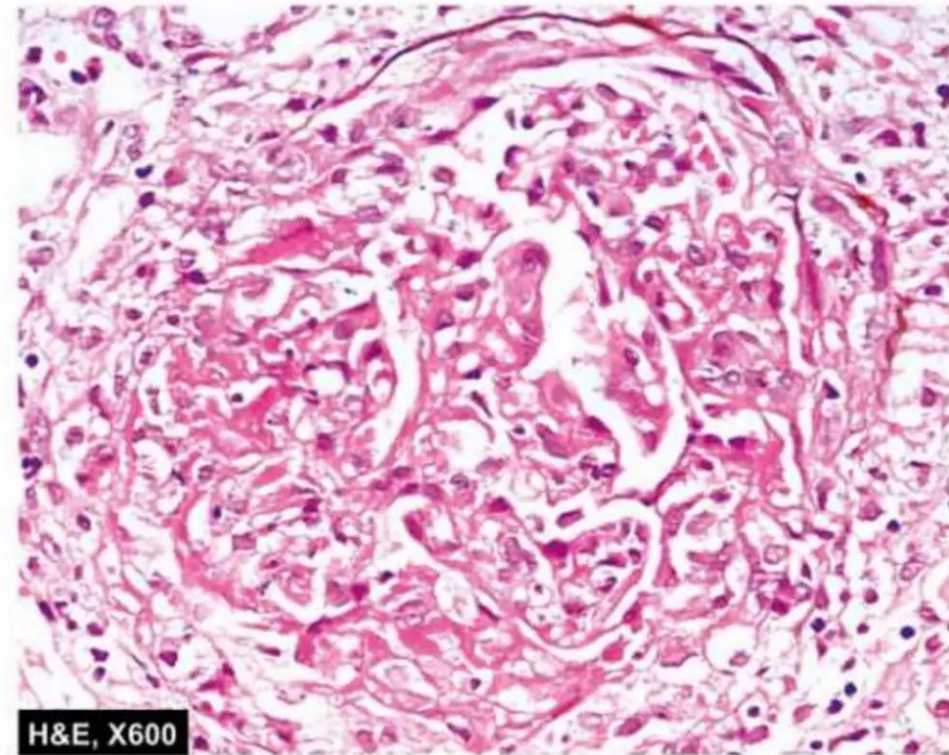
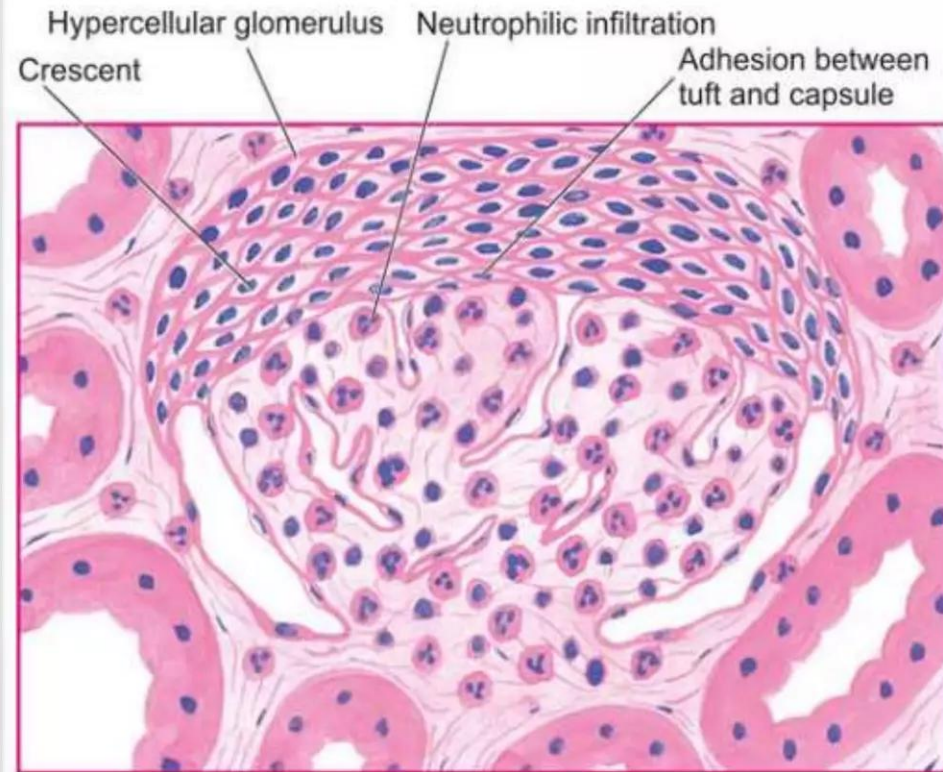
- RPGN dikategorikan sebagai kondisi gawat darurat ginjal (renal emergency) karena penurunan drastis fungsi ginjal (penurunan LFG sebesar 50% dalam waktu tiga bulan)

- Gambaran Histopatologi Khas:

Ditemukannya struktur bulan sabit (crescent) yang mengobliterasi (menyumbat) ruang Bowman pada sebagian besar glomerulus.

Crescent ini terbentuk akibat cedera membran basal glomerulus (GBM) yang berat, sehingga leukosit, fibrin, makrofag, dan fibroblas masuk serta berproliferasi di ruang Bowman

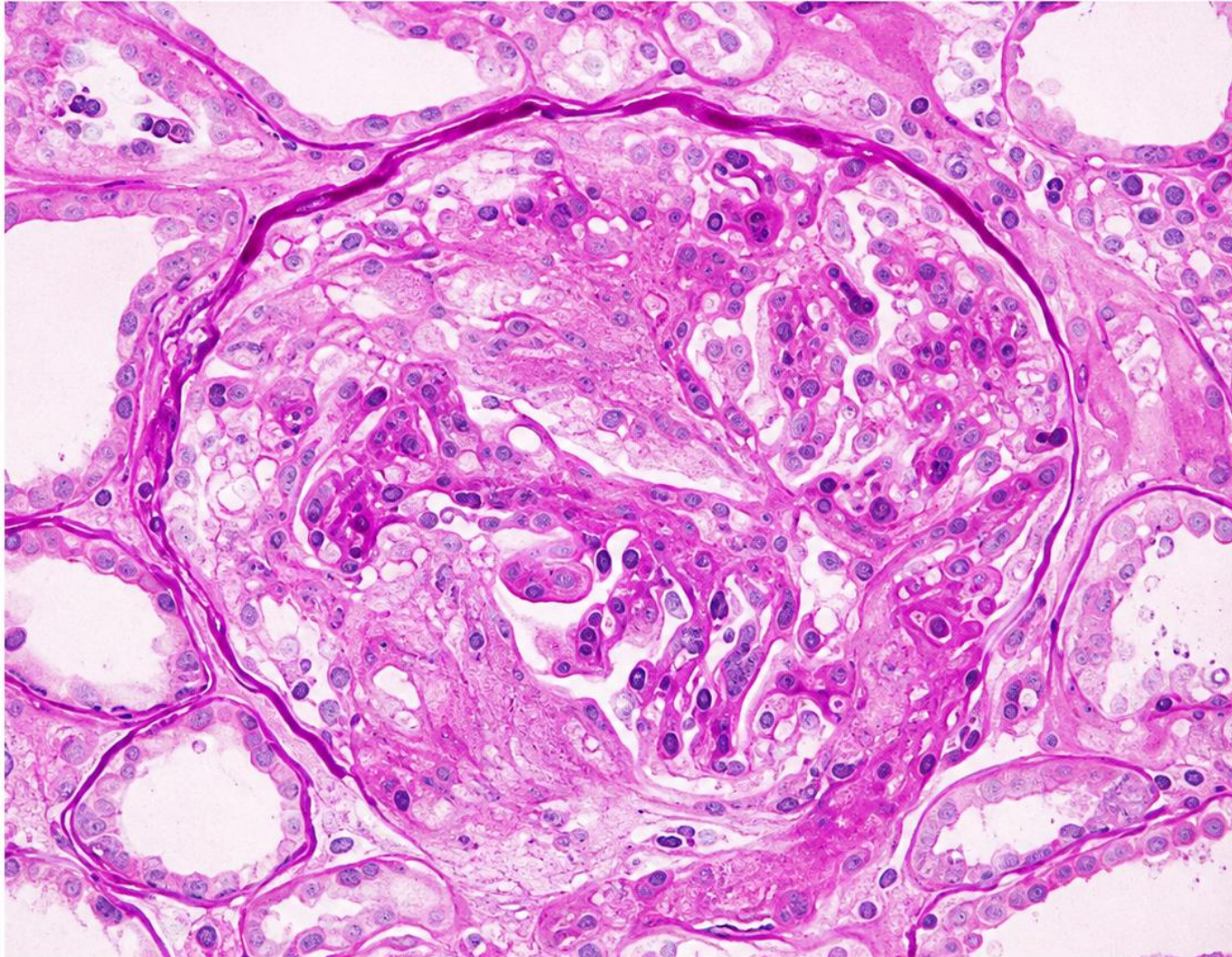
RPGN (post-infectious type), Light microscopic appearance.



There are crescents in Bowman's space forming adhesions between the glomerular tuft and Bowman's capsule.

The tuft shows hypercellularity & leucocytic infiltration.

Rapidly Progressive Glomerulonephritis (RPGN)



Pewarnaan Histologi:
PAS (Periodic Acid–Schiff) Stain,
ditandai dengan membran basalis
glomerulus dan kapsula Bowman
yang terwarnai merah
keunguan/merah muda terang
(magenta).

Temuan Khas (Crescentic Shape):
Terlihat pembentukan "Crescent"
(bulan sabit) di dalam ruang
Bowman, yang disebabkan oleh
proliferasi sel epitel parietal serta
penumpukan makrofag dan
fibrin. Crescent ini menekan tuft
kapiler glomerulus di tengahnya.

Glomerulonefritis Akut (GNA / GNAPS)

Berbeda dengan RPGN yang bersifat progresif merusak, GNA (khususnya Glomerulonefritis Akut Pasca-Streptokokus / GNAPS) sering kali bersifat self-limiting (bisa sembuh sendiri) jika ditangani dengan baik.

- Patofisiologi: Reaksi imunologi tipe III di mana kompleks antigen-antibodi (seperti protein ekskretori streptokinase dari bakteri *Streptococcus* β -hemolyticus grup A) terperangkap di membran basalis glomerulus. Hal ini mengaktifkan komplemen C3a & C5a, memicu pelepasan sitokin proinflamasi (TNF- α , IL-1), dan mendatangkan sel radang.
- Gambaran Histopatologi: Hipertrofi glomerulus yang ditandai dengan hiperselularitas (pembengkakan sel, proliferasi sel endotel, sel epitel, dan sebaran sel radang) yang berujung pada penyempitan ruang Bowman serta hemoragik

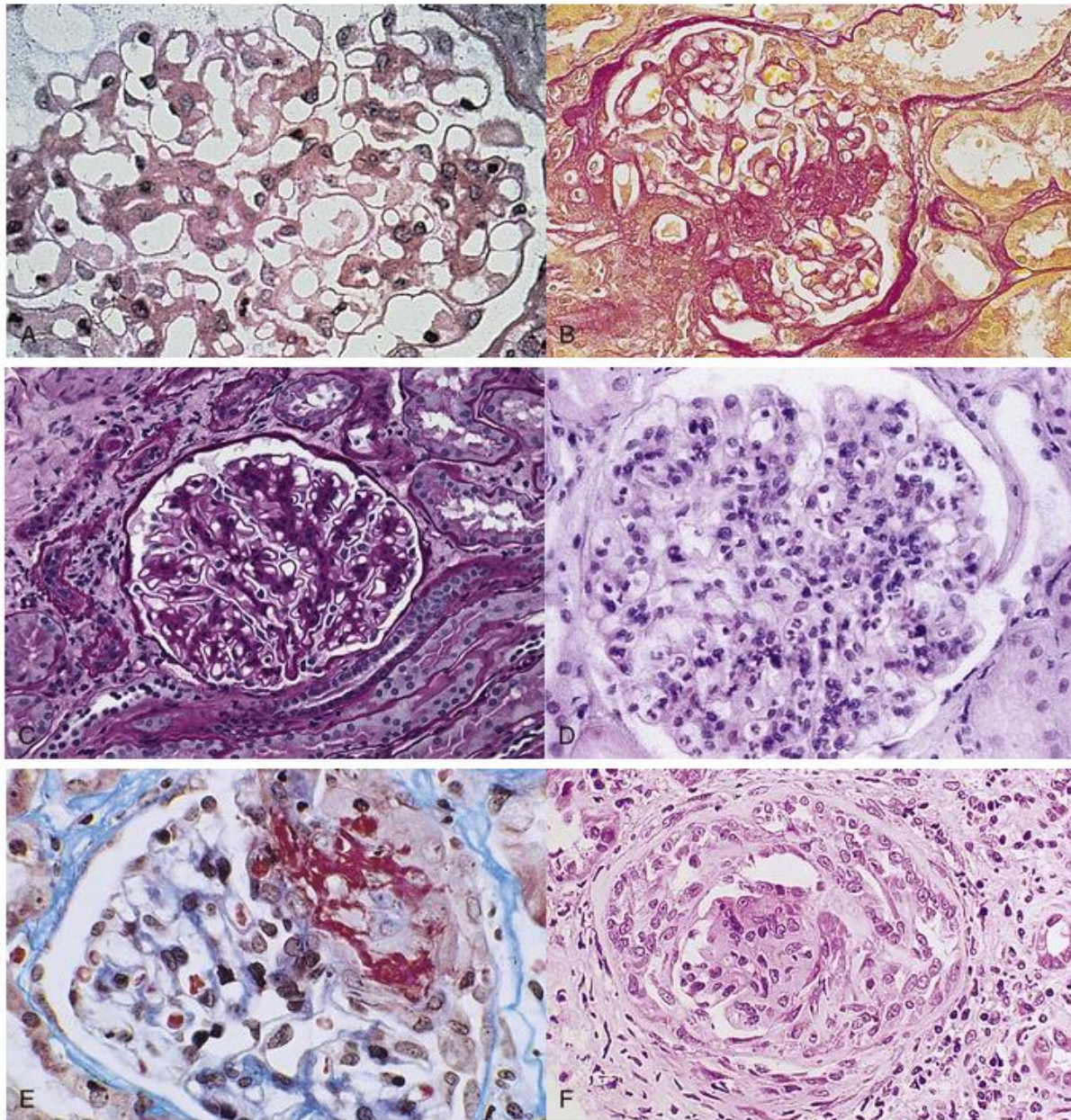
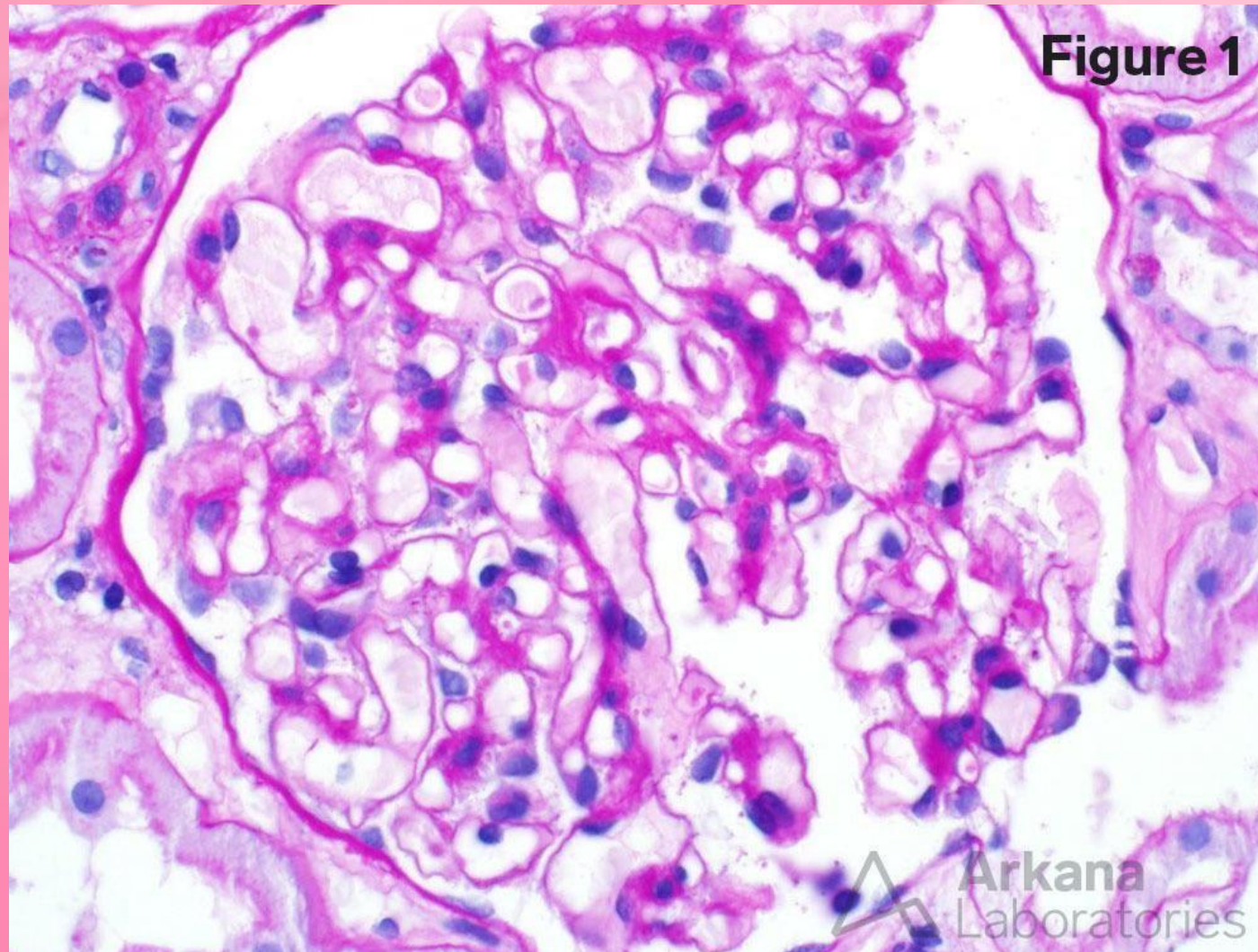


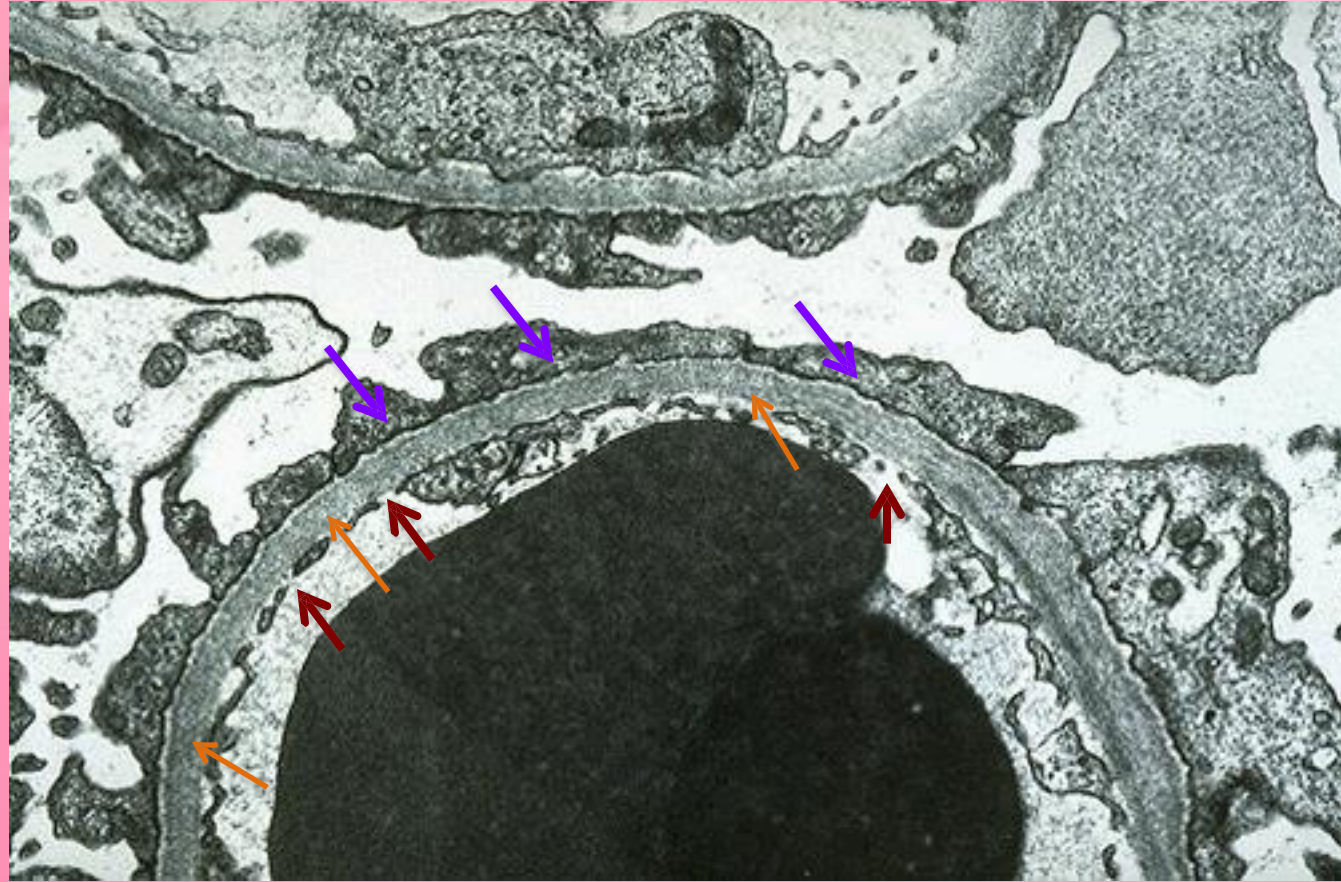
FIGURE 16-1 Pathology of glomerular disease: light microscopy. Characteristic patterns of glomerular disease illustrating the range of histologic appearances and the descriptive terms used. **A**, Normal glomerulus: minimal change disease. **B**, Segmental sclerosis: focal segmental glomerulosclerosis. **C**, Diffuse mesangial hypercellularity: IgA nephropathy. **D**, Diffuse endocapillary hypercellularity: poststreptococcal glomerulonephritis. **E**, Segmental necrosis: renal vasculitis. **F**, Crescent formation: anti-glomerular basement membrane disease. (A and B, Hematoxylin-eosin; C, D, and F, periodic acid-Schiff; E, trichrome.)

Patologi of glomerular disease (light microscopy). **Karakteristik utama** of glomerular lesions featuring the range of morphologic patterns and the pathologic lesions seen.

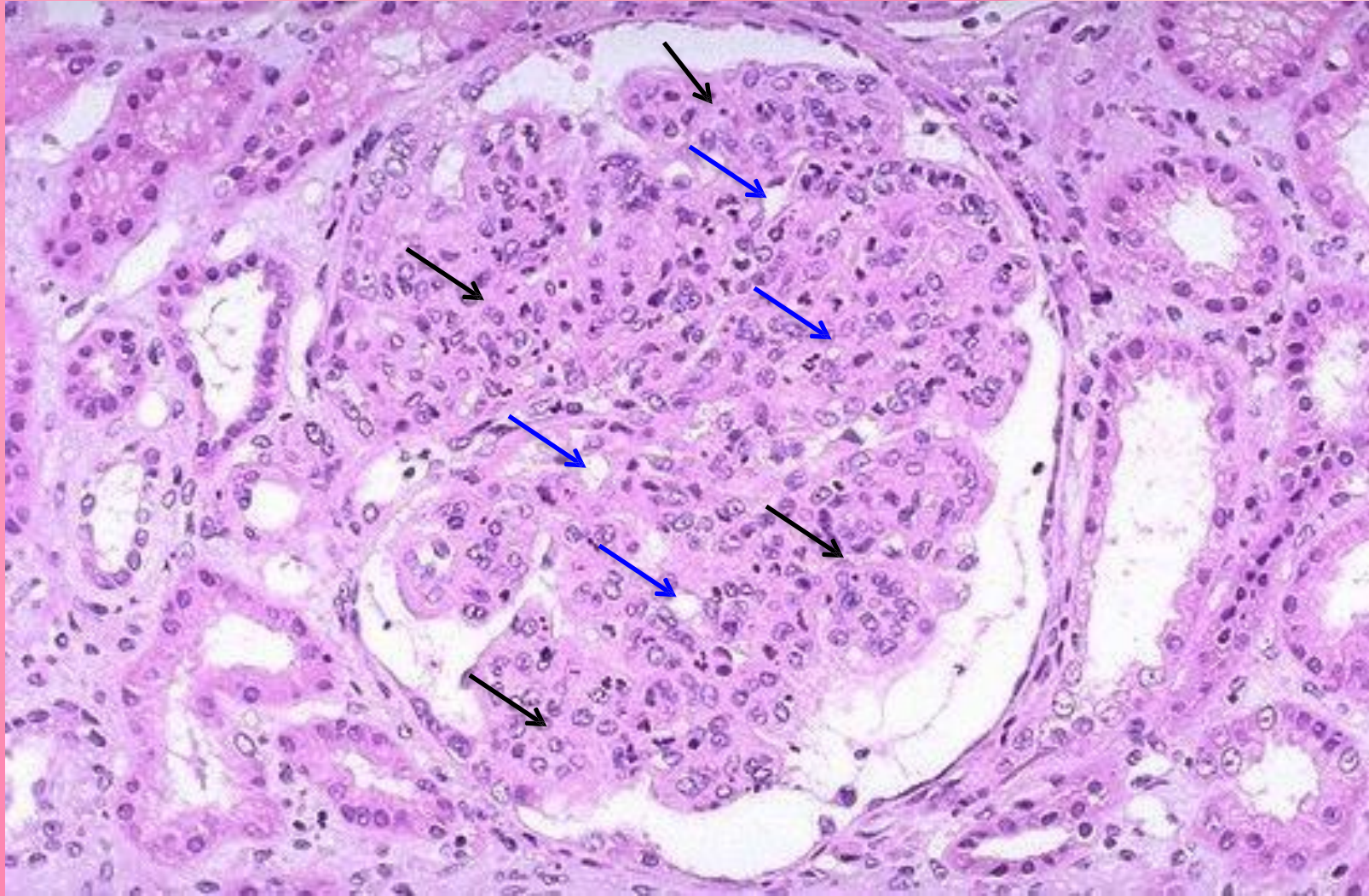
- A. Normal glomerulus:** normal design feature.
- B. Segmental sclerosis:** focal segmental glomerulosclerosis.
- C. Diffuse mesangial hypercellularity:** IgA nephropathy.
- D. Diffuse endocapillary hypercellularity:** postinfectious glomerulonephritis.
- E. Segmental necrosis:** crescentic.
- F. Global sclerosis:** early glomerular/basement membrane disease.



MINIMAL CHANGE DISEASE (MCD). By light microscopy, the glomerulus is normal with MCD.



This is **MINIMAL CHANGE DISEASE (MCD)** In this electron micrograph, the capillary loop in the lower half contains two electron dense RBC's. **Fenestrated endothelium** is present, and the **basement membrane** is normal. However, **overlying epithelial cell foot processes are effaced (giving the appearance of fusion)** and run together.



This glomerulus is **hypercellular** and **capillary loops are poorly defined**. This is a type of **PROLIFERATIVE GLOMERULONEPHRITIS** known as **POST-INFECTIOUS GLOMERULONEPHRITIS**. This case followed a **group A beta hemolytic streptococcal infection** of the pharynx 3 weeks earlier, and thus it could be termed '**POST-STREPTOCOCCAL GLOMERULONEPHRITIS**'.

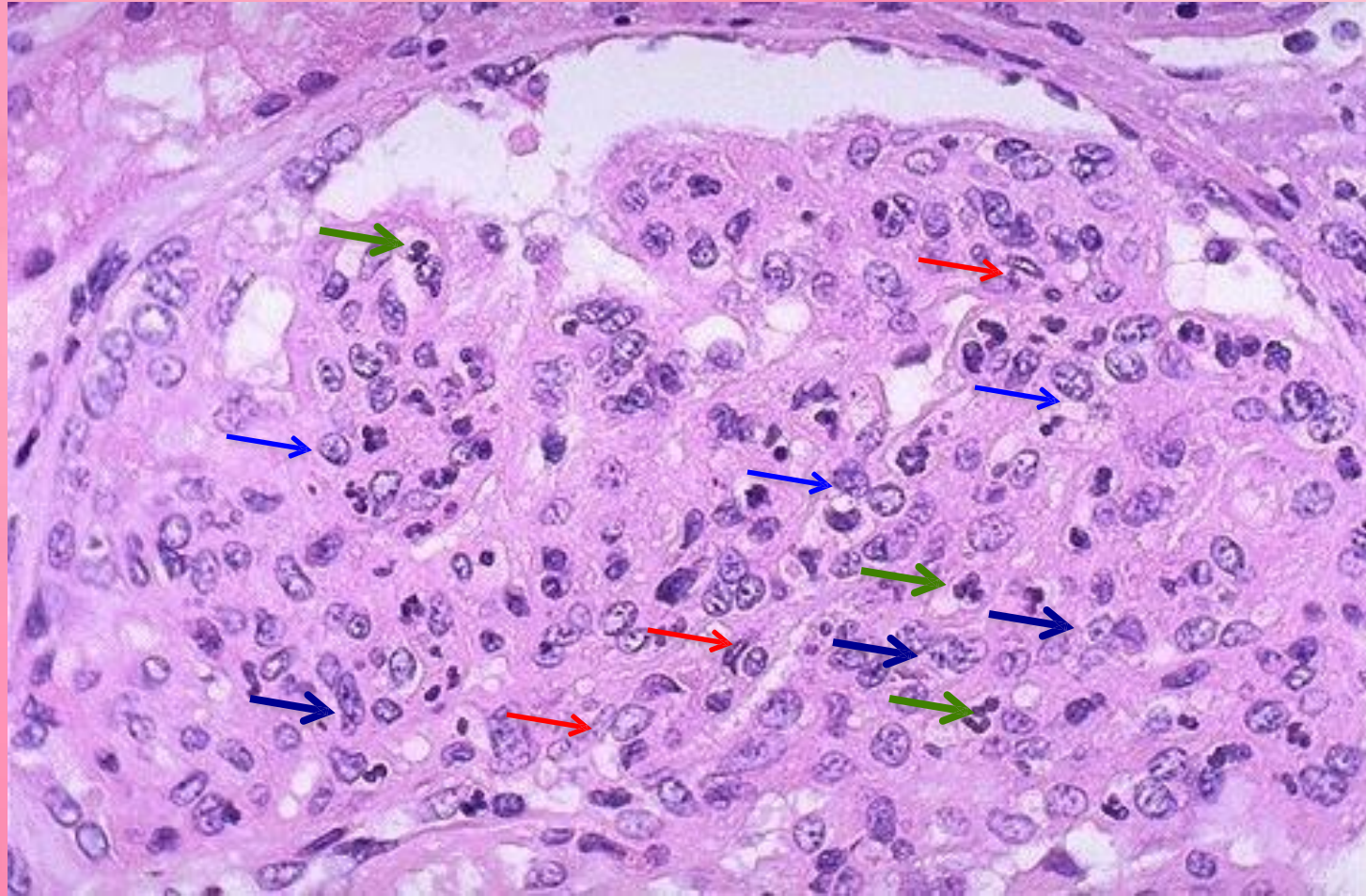
Nephrons

Streptococci

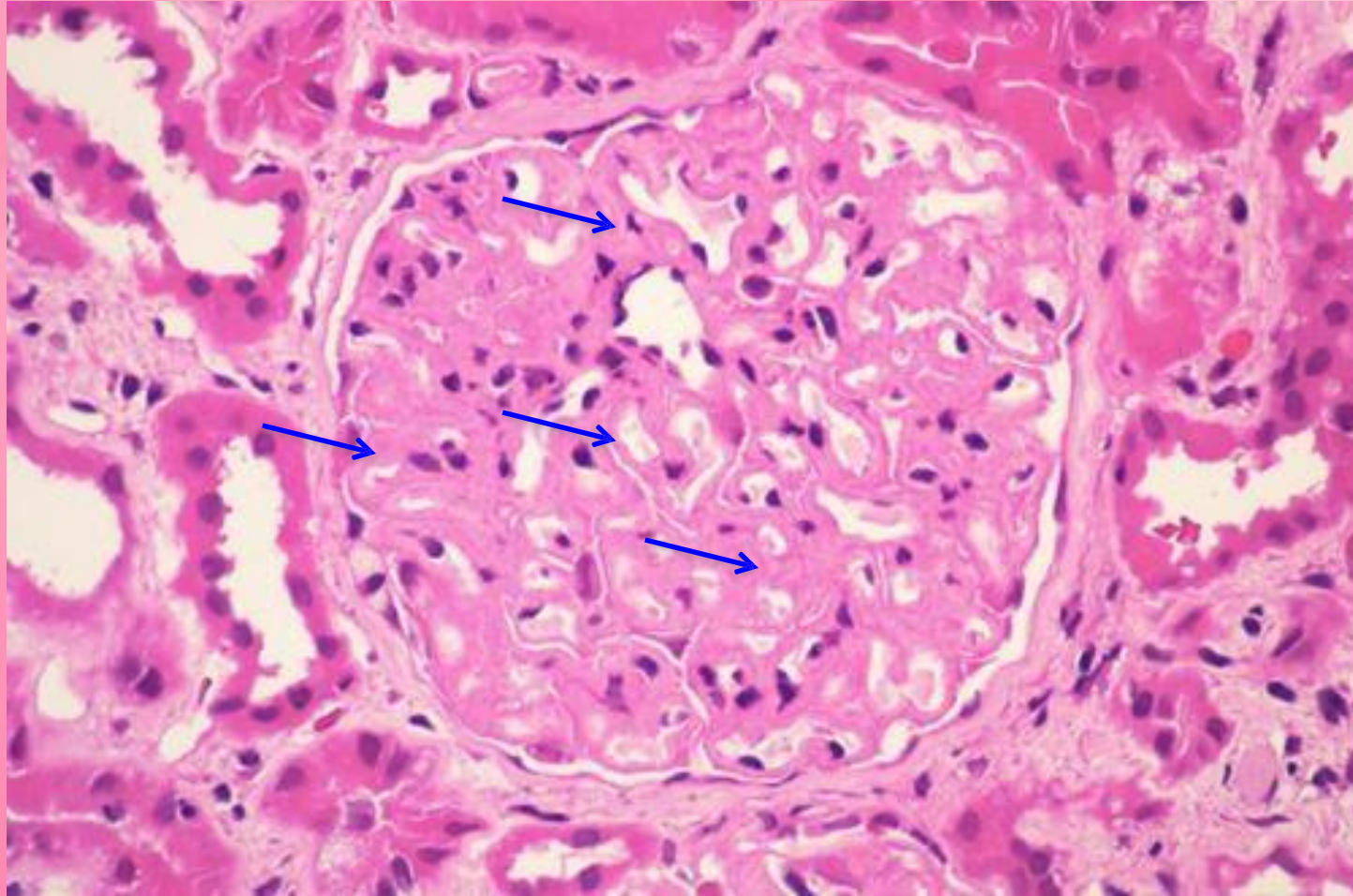


Antibodies

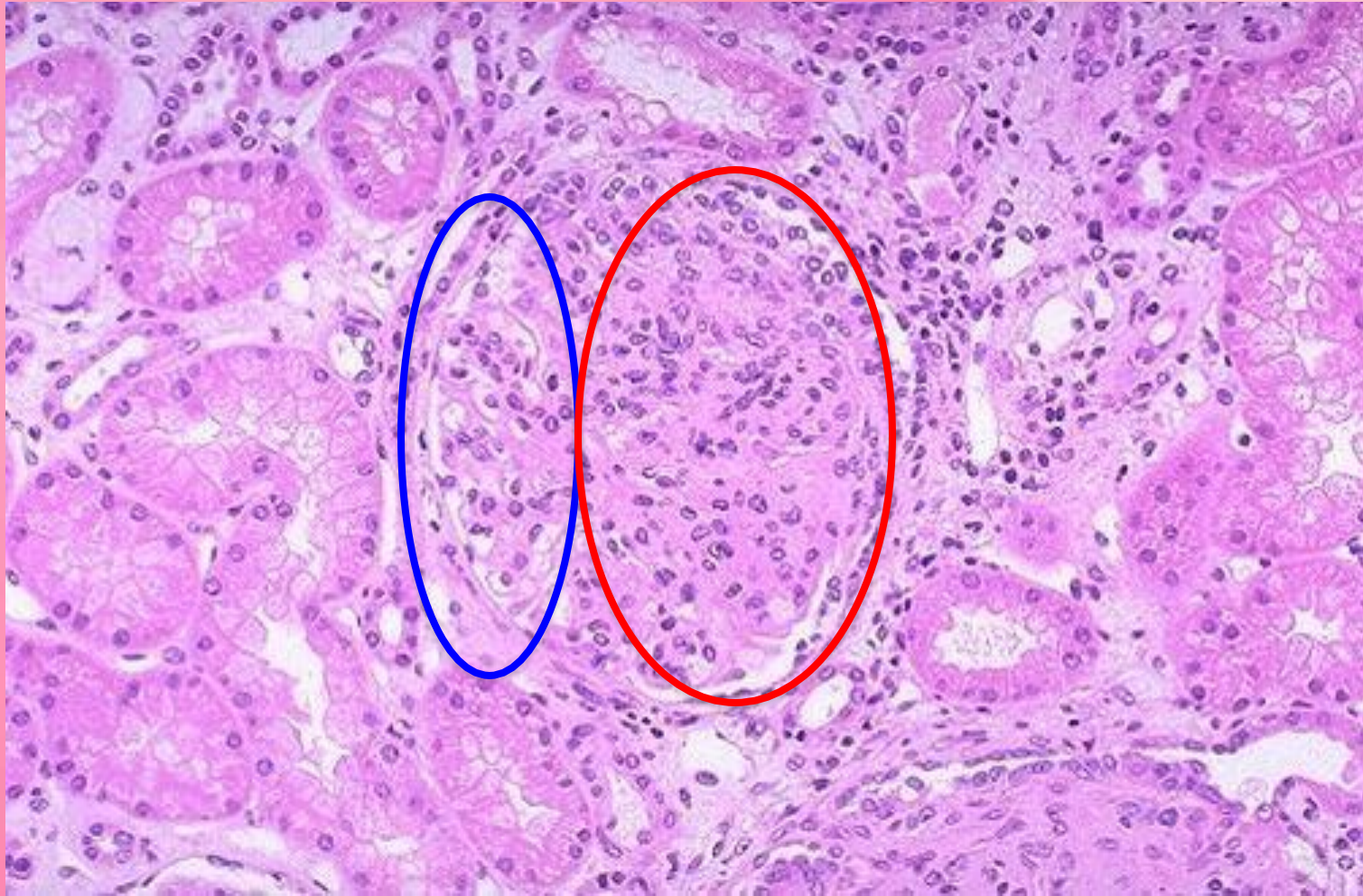




The hypercellularity of post-infectious glomerulonephritis is due to increased numbers of **epithelial**, **endothelial**, and **mesangial cells** as well as **neutrophils** in and around the glomerular capillary loops. This disease may follow several weeks after infection with certain strains of group A beta hemolytic streptococci. Patients who have had a strep infection typically have an elevated anti-streptolysin O (ASO) titer.



Here is the light microscopic appearance of membranous nephropathy in which **the capillary loops are thickened and prominent**, but the cellularity is not increased. **MEMBRANOUS GLOMERULONEPHRITIS (MGN)** is the most common cause for nephrotic syndrome in adults. In most cases there is no underlying condition present (idiopathic). However, some cases of membranous GN can be linked to a chronic infectious disease such as hepatitis B, a carcinoma, or SLE.

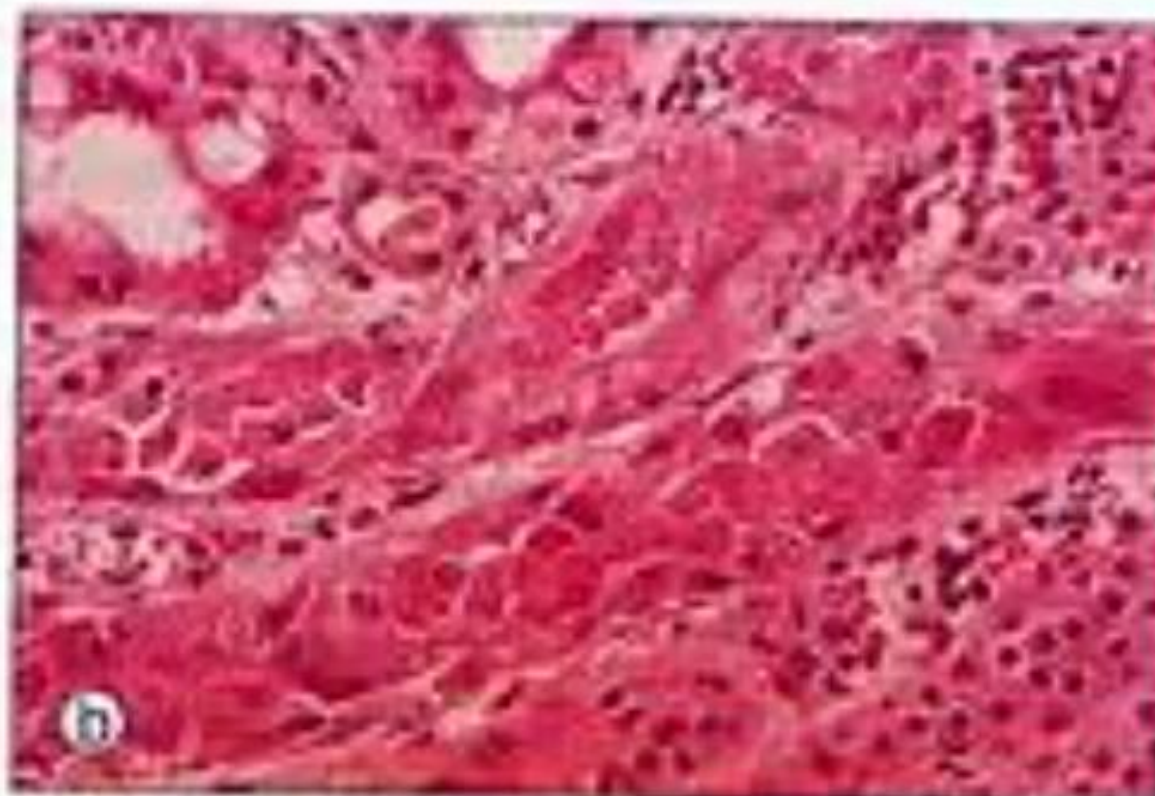


This is **MEMBRANOPROLIFERATIVE GLOMERULONEPHRITIS (MPGN)**. Those cases that are idiopathic and 'primary'. The pathologic findings shown here include increased glomerular overall cellularity, mainly increased mesangial cellularity. Some cases of MPGN are 'secondary' to underlying conditions such as autoimmune diseases, infections, and malignancies, among others.

Nekrosis Tubular Akut (NTA / ATN - Acute Tubular Necrosis) Nefrotoksik

- ATN merupakan penyebab gagal ginjal akut renal yang sangat penting dipelajari, salah satunya dipicu oleh paparan zat kimia atau obat-obatan dosis toksik (misalnya overdosis Parasetamol).
- Patogenesis: Cedera langsung pada sel epitel tubulus kontortus proksimal (TKP) akibat akumulasi metabolit reaktif (NAPQI) saat cadangan antioksidan glutathione habis. Mekanisme kerusakannya mencakup apoptosis sel epitel akibat stres retikulum endoplasma (ER stress) serta vasokonstriksi ginjal yang menyebabkan iskemia dan hipoksia tubulus.
- Gambaran Histopatologi: Korteks ginjal tampak pucat dan edem, terjadi vakuolisasi sitoplasma (degenerasi hidropik) pada sel epitel tubulus proksimal, destruksi sel epitel, hingga hancur dan menutupnya lumen tubulus proksimal.

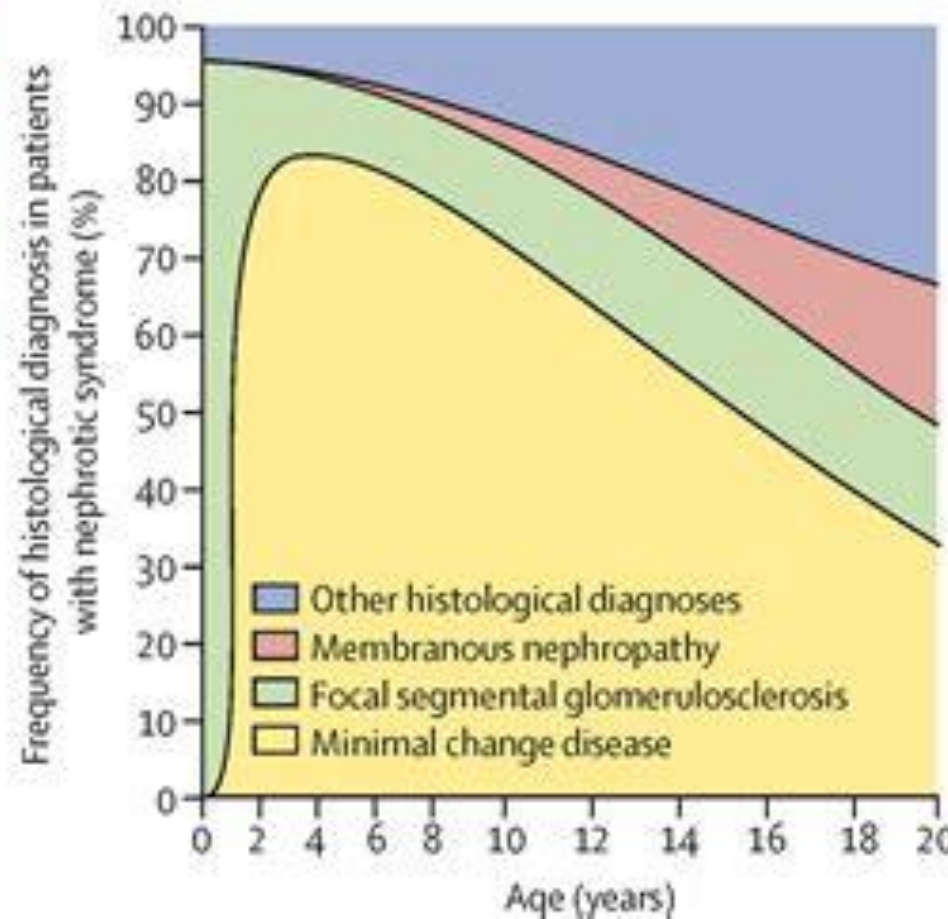
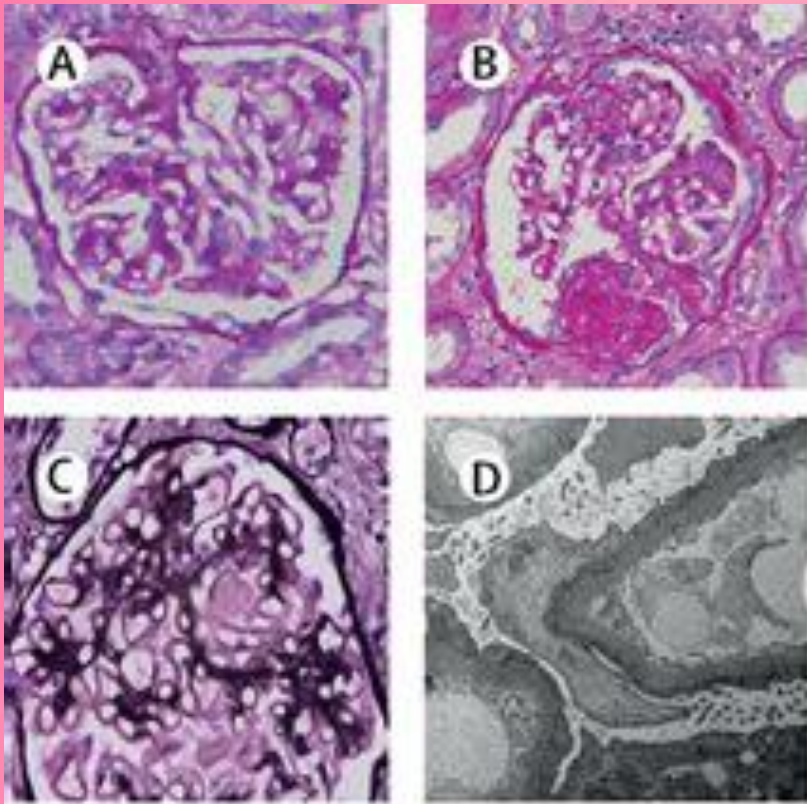
Acute Tubular Necrosis (ATN) Histopathology





Sindrom Nefrotik (*Nephrotic Syndrome*)

- Mahasiswa harus mampu membedakan sindrom nefritik (inflamasi) dengan sindrom nefrotik (gangguan filtrasi protein)
- Patofisiologi: Terjadi kerusakan atau peningkatan permeabilitas membran kapiler glomerulus
- Manifestasi Klinis Utama: Kehilangan protein darah secara masif melalui urin (proteinuria masif $>3,5$ gram/24 jam), yang menyebabkan penurunan albumin darah (hipoalbuminemia), edema anasarka, hiperlipidemia (perubahan profil lipid), hipertensi, dan kelemahan fisik.



Penyebab utama Sindrom Nefrotik pada usia 0–20 tahun

1. Grafik Distribusi Berdasarkan Usia

- Minimal Change Disease (Kuning): Penyebab paling dominan pada anak-anak/balita (mencapai >80% pada usia 2–6 tahun), lalu proporsinya menurun seiring bertambahnya usia.
- FSGS & Membranous Nephropathy (Hijau & Merah Muda): Frekuensinya meningkat bertahap pada usia remaja dan dewasa muda.

2. Panel Histopatologi (A–D)

A (Minimal Change Disease): Glomerulus tampak relatif normal pada mikroskop cahaya biasa.

B (Focal Segmental Glomerulosclerosis / FSGS): Terlihat jaringan parut (sklerosis) pada sebagian segmen glomerulus.

C (Membranous Nephropathy): Penebalan difus pada dinding kapiler glomerulus.

D (Mikroskop Elektron): Penipisan/penyatuan tonjolan kaki sel podosit (effacement of foot processes).

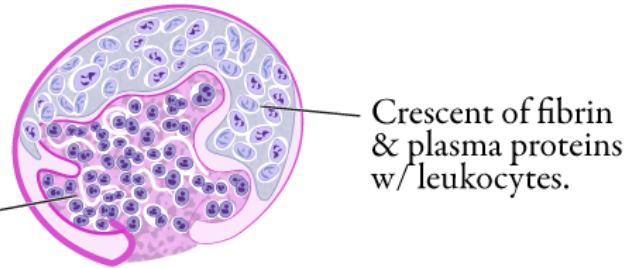
Nephritic Syndrome

HISTOPATH PATTERNS

- Focal or diffuse
- Segmental or global
- Membranoproliferative or proliferative

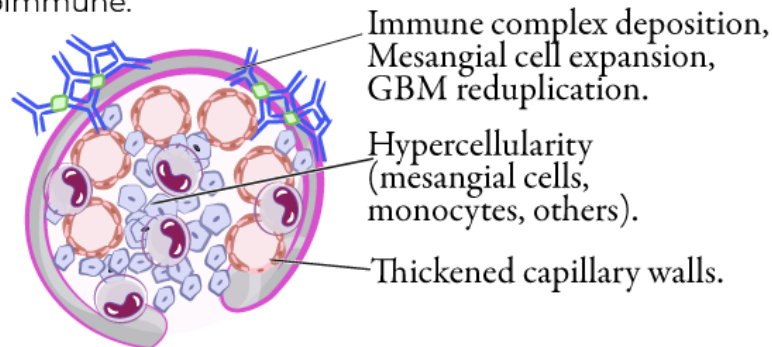
RAPIDLY PROGRESSIVE GN/ CRESCENTIC GN

- Rapidly deteriorating renal function.
- Poor prognosis.



MEMBRANOPROLIFERATIVE

- Mesangial hypercellularity, capillary wall thickening. GBM Splitting.
- Chronic infections, autoimmune.



DIFFUSE PROLIFERATIVE

- Mesangial or capillary proliferation.
- Wire looping caps.
- Diffuse = > 50% of glomeruli are involved. IgA, SLE.



Tiga jenis kerusakan ginjal pada Sindrom Nefritik:

- Crescentic GN (RPGN): Ditandai bentukan bulan sabit (crescent) dari fibrin dan sel radang; fungsi ginjal turun sangat cepat dan prognosisnya buruk.
- Membranoproliferative (MPGN): Ditandai penebalan/pemisahan membran ginjal (GBM splitting) dan penumpukan sel mesangial akibat endapan kompleks imun (sering karena infeksi kronis atau autoimun).
- Diffuse Proliferative: Ditandai perbanyakkan sel kapiler/mesangial luas (>50% glomerulus) dengan ciri khas wire looping; umumnya terkait Lupus (SLE) dan nefropati IgA.

Gagal Ginjal Akut (AKI) dan Penyakit Ginjal Kronik (CKD)

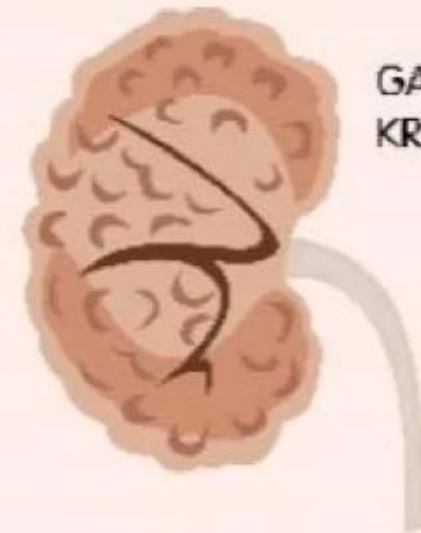
- Gagal Ginjal Akut (AKI / Acute Kidney Injury): Penurunan fungsi filtrasi nefron secara mendadak, ditandai dengan oliguria/anuria serta akumulasi senyawa nitrogen beracun di darah (uremia / azotemia) berupa peningkatan BUN dan kreatinin. Penyebabnya dikelompokkan menjadi prerenal (hipovolemia), renal (seperti RPGN atau ATN), dan postrenal (obstruksi urin).
- Penyakit Ginjal Kronik (CKD / Chronic Kidney Disease): Penurunan laju filtrasi glomerulus ($GFR < 60 \text{ mL/menit/1,73 m}^2$) yang berlangsung secara bertahap selama minimal 3 bulan. Kerusakan nefron yang progresif menyebabkan glomerulosklerosis yang bersifat ireversibel



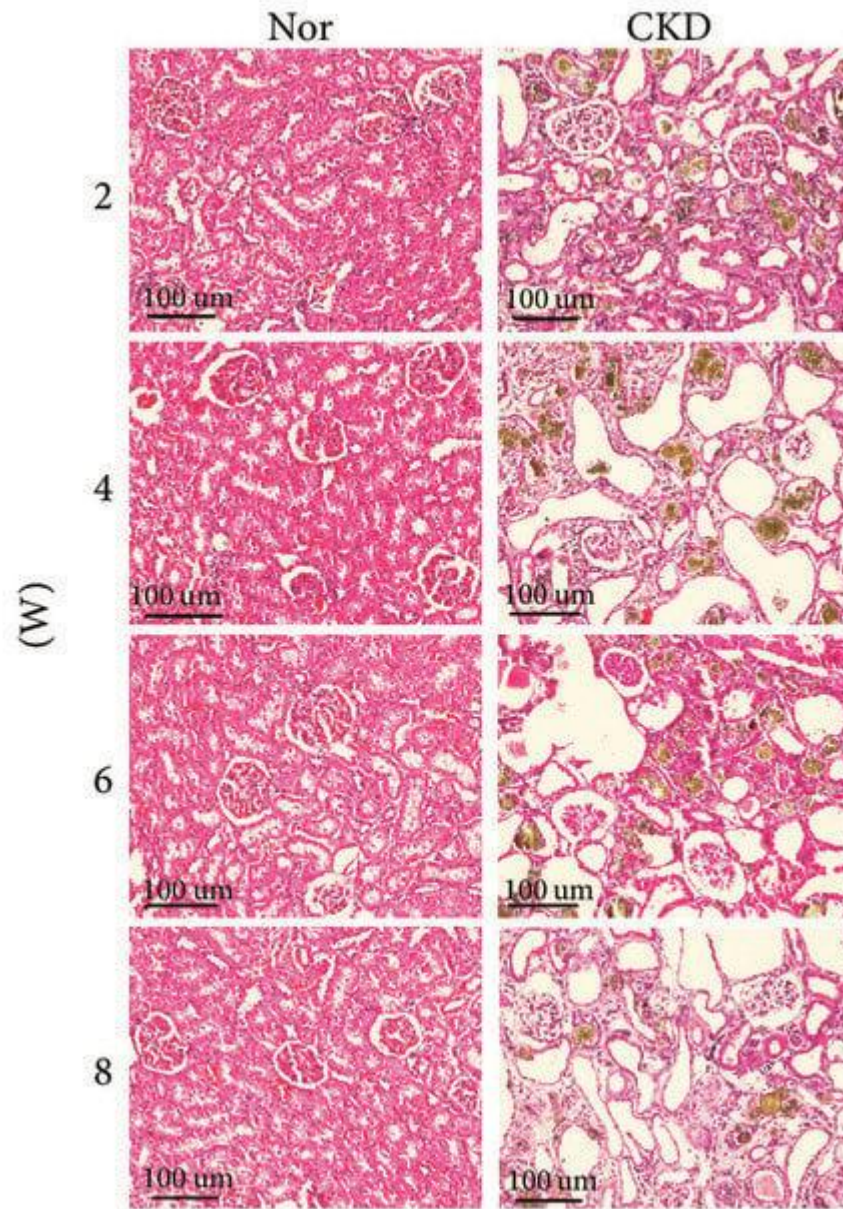
NORMAL



GAGAL GINJAL
AKUT



GAGAL GINJAL
KRONIS



Gambar perbandingan struktur histopatologi jaringan ginjal normal (Nor) dan model penyakit ginjal kronis (CKD) dari minggu ke-2 hingga minggu ke-8 (W2–W8):

- Kelompok Normal (Nor): Struktur glomerulus dan tubulus ginjal tampak padat, teratur, utuh, dan tidak mengalami perubahan signifikan seiring berjalannya waktu.
- Kelompok CKD (Chronic Kidney Disease): Menunjukkan kerusakan struktural progresif seiring bertambahnya minggu (W2 - W8), yang ditandai oleh:
 - Dilatasi dan atrofi tubulus: Saluran tubulus melebar secara abnormal dengan lumen yang kosong (area putih meluas).
 - Endapan kristal/massa kecokelatan: Tampak deposit material patologis di dalam tubulus (khas pada model CKD akibat induksi zat seperti adenin).
 - Fibrosis dan peradangan interstisial: Matriks jaringan penyokong rusak dan menipis seiring waktu.

Batu Ginjal (Nefrolitiasis / *Renal Stones*)

- Patogenesis: Terbentuk akibat konsentrasi mineral atau garam di dalam urin yang melebihi ambang batas jenuh (supersaturasi), sehingga zat tersebut mengendap dan mengkristal di dalam tubulus ginjal atau ureter.
- Jenis Batu & Faktor Risiko: Sebagian besar berupa batu kalsium (oksalat dan fosfat).
- Faktor pemicunya meliputi hiperkalsiuria, hiperoksalouria, pH urin abnormal, dehidrasi, serta asupan tinggi makanan tinggi oksalat (seperti bayam, kacang-kangan, dan ubi).

Kidney Stones

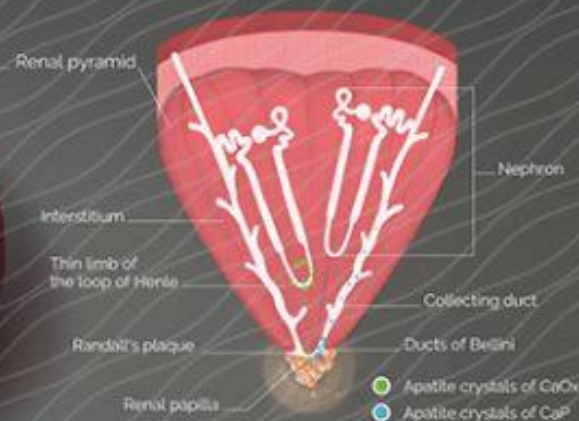
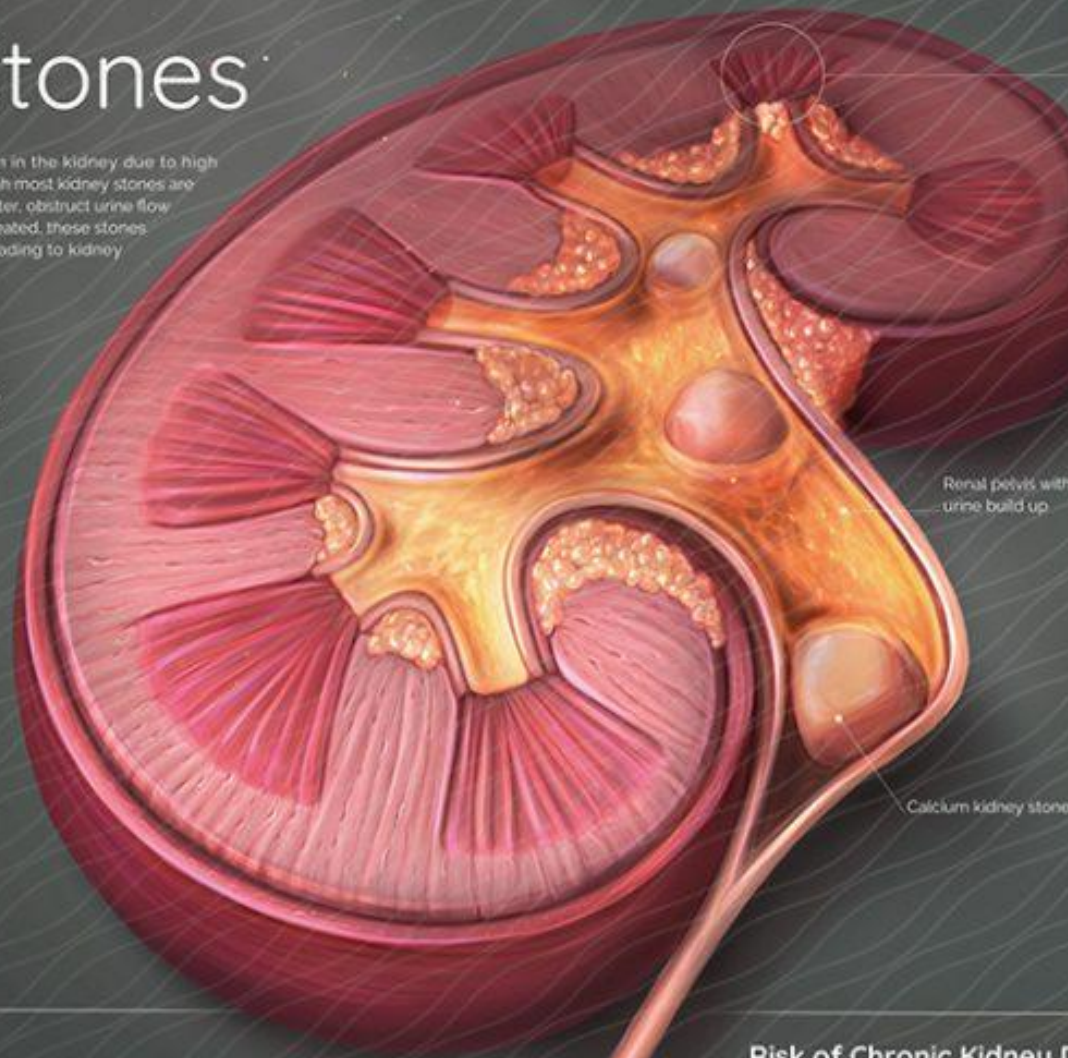
Kidney stones are crystalline deposits that form in the kidney due to high concentrations of certain salts in the urine. Although most kidney stones are small, large kidney stones (5-7mm) can plug the ureter, obstruct urine flow and cause severe pain in the abdomen. If left untreated, these stones may cause irreversible damage to the kidney, leading to kidney failure in the worst case scenario.

Calcium Stone Formation

Calcium stones, which are the most common type of kidney stones, occur when there is a high concentration of calcium in the urine.

There are two types of calcium stones: Calcium Oxalate (CaOx) and Calcium Phosphate (CaP) stones.

Both stones form when calcium crystals overgrow on plaques composed of minerals called **apate**. However, CaOx and CaP stones have different mechanisms and locations of formation.



Calcium Oxalate Formation

Apatite crystals emerge on the outer wall of the thin limb of the loop of Henle, and move through the interstitium to the renal papilla.

Randall's plaque is formed by apatite crystals at the renal papilla. The calcium oxalate crystals grow on the plaque to create CaOx stone.

The stones eventually fall off from the growth site, and large stones block the ureter, which obstructs urine flow and increases pressure and constricts blood vessels in the kidney.

Symptoms include severe pain the abdomen, nausea and vomiting, painful urination and blood in the urine.

Calcium Phosphate Formation

Apatite crystals fill the collecting tubule to the ducts of Bellini. Calcium phosphate crystals grow on the apatite plaque to form CaP stone.

Apatite crystals appear in the lumen of the terminal collecting duct.

Types of Kidney Stones



Name
Cause

Calcium (80%)
High calcium concentration in urine



Struvite (10%)
Urinary tract infection by micro-organism with urease



Uric Acid (9%)
High ammonia concentration in urine and low urine pH level



Cystine (1%)
Increased cystine urine absorption causes high cystine concentration

(% represents incidence)

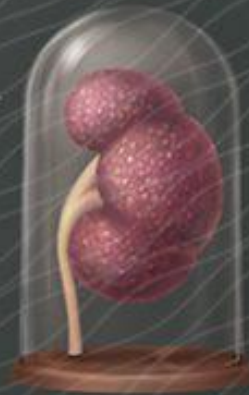
Risk of Chronic Kidney Disease

Patients who've had a kidney stone have an increased risk of **chronic kidney disease (CKD)**, which is a progressive loss of functional nephrons and kidney function.

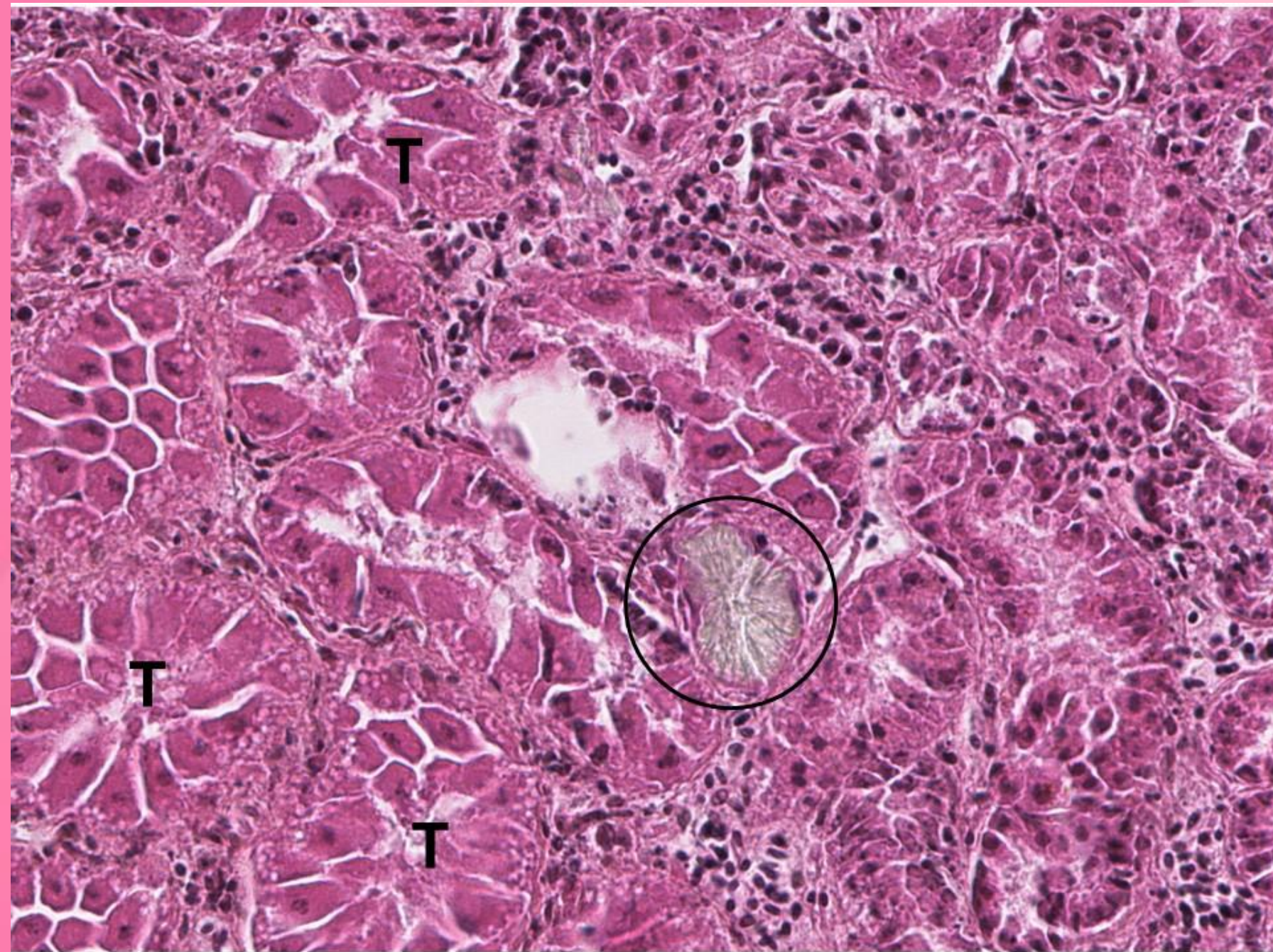
The mechanism by which kidney stone disease leads to CKD is largely unknown. However, it has been hypothesized that the obstruction of urine flow increases

the pressure within the kidney, causing renal vasoconstriction and renal blood flow reduction. This makes the kidney ischemic, and if blood flow restriction is severe, it may lead to CKD.

Due to damaged blood vessels, kidneys with CKD are reddish-brown and have a granular surface.



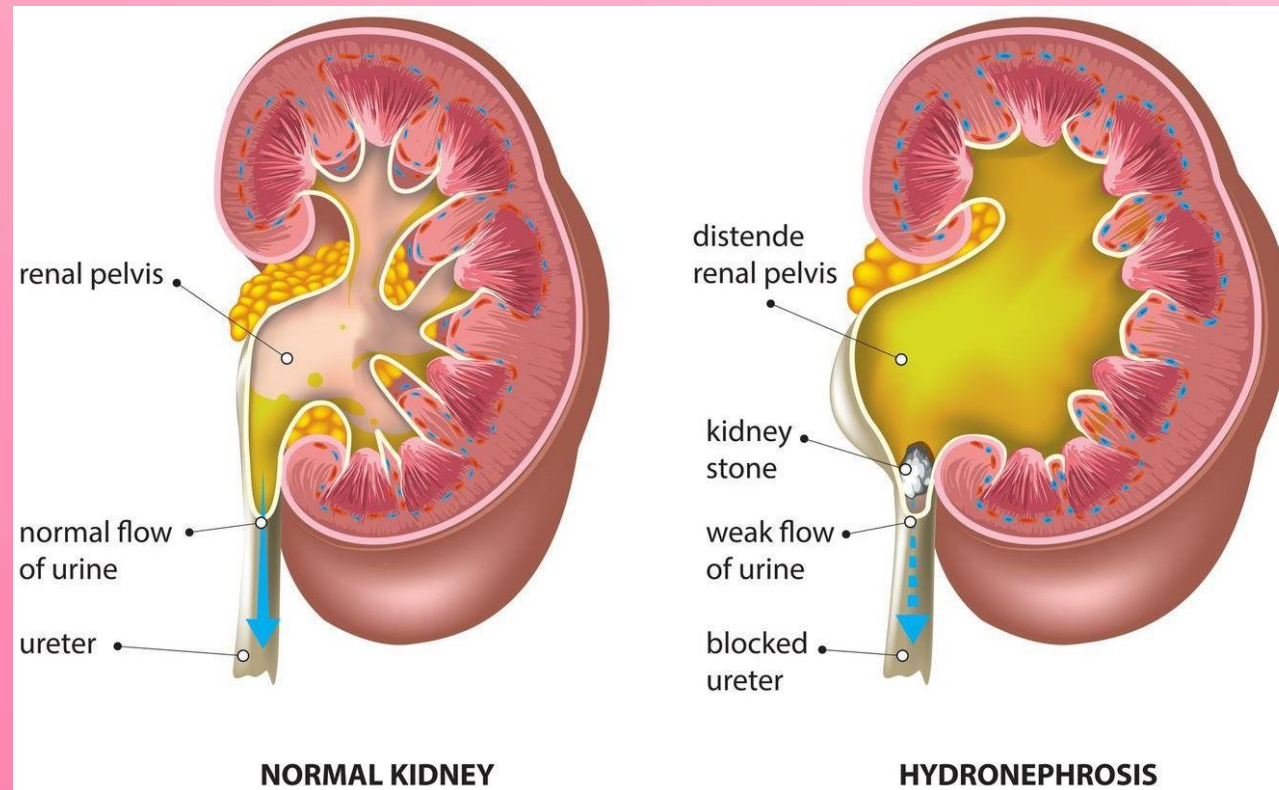
Kidney with CKD

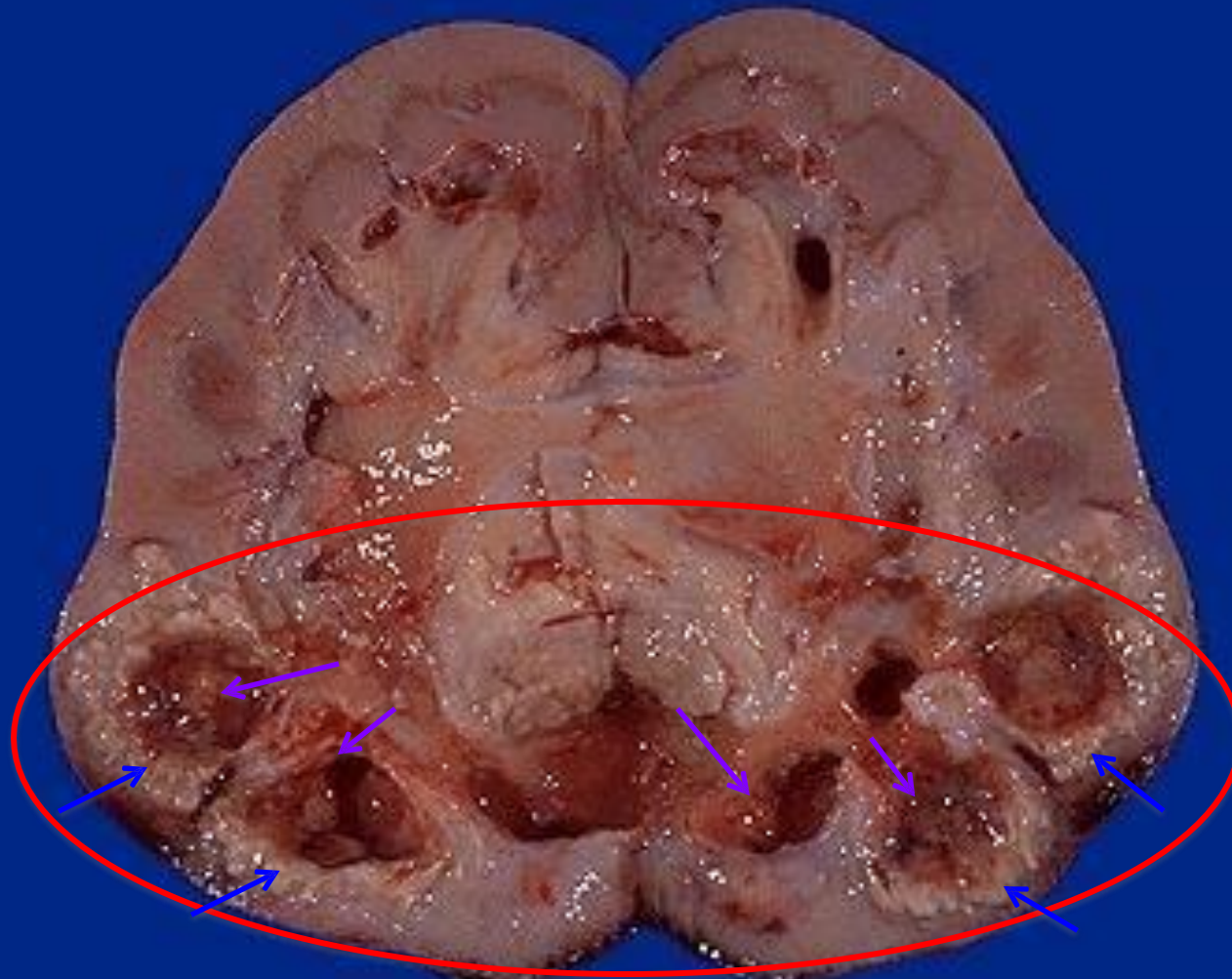


Gambar histopatologi ini menunjukkan kondisi Nefropati Oksalat (Oxalate Nephropathy) atau pengendapan kristal kalsium oksalat pada tubulus ginjal:

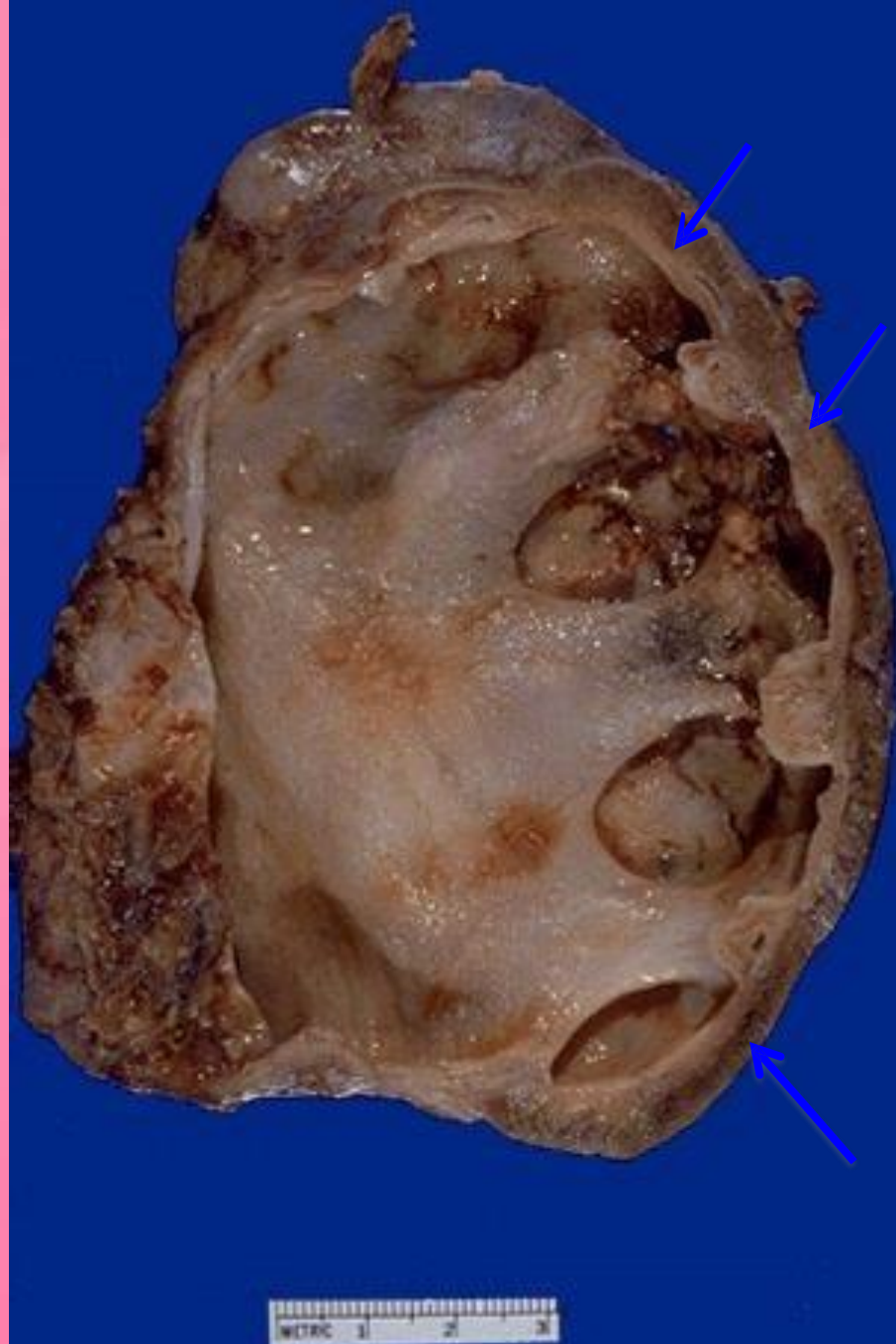
- **Lingkar Hitam (Deposit Kristal):** Menunjukkan agregat kristal kalsium oksalat di dalam lumen tubulus ginjal. Kristal ini tampak pucat/kekuningan transparan dengan susunan radial khas (biasanya sangat birefringen di bawah mikroskop cahaya terpolarisasi).
- **Huruf "T" (Tubules):** Menunjukkan tubulus-tubulus proksimal yang mengalami kerusakan, pembengkakan sel epitel, serta nekrosis tubular akut (Acute Tubular Necrosis / ATN) akibat obstruksi dan toksisitas kristal.
- **Konteks Klinis:** Temuan ini sering menjadi penyebab cedera ginjal akut (Acute Kidney Injury) akibat keracunan etilen glikol (zat antibeku), konsumsi vitamin C dosis sangat tinggi, atau malabsorpsi usus (seperti pada penyakit usus inflamasi/riwayat operasi bypass lambung).

OBSTRUCTIVE AND VASCULAR DISEASE OF KIDNEY





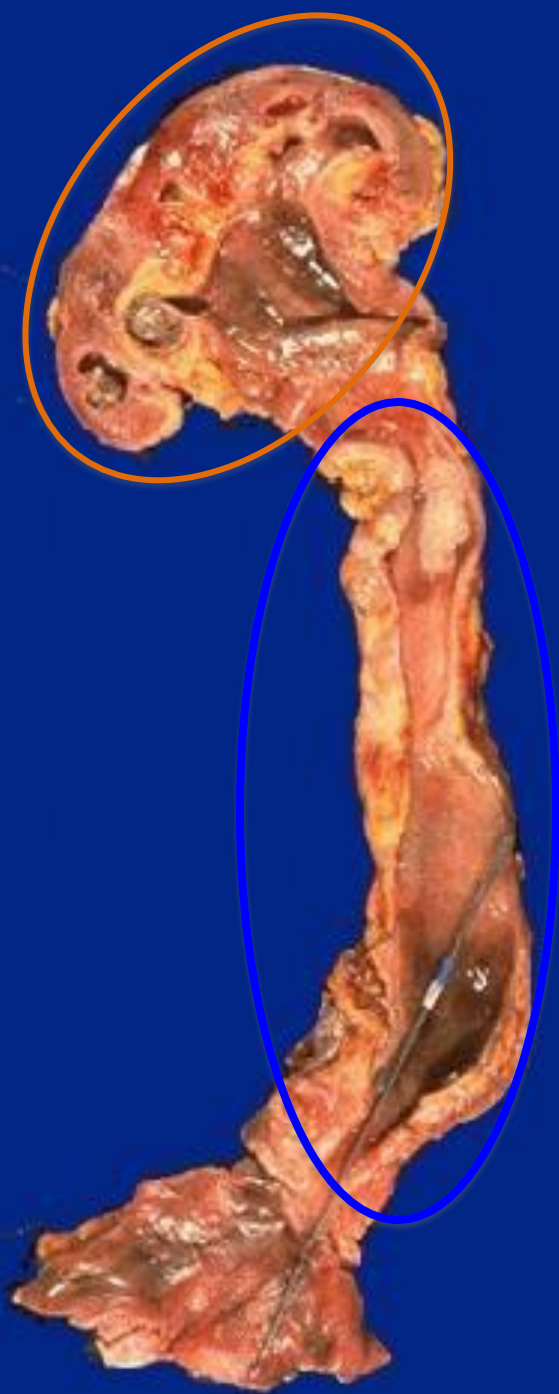
There was a large renal calculus (stone) that obstructed the calyces of the lower pole of this kidney, leading to a **FOCAL HYDRONEPHROSIS** (dilation of the collecting system). The stasis from the obstruction and dilation led to infection. The infection with inflammation is characterized by the **pale yellowish-tan areas** next to the dilated calyces with **hyperemic mucosal surfaces**. The upper pole is normal and shows good corticomedullary demarcations.



Here is a kidney with much more advanced **HYDRONEPHROSIS** in which there is **only a thin rim of remaining renal cortex**. Such a kidney is non-functional and a source for ongoing infection. If this process is unilateral, then the problem originates from the ureteral orifice up to the pelvis. In this case, a large "staghorn" calculus (so named because the prominent projections of the stone into the calyces resemble deer antlers) was present that filled up the pelvis and calyceal system. If this process were bilateral, then the problem would originate in the bladder trigone or urethra (or the prostate around the urethra) or some process (such as a large neoplasm) that could impinge on both ureters.



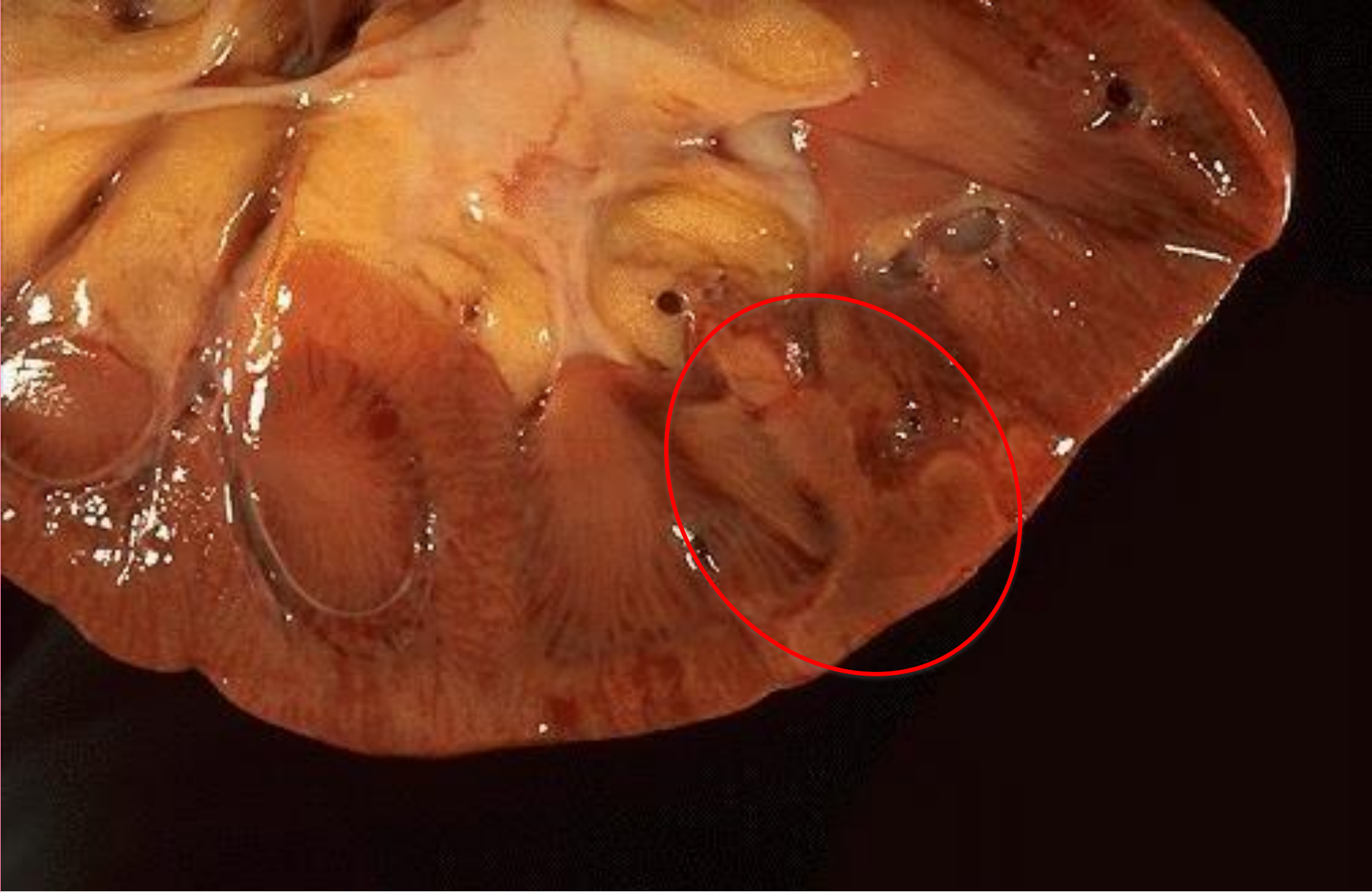
The arrow points to the culprit in this case of **HYDRONEPHROSIS**--a **ureteral calculus at the ureteropelvic junction**. This kidney demonstrates marked hydronephrosis with nearly complete loss of cortex.



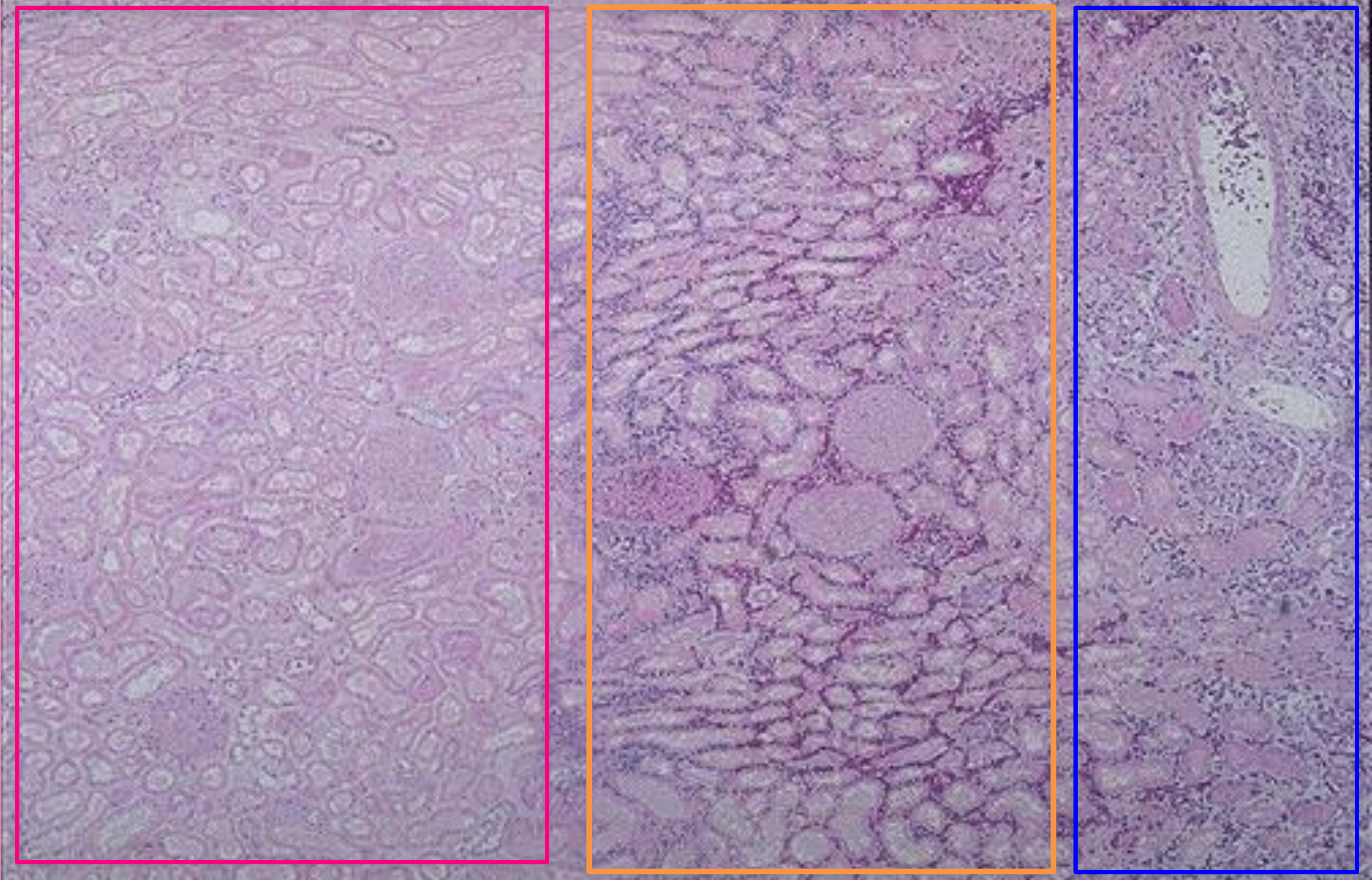
A long-standing
**OBSTRUCTION (PROBABLY
CONGENITAL) AT THE
URETERAL ORIFICE**
through which the metal
probe passes led to the
marked **hydroureter** and
hydronephrosis seen here.



This is an **ACUTE RENAL INFARCTION**. Note the wedge shape of this **zone of coagulative necrosis** resulting from loss of blood supply with resultant tissue ischemia that produces the pale infarct. The small amount of blood supply from the capsule supplies the immediate subcortical zone. The remaining cortex is congested, as is the medulla.

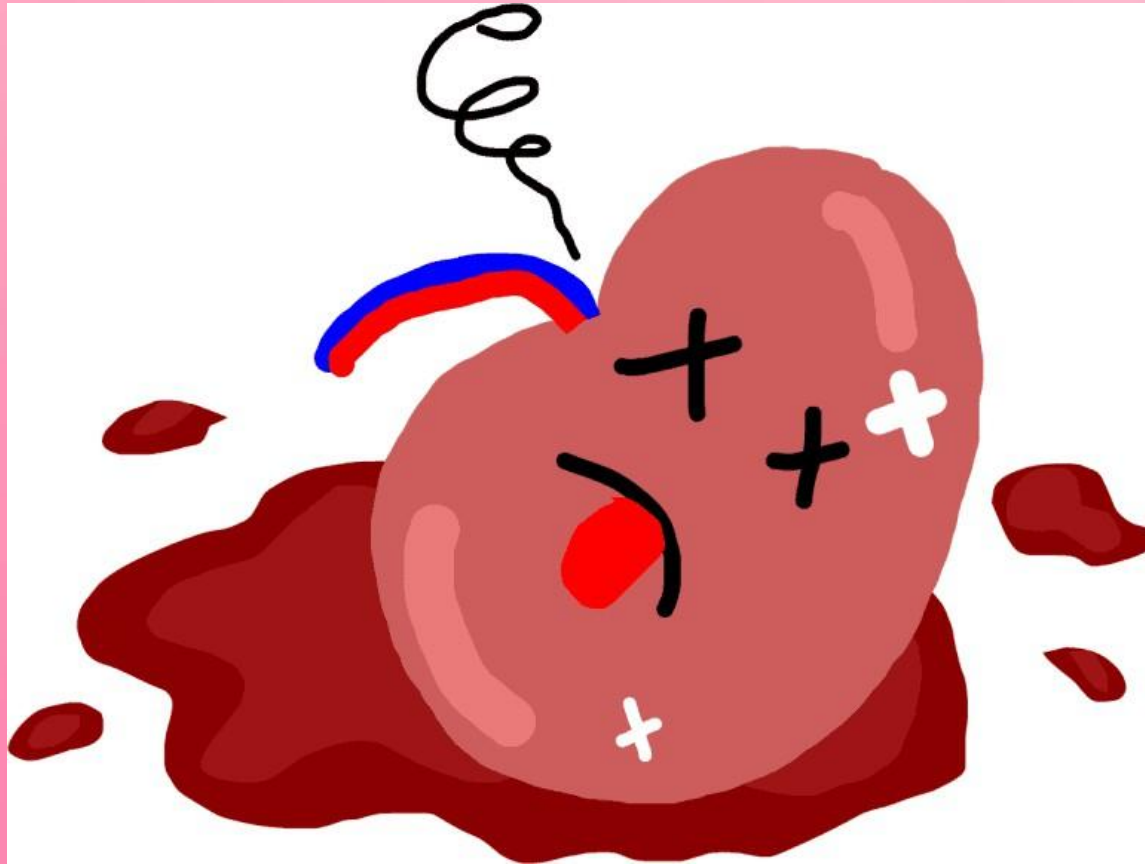


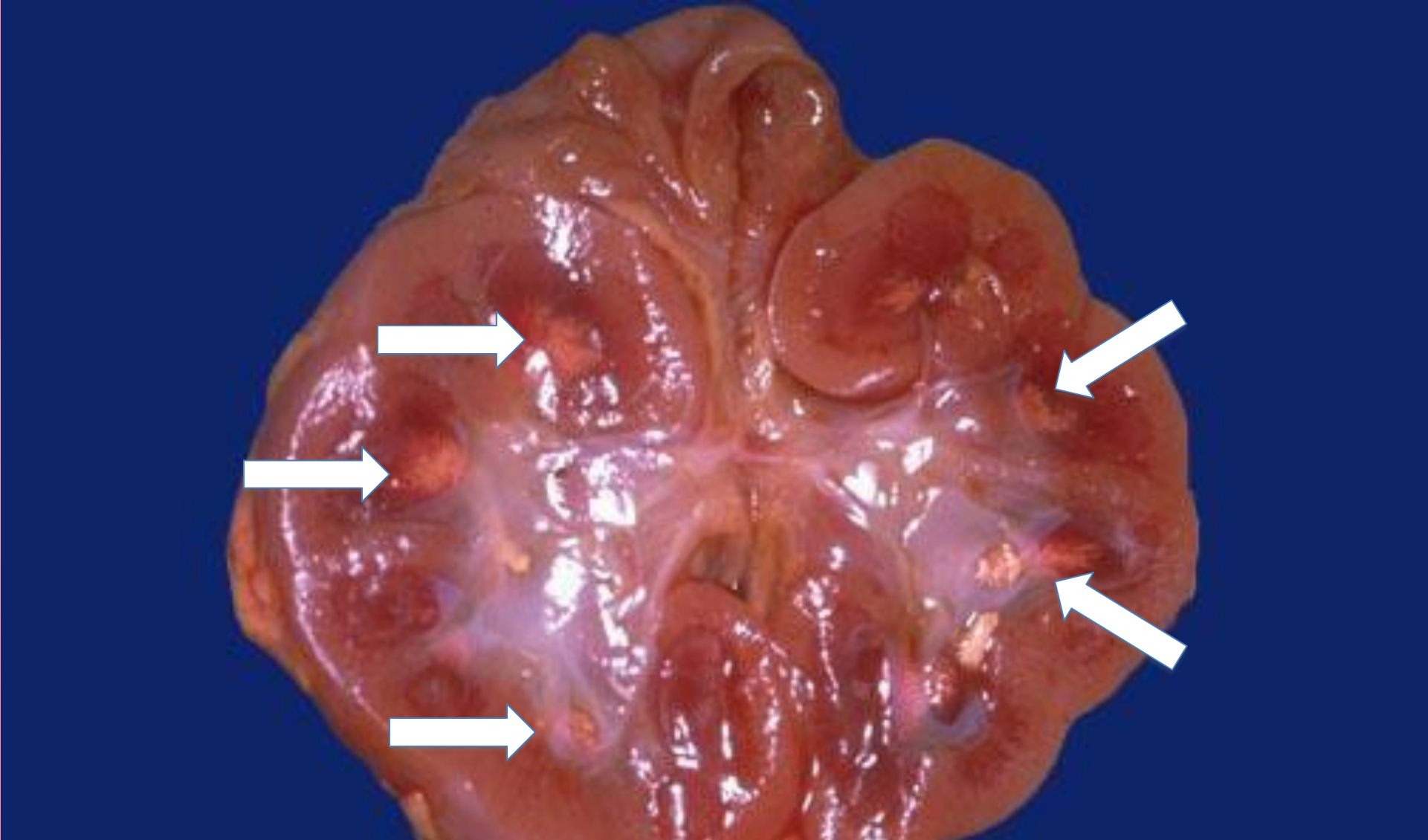
This **ACUTE RENAL INFARCTION** is pale, typical of coagulative necrosis. It is roughly wedge-shaped. Renal infarctions usually result from embolization of cardiac valvular vegetations. Sometimes a renal arterial vasculitis can lead to infarction.



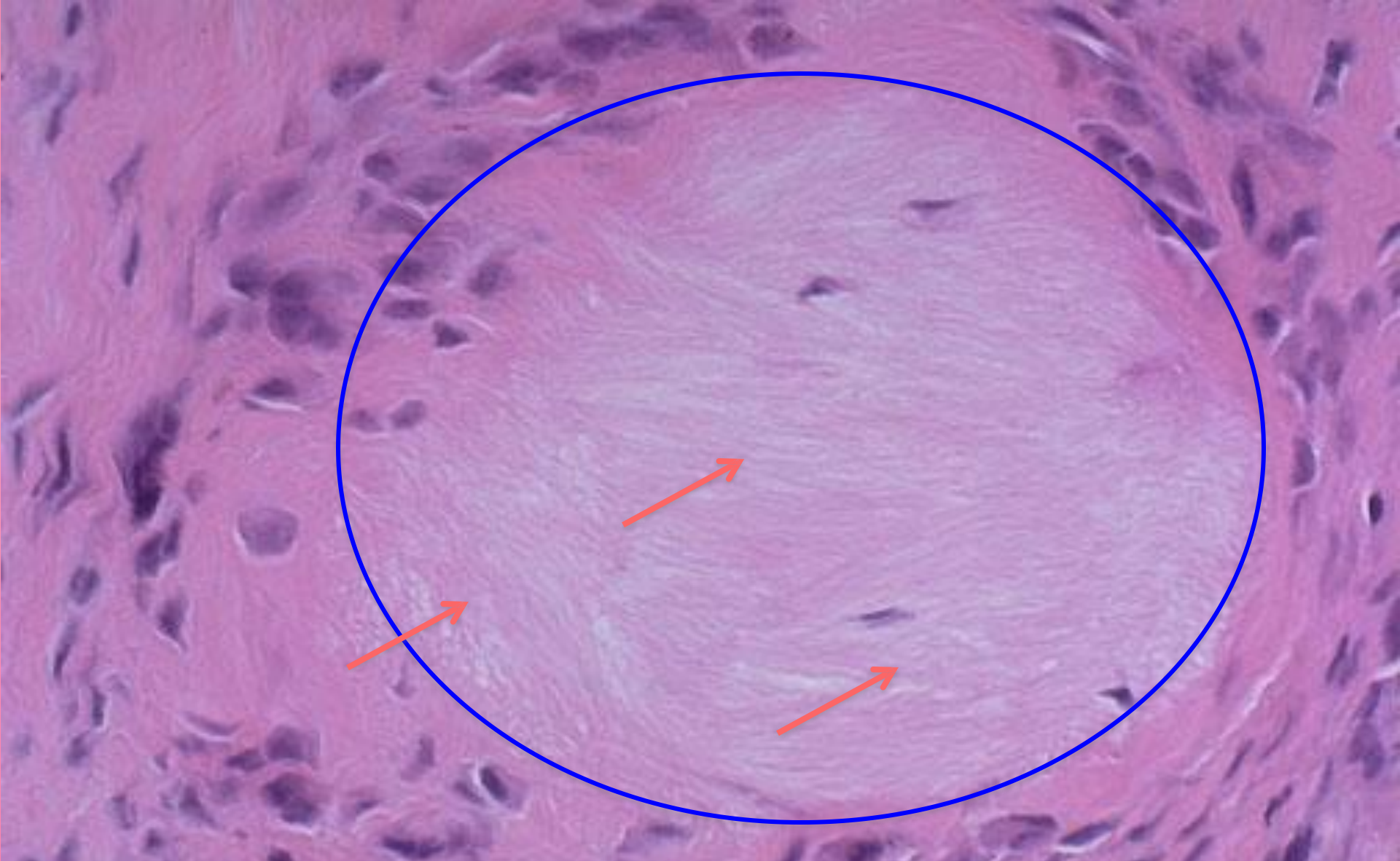
This is the microscopic appearance of an **ACUTE RENAL INFARCT**. At the far right is **normal kidney**, then to the left of that **hyperemic kidney that is dying**, then to the left of that **pale pink infarcted kidney in which both tubules and glomeruli are dead**.

INTERSTITIAL DISEASE

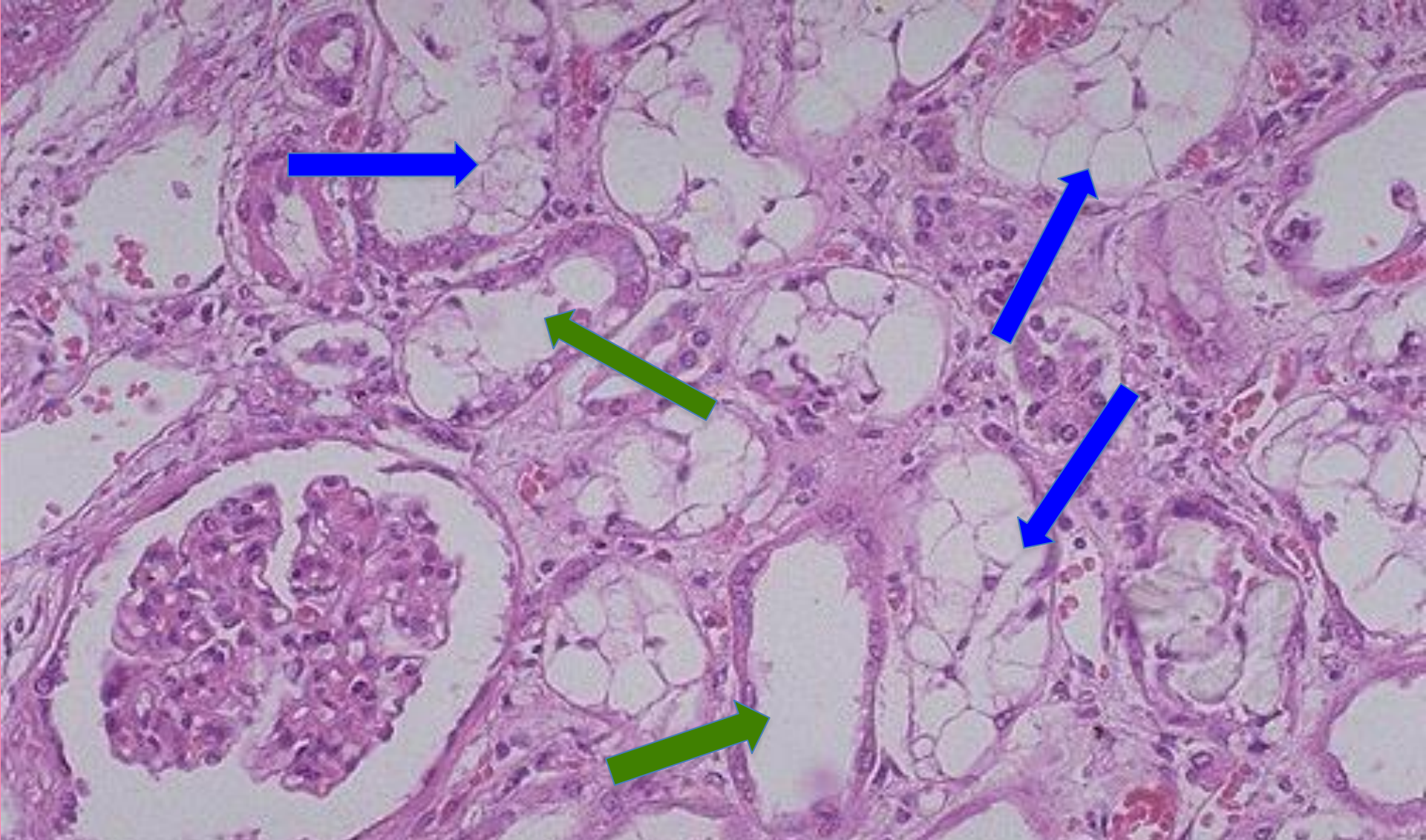




This is **CHRONIC URATE NEPHROPATHY** with pale yellowish tan **tophaceous deposits** in the medulla. There is also an acute urate nephropathy that can occur with a "lysis" syndrome resulting from massive cellular necrosis of leukemia or lymphoma cells with chemotherapy. The metabolic breakdown of the cell nuclei yields large amounts of urate which, when excreted, plug renal tubules.



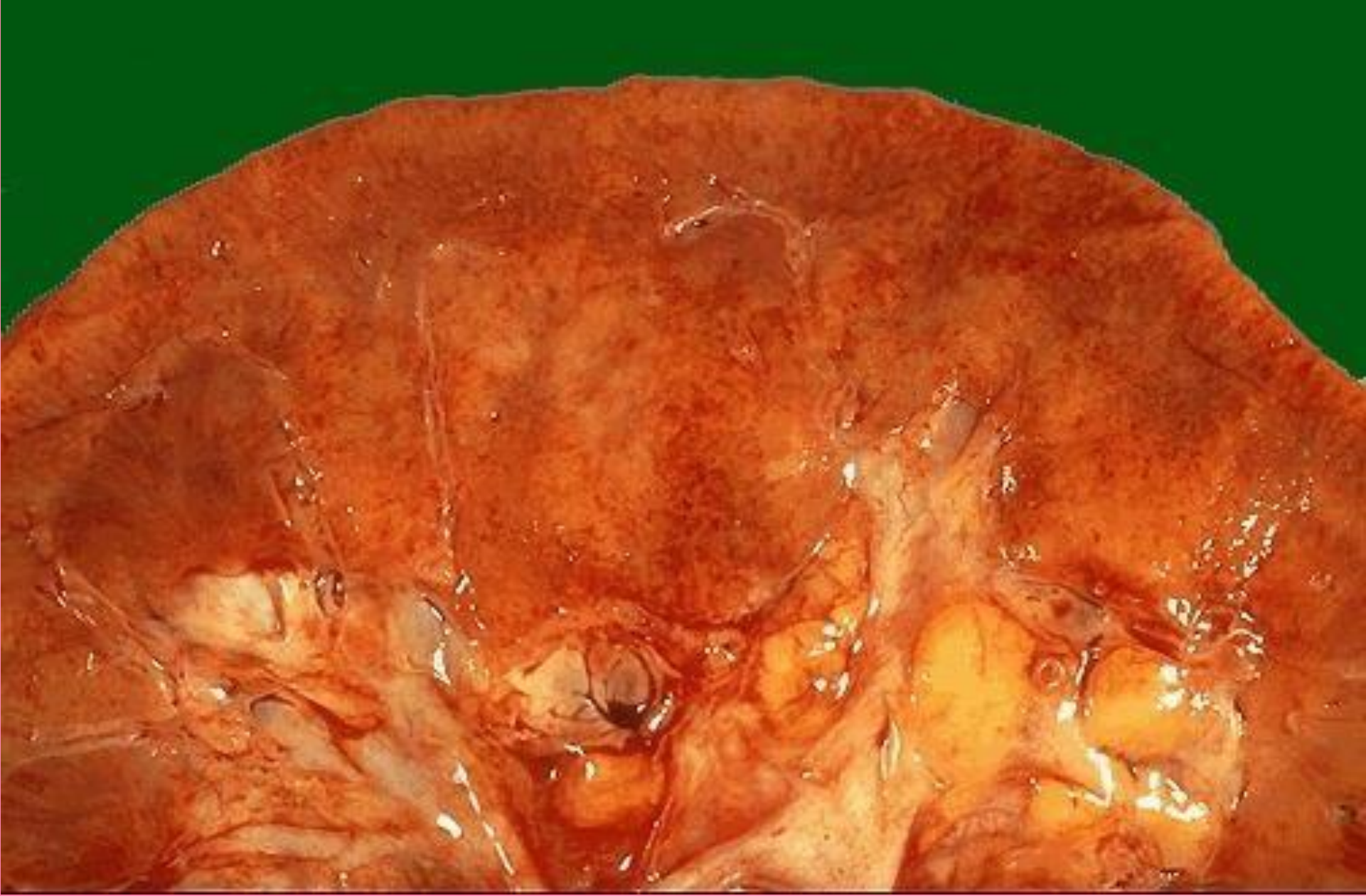
CHRONIC URATE NEPHROPATHY leads to deposition of **uric acid crystals** in interstitium, forming **tophi** with surrounding foreign body inflammation, mononuclear cell infiltrates, and fibrosis. The long, needle-shaped crystals form the pale mass shown here at high magnification. Patients with hyperuricemia may also have nephrolithiasis with uric acid stone



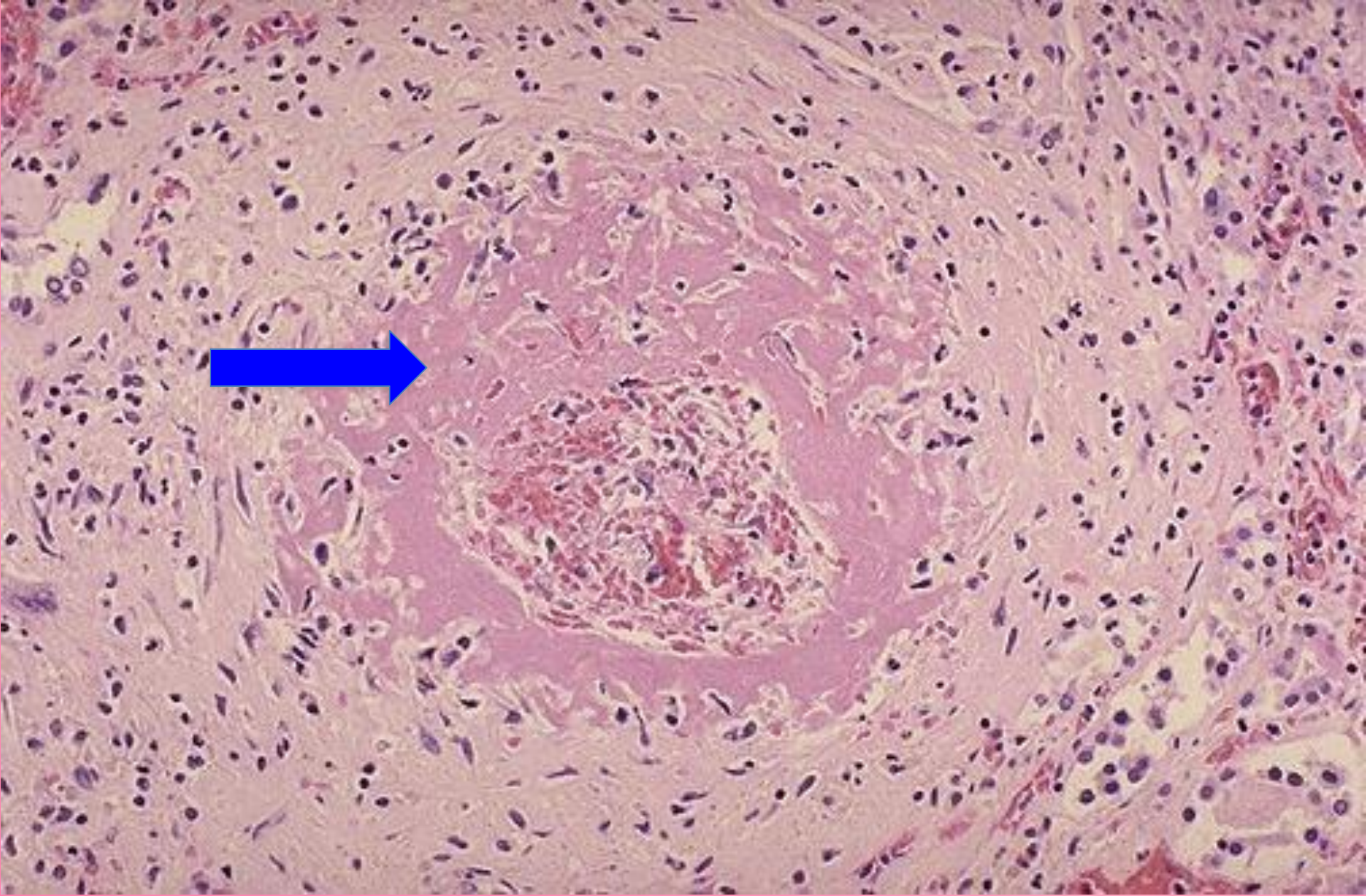
The **tubular vacuolization** and **dilation** here is a result of ethylene glycol poisoning. This is representative of **ACUTE TUBULAR NECROSIS (ATN)**, which has many causes. ATN resulting from toxins usually has diffuse tubular involvement, whereas ATN resulting from ischemia (as in profound hypotension from cardiac failure) has patchy tubular involvement.



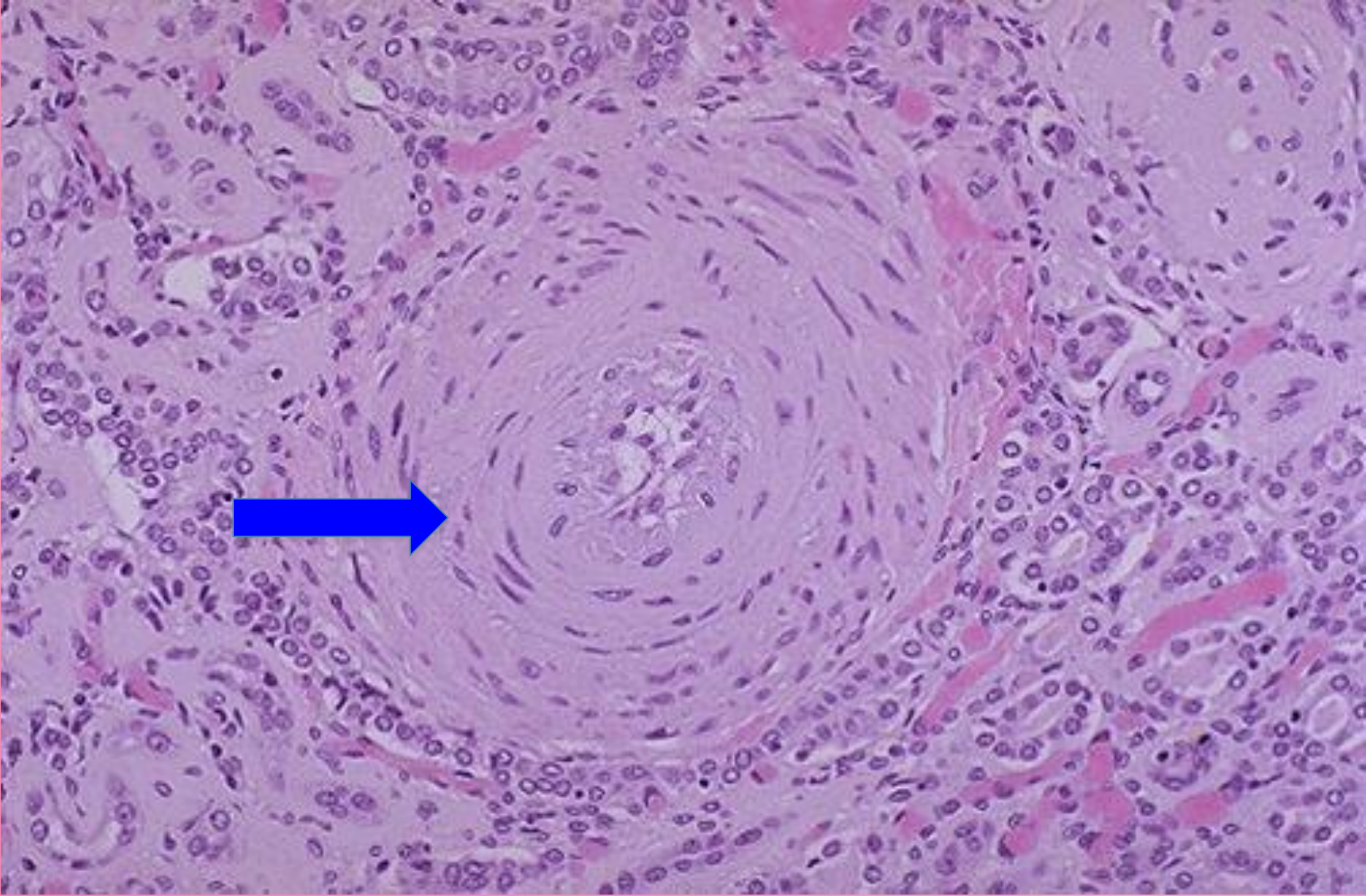
Here is an example of renal vascular disease known as **BENIGN NEPHROSCLEROSIS**. The smaller arteries in the kidney have become thickened and narrowed. Hyaline arteriolosclerosis with hypertension or diabetes mellitus is usually present. This leads to patchy ischemic atrophy with focal loss of parenchyma that gives **the surface of the kidney the characteristic granular appearance** as seen here.



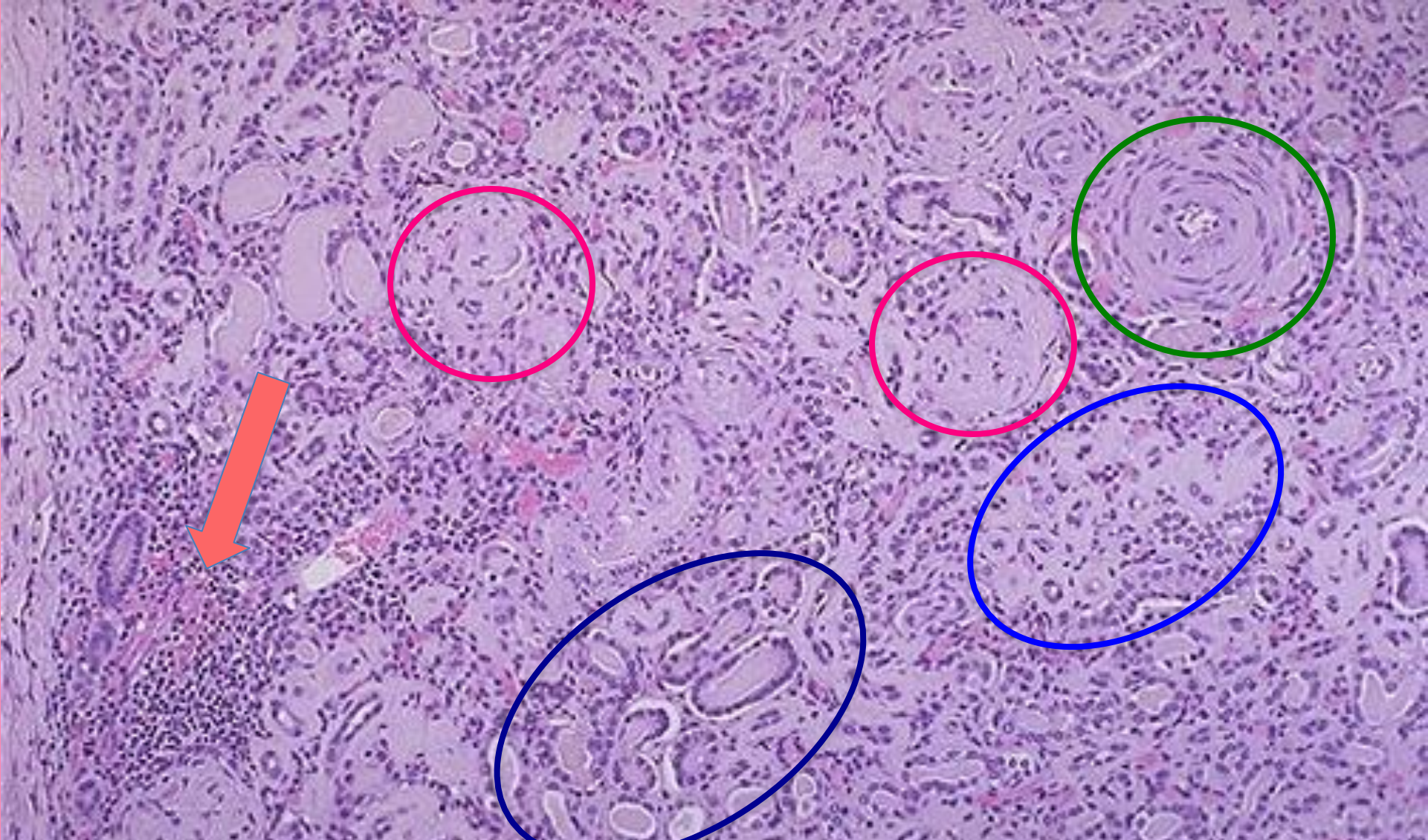
In **MALIGNANT NEPHROSCLEROSIS**, the kidney demonstrates focal small hemorrhages. This is due to an accelerated phase of hypertension in which blood pressures are very high (such as 300/150 mm Hg).



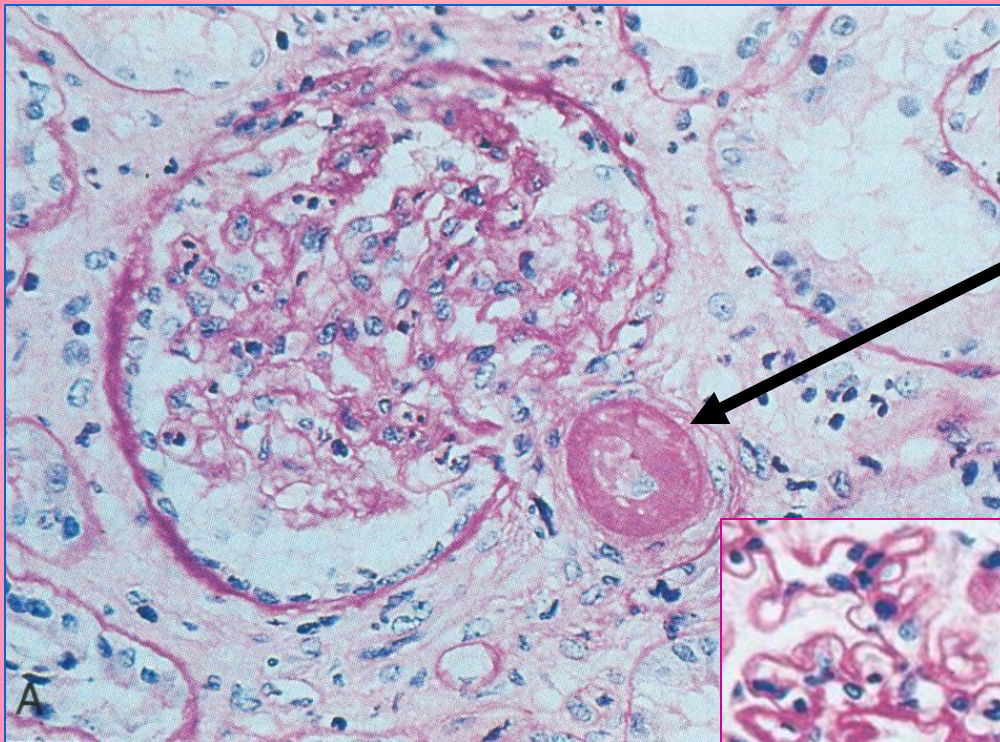
MALIGNANT HYPERTENSION leads to fibrinoid necrosis of small arteries as shown here. The damage to the arteries leads to formation of pink fibrin--hence the term "fibrinoid" (**necrosis fibrinoid**).



Thickening of the arterial wall with **MALIGNANT HYPERTENSION** also produces a **hyperplastic arteriolitis**. The arteriole has an "**onion skin**" appearance.



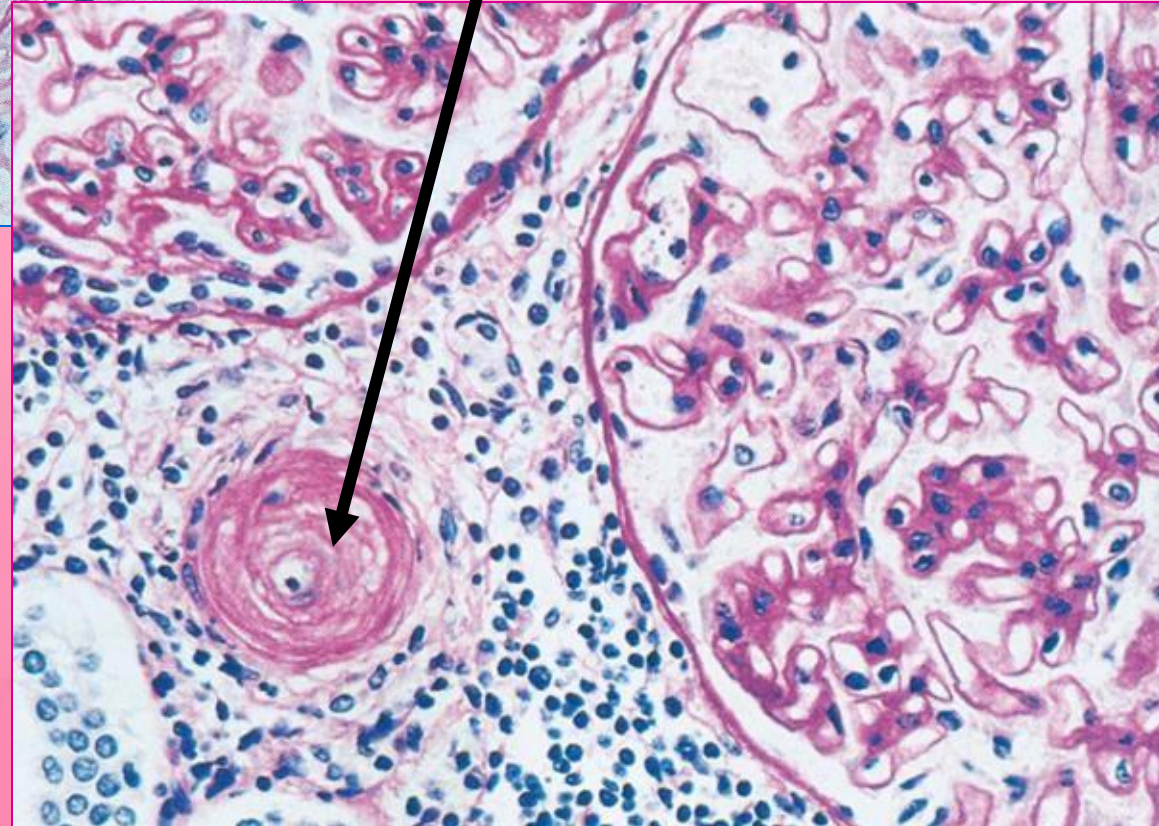
The microscopic appearance of the "**END STAGE KIDNEY**" is similar regardless of cause, which is why a biopsy in a patient with chronic renal failure yields little useful information. The **cortex is fibrotic**, **the glomeruli are sclerotic**, there are **scattered chronic inflammatory cell infiltrates**, and **the arteries are thickened**. Tubules are often dilated and filled with pink casts and give an appearance of "thyroidization."

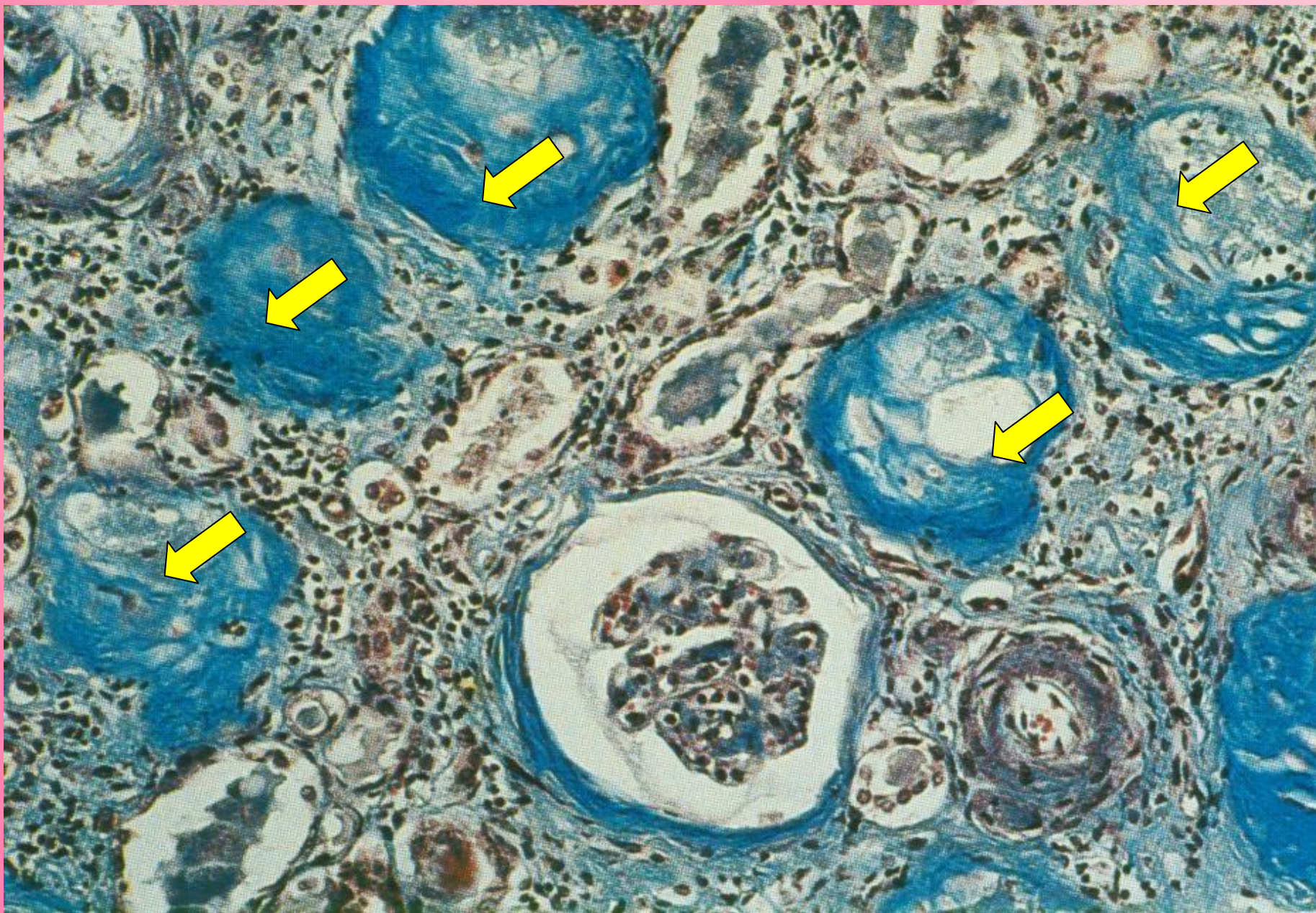


Malignant hypertension:

A: Fibrinoid necrosis of afferent arteriole

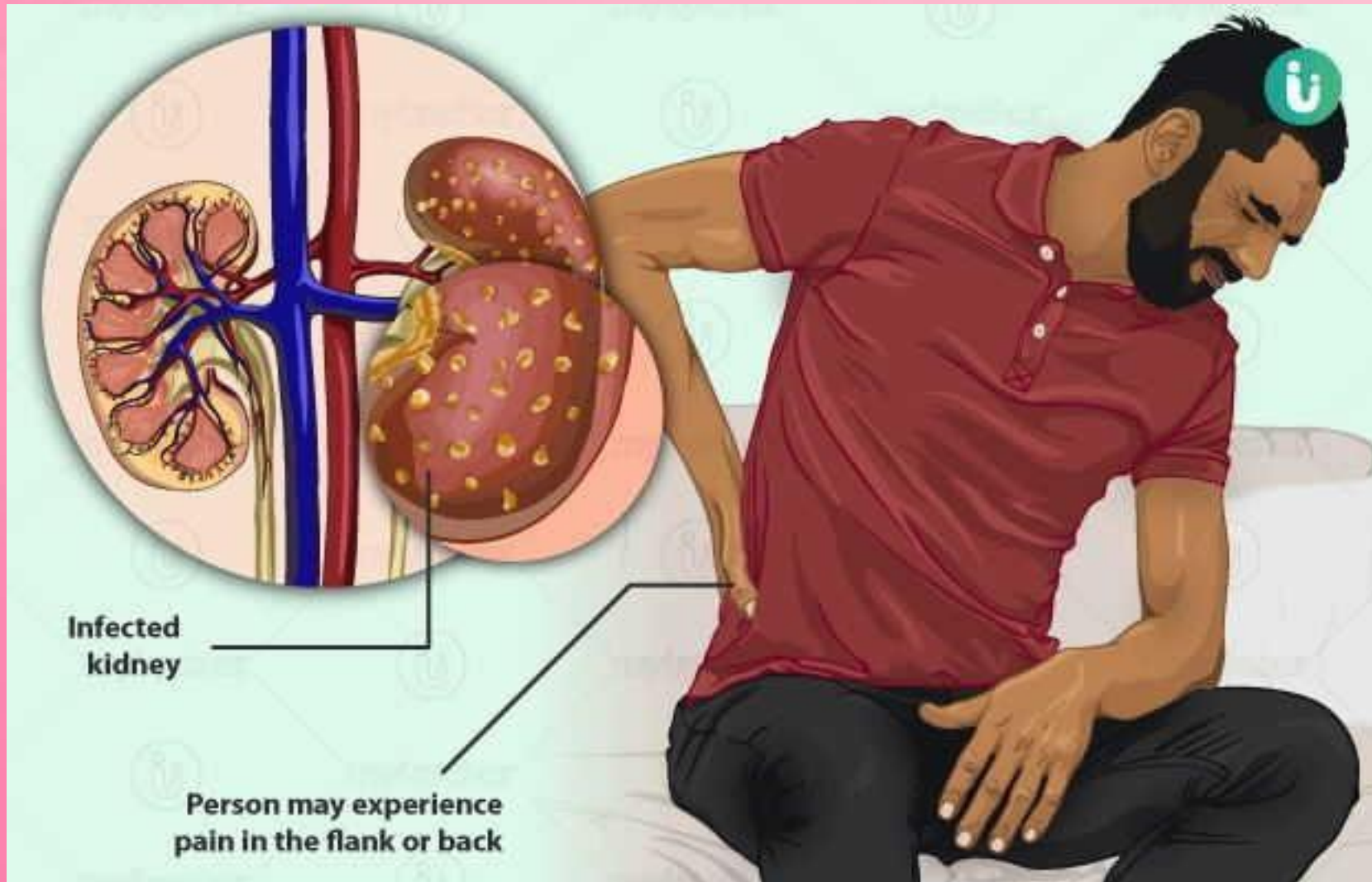
B: Hyperplastic arteriolitis (onion-skin lesion)





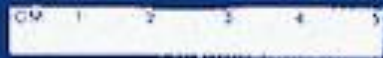
CHRONIC GLOMERULONEPHRITIS. Masson trichrome staining
(blue staining collagen) Indicating **glomerulosclerosis**

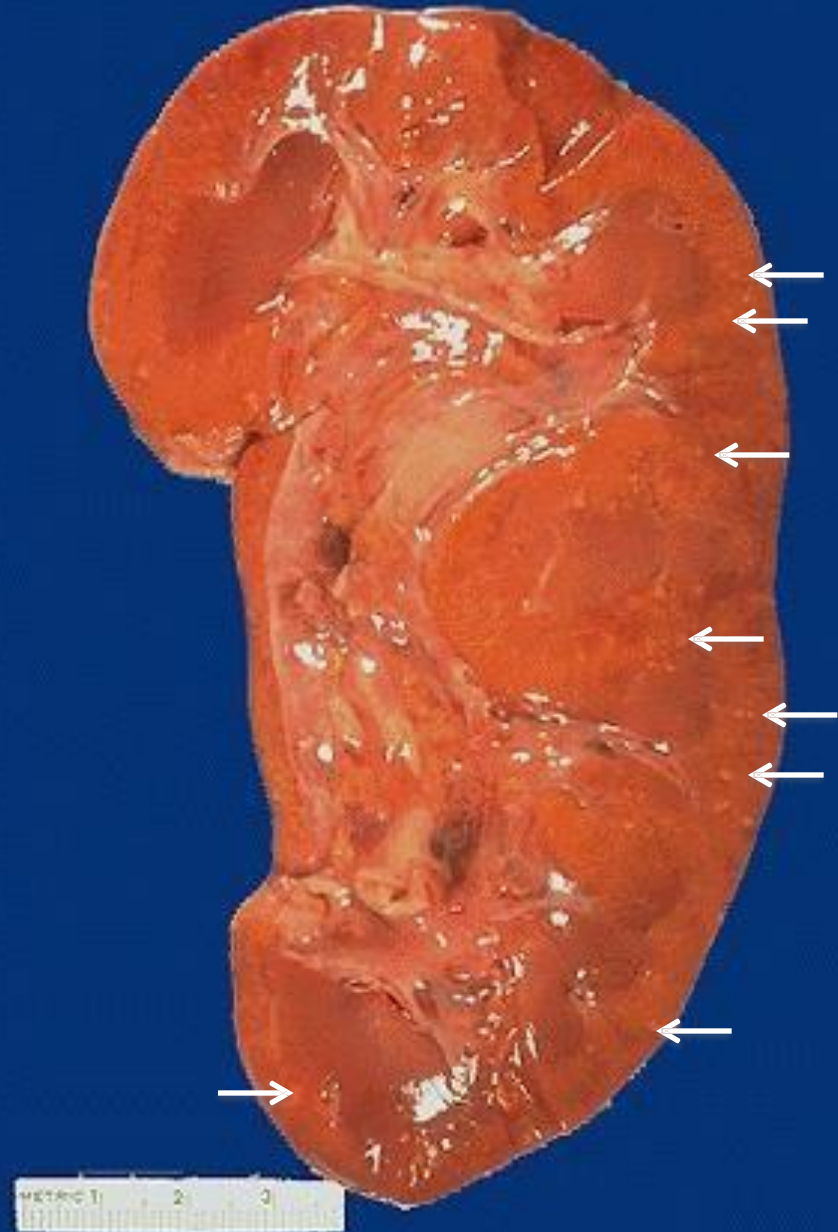
INFECTIOUS DISEASE OF KIDNEY



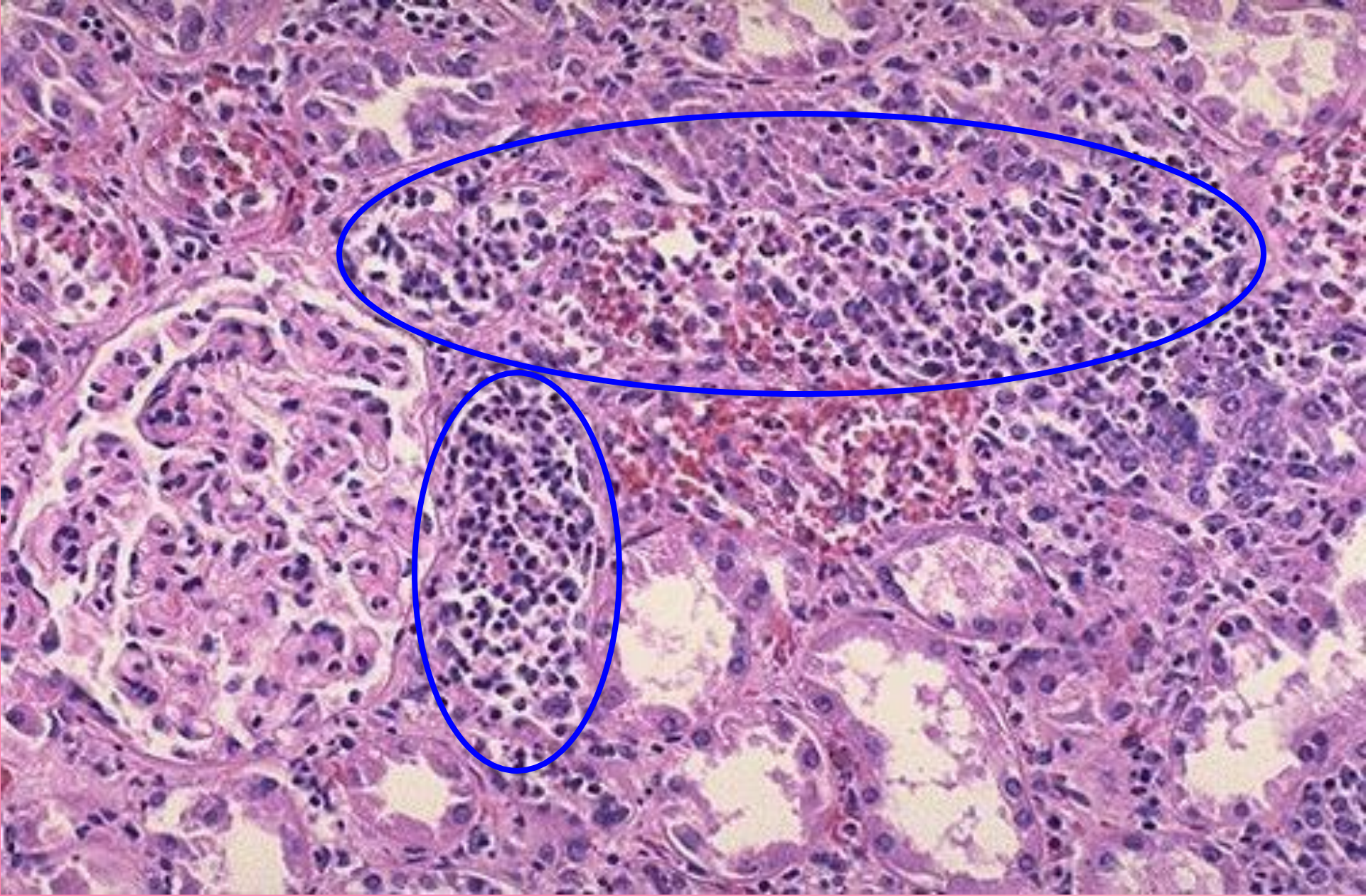
KIDNEY ABSCESS

- In the lower pole of this kidney is a 1 cm
- pale yellow abscess. Infections can reach the kidney either **by ascending up the urinary tract (from a bladder infection, for example)** or by **hematogenous spread with sepsis**. This lone abscess was probably **hematogenous in origin**.

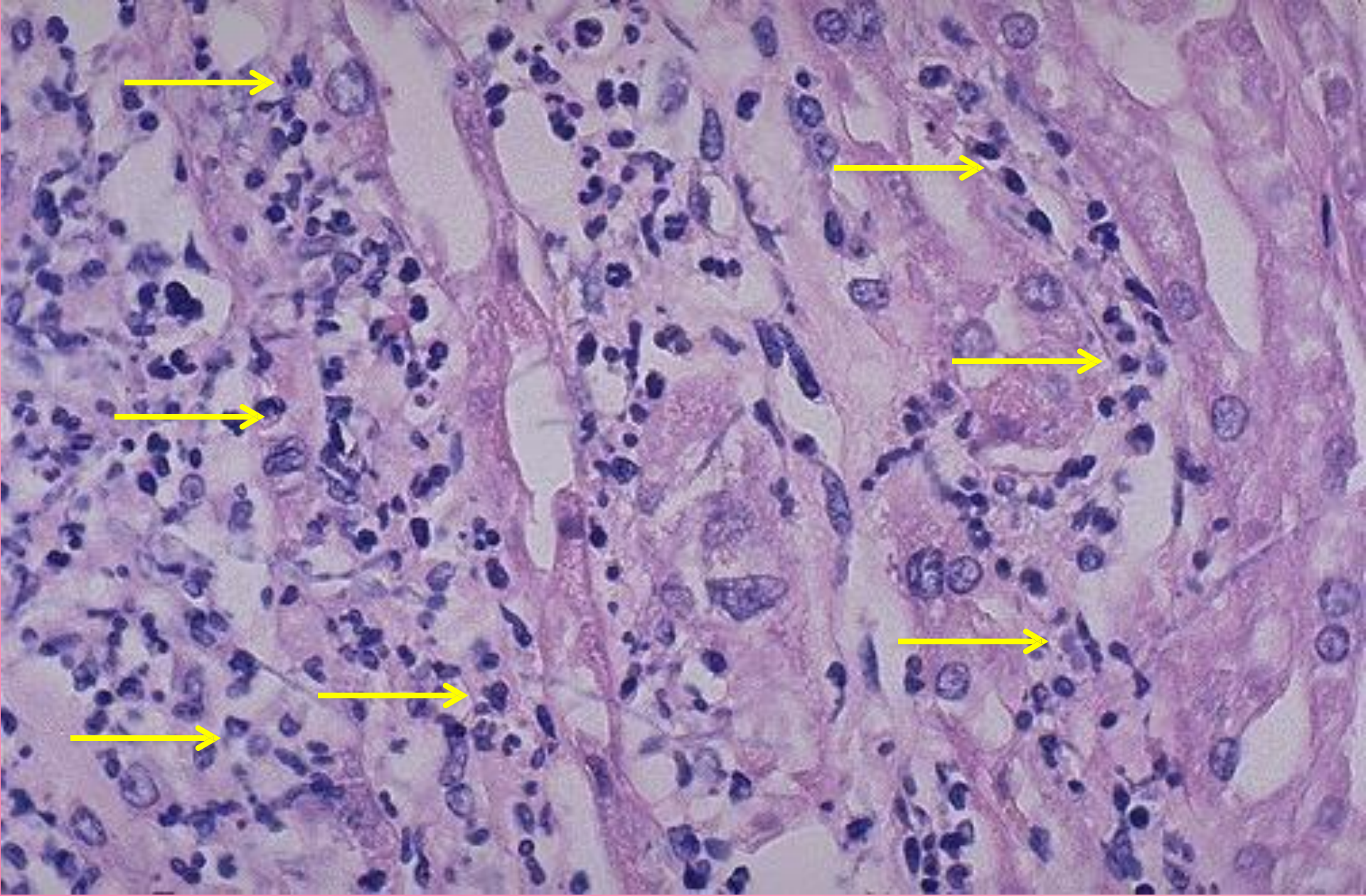




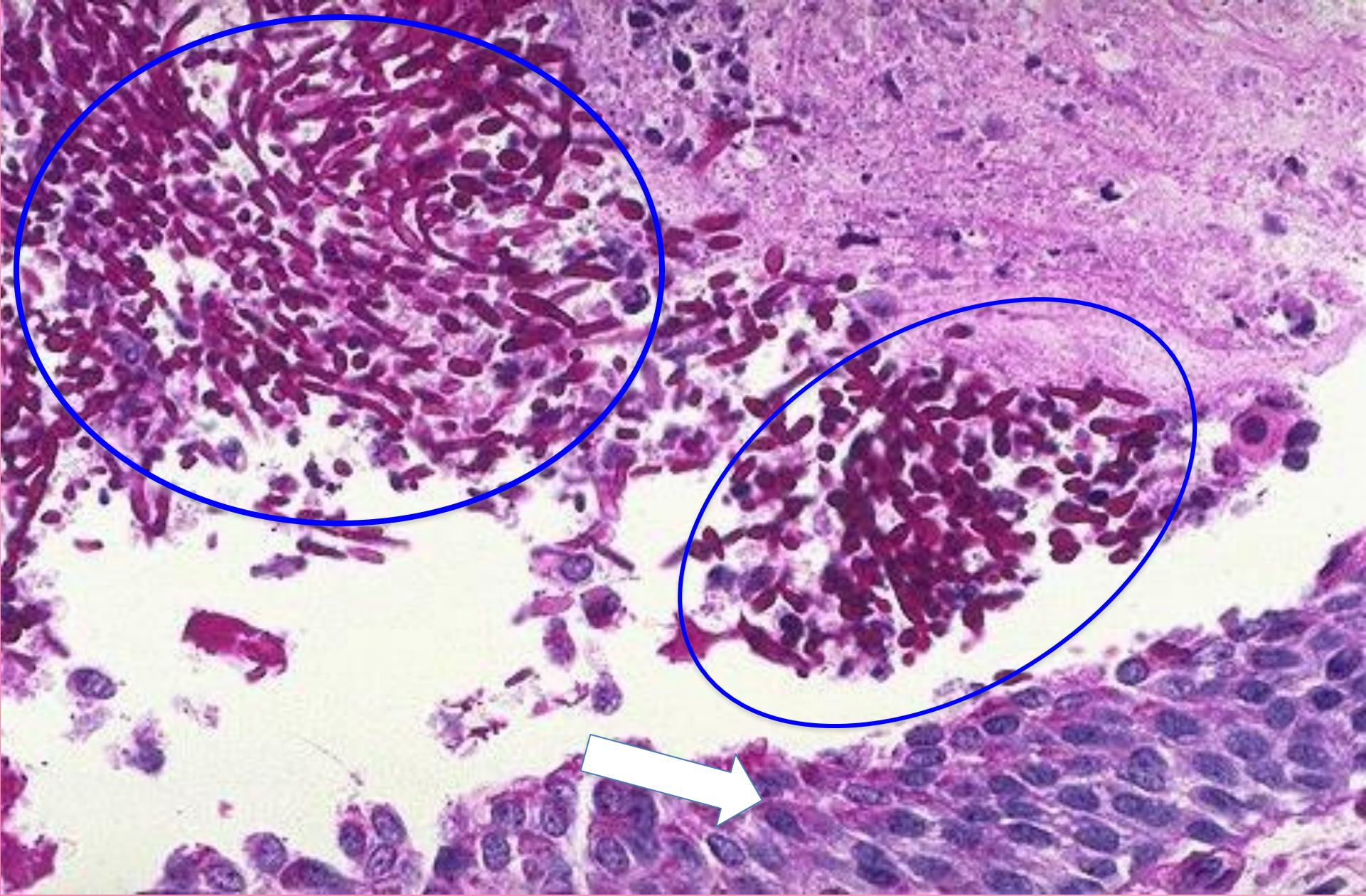
The cut surface of this kidney demonstrates many small yellowish MICROABSCESSES



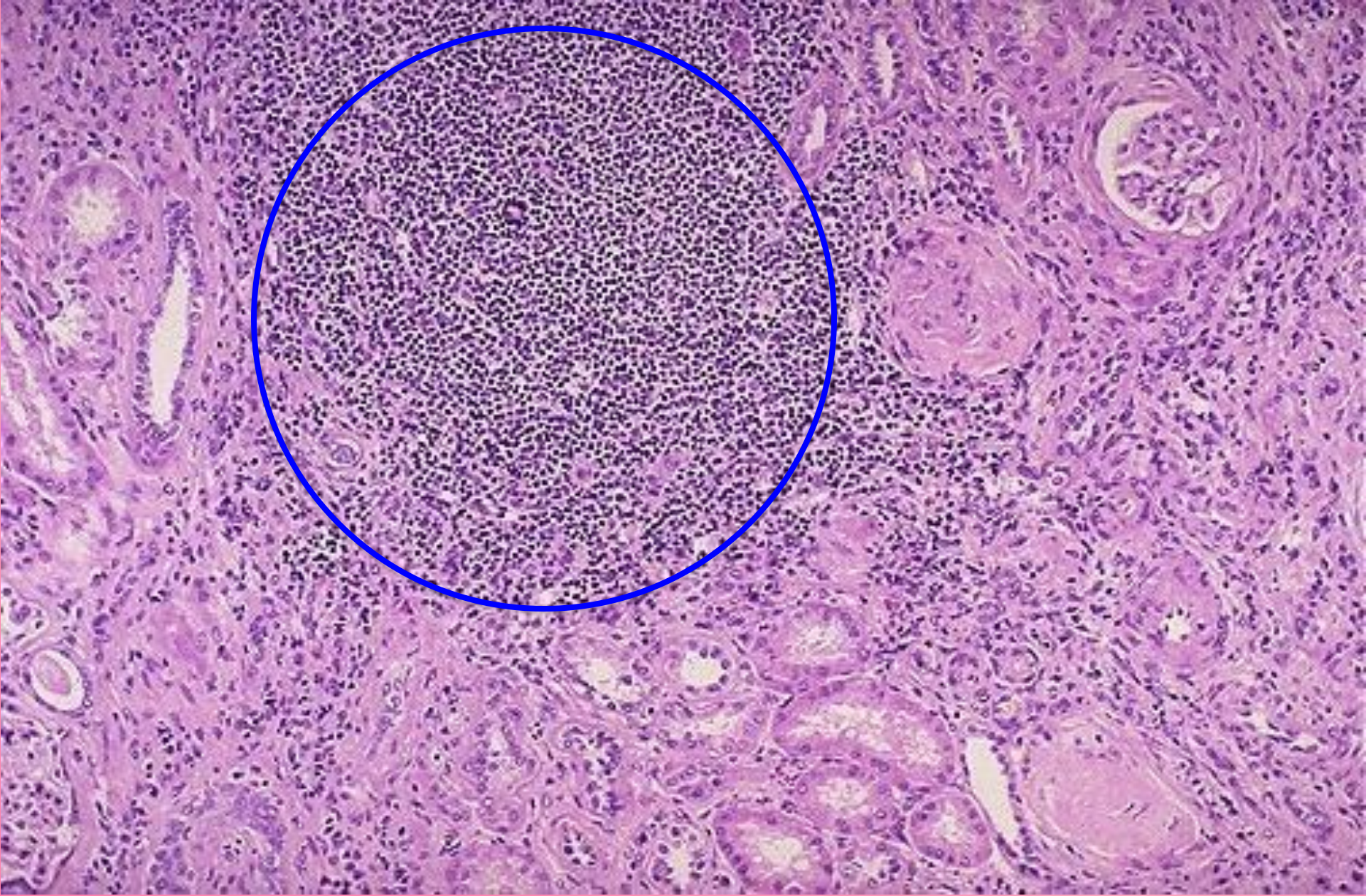
This is an ascending bacterial infection leading to **ACUTE PYELONEPHRITIS**. Numerous PMN's are seen filling renal tubules across the center and right of this picture.



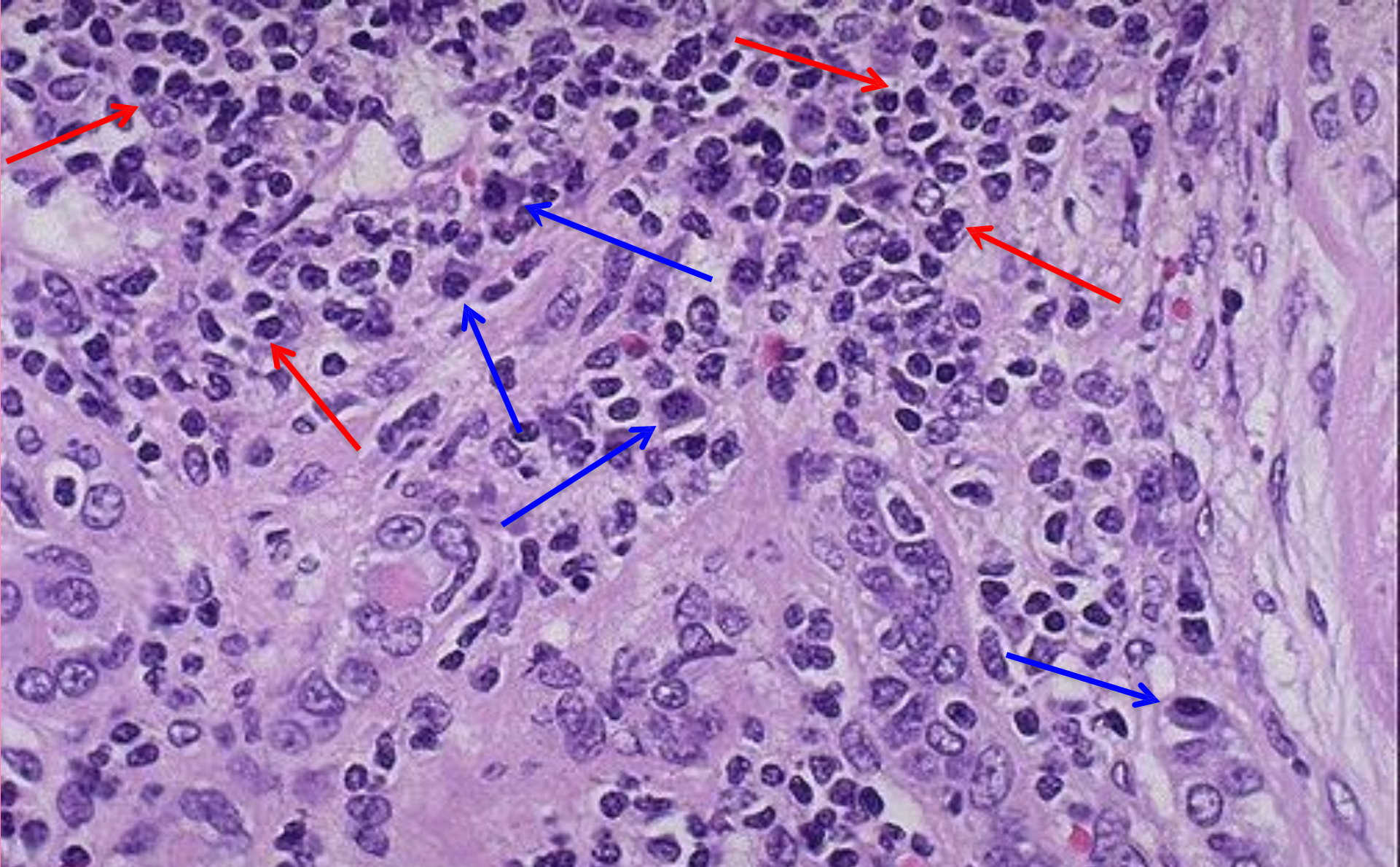
At high magnification, many **neutrophils** are seen in the tubules and interstitium in a case of **ACUTE PYELONEPHRITIS**.



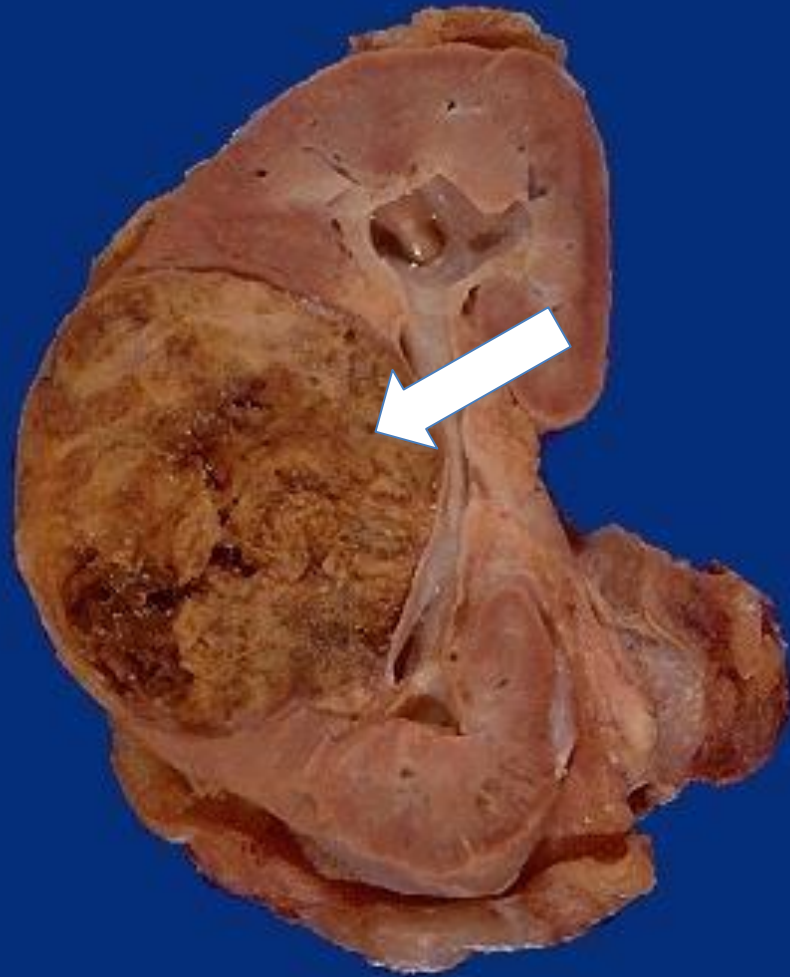
This PAS stain of a renal papilla with **a portion of transitional lining epithelium** at the lower right demonstrates many **BUDDING CELLS AND PSEUDOHYPHAE OF CANDIDA ALBICANS**.



The large collection of **chronic inflammatory cells** here is in a patient with a history of multiple recurrent urinary tract infections. This is **CHRONIC PYELONEPHRITIS**.



Both **lymphocytes** and **plasma cells** are seen at high magnification in this case of **CHRONIC PYELONEPHRITIS**. It is not uncommon to see lymphocytes accompany just about any chronic renal disease: glomerulonephritis, nephrosclerosis, pyelonephritis. However, the plasma cells are most characteristic for chronic pyelonephritis.

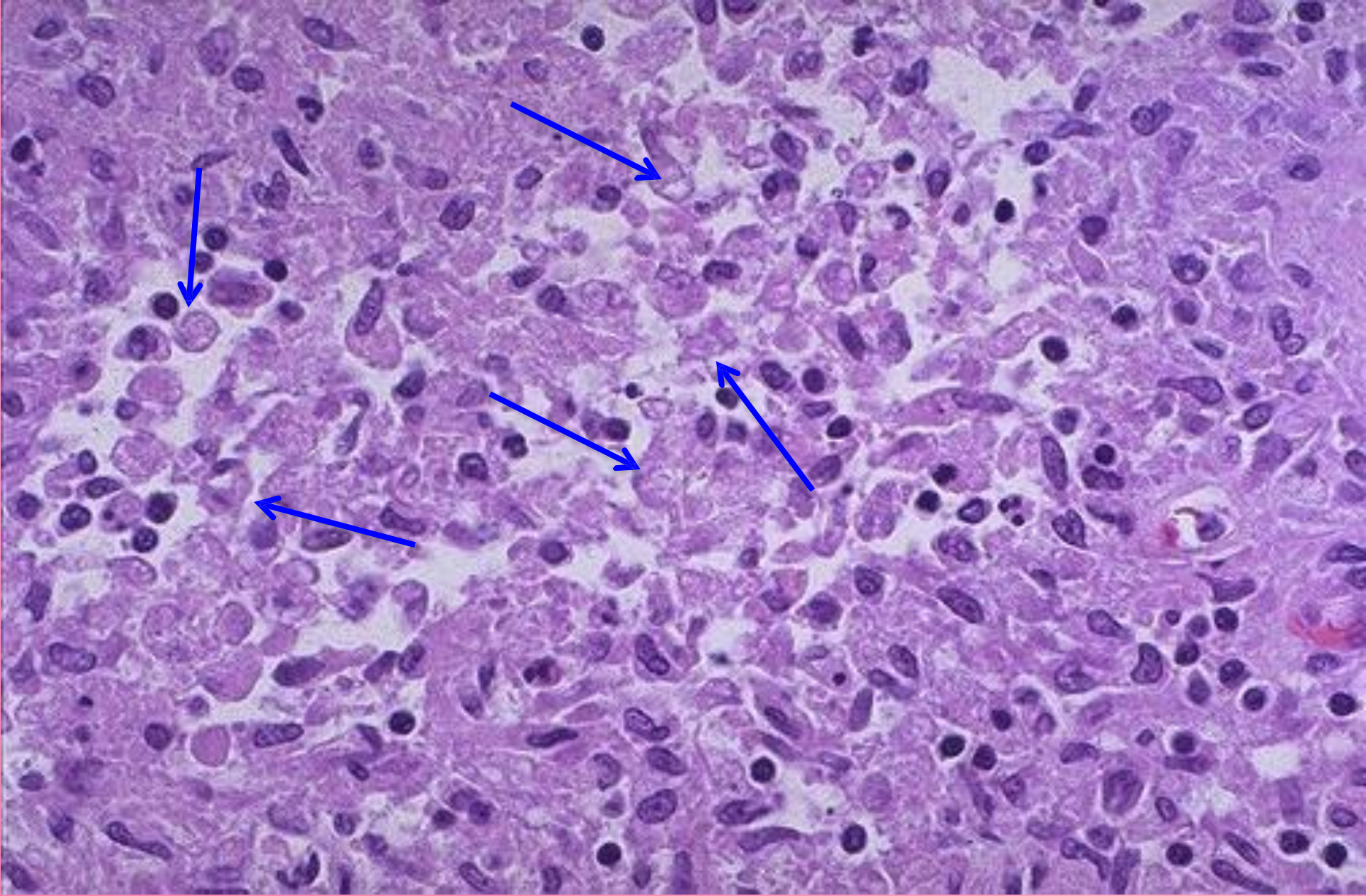


Sometimes long-standing infection may be localized and form a mass-like lesion.

This is a disease known as

XANTHOGRANULOMATOUS

PYELONEPHRITIS. It is uncommon, but may mimic a neoplasm.



The microscopic appearance of **XANTHOGRANULOMATOUS PYELONEPHRITIS** shows many pale to foamy macrophages from breakdown of renal parenchyma with ongoing inflammation.

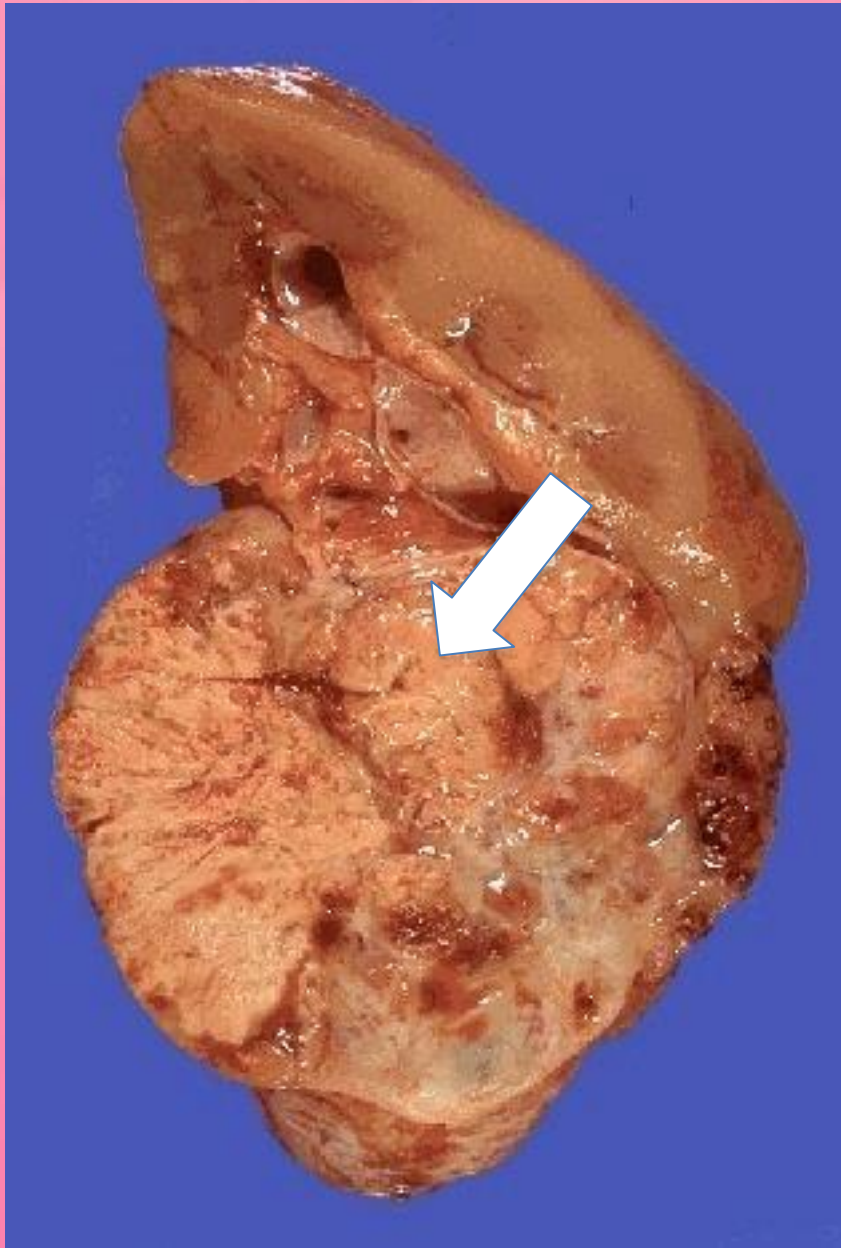
LESI NEOPLASTIK PADA GINJAL DAN SALURAN KEMIH

1. Lesi Neoplastik pada Ginjal (Renal cell carcinoma)
2. Lesi Neoplastik pada kandung kemih (Urothelial carcinoma)
3. Lesi neoplastik pada testis
4. Lesi neoplastik pada prostat

1. Lesi Neoplastik pada Ginjal (Renal cell carcinoma)

Pendahuluan & Klasifikasi Keganasan Renal

- Kanker ginjal merupakan salah satu keganasan urologi yang memiliki angka morbiditas signifikan. Karsinoma Sel Ginjal (*Renal Cell Carcinoma* / RCC) adalah jenis keganasan primer yang paling sering terjadi pada parenkim ginjal dewasa.
- Di antara berbagai sub tipe histologis RCC, ***Clear Cell Renal Cell Carcinoma (ccRCC)*** merupakan varian epitelial renal yang paling dominan, mencakup sekitar 75% hingga 85% dari keseluruhan kasus RCC, dan dikenal memiliki sifat klinis yang sangat agresif.
- **Predominansi Laki-Laki:** ccRCC menunjukkan dominasi yang sangat kuat pada pasien laki-laki dengan proporsi mencapai **70% (rasio pria berbanding wanita sebesar 2.3:1)**. Hal ini sangat dikaitkan dengan faktor risiko gaya hidup yang dapat dimodifikasi, terutama tingginya paparan asap rokok.
- Pada anak-anak tumor ginjal yang paling sering Willms Tumor.



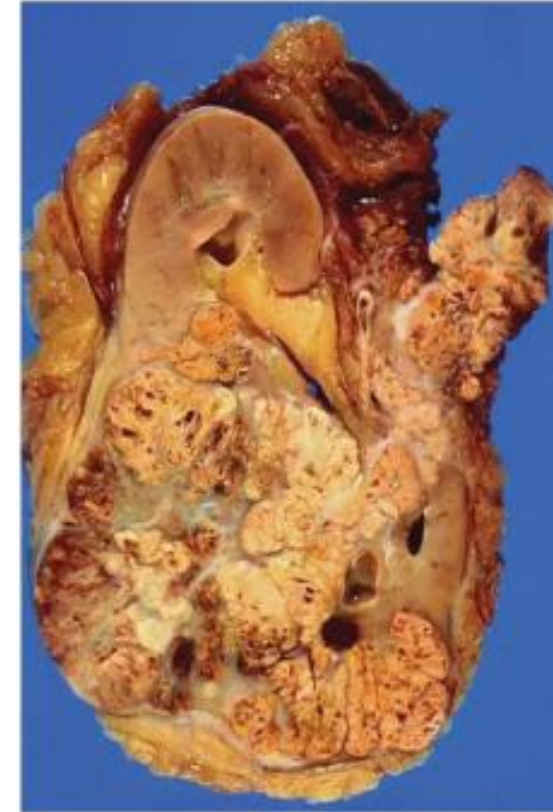
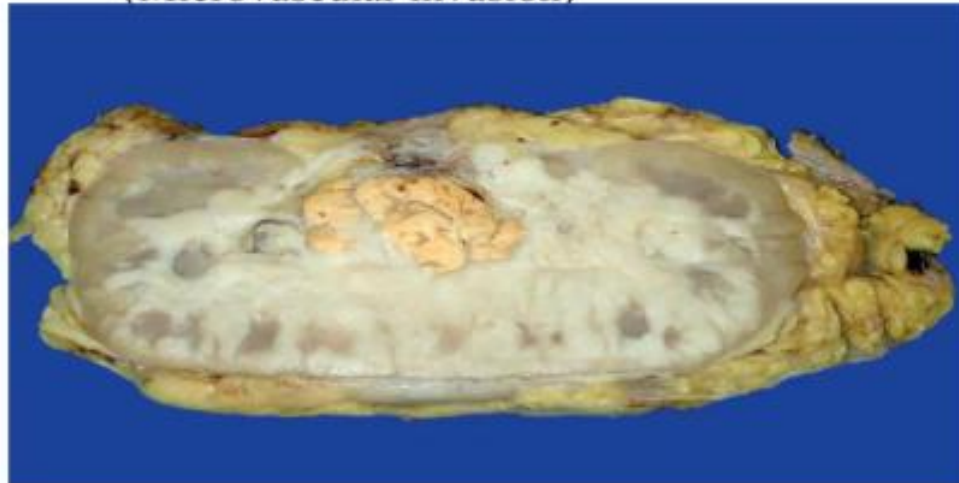
This is a **RENAL CELL CARCINOMA** arising in the lower pole of the kidney. It is fairly circumscribed. The cut surface demonstrates a variegated appearance with yellowish areas, white areas, brown areas, and hemorrhagic red areas. Though these neoplasms are usually slow-growing, they can often reach a considerable size before detection because there is a lot of room to enlarge in the retroperitoneum, and there is another kidney to provide renal function.

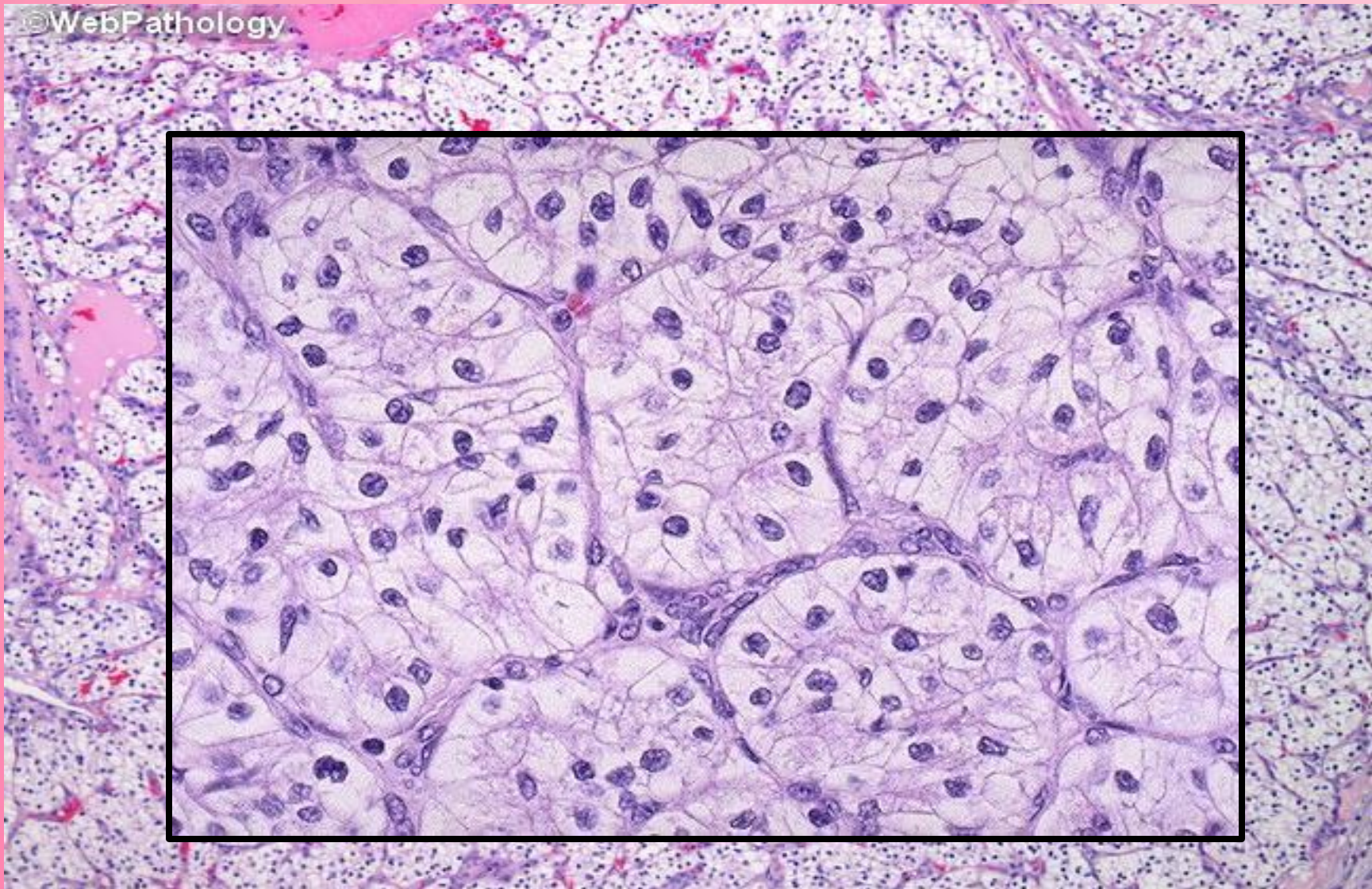
Thus, presenting symptoms and signs usually include: Flank pain, Hematuria (often microscopic), Weight loss, Mass on palpation & Hypertension

Makroskopik

Prognostic factors in RCC

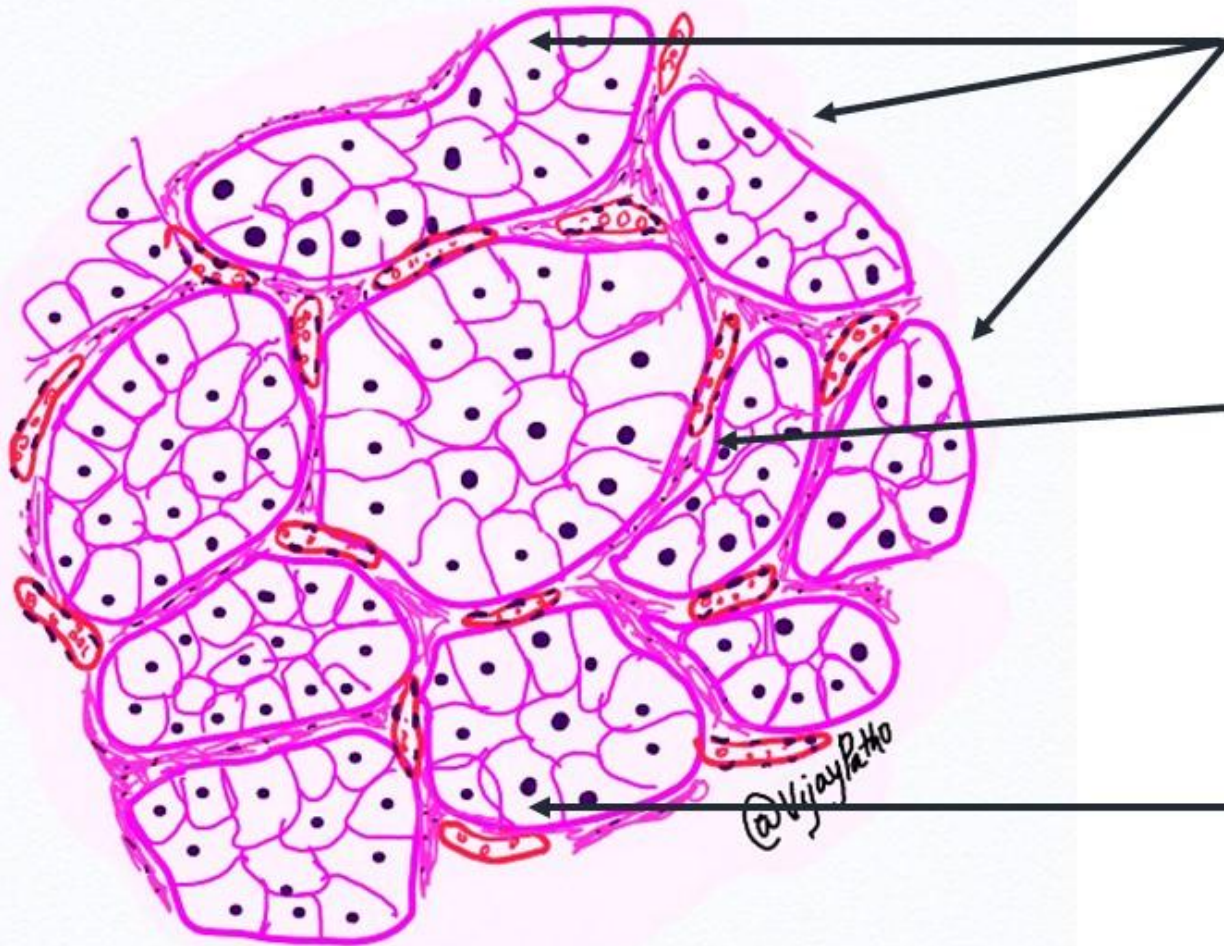
1. Pathologic stage
2. Tumor WHO/ISUP grade
3. Morphologic type
4. Sarcomatoid-rhabdoid differentiation
5. Tumor necrosis
(Microvascular invasion)





This medium power view is the typical appearance of **CLEAR CELL RENAL CELL CARCINOMA**, the most common type of renal cell carcinoma.

RENAL CELL CARCINOMA(Clear cell type)

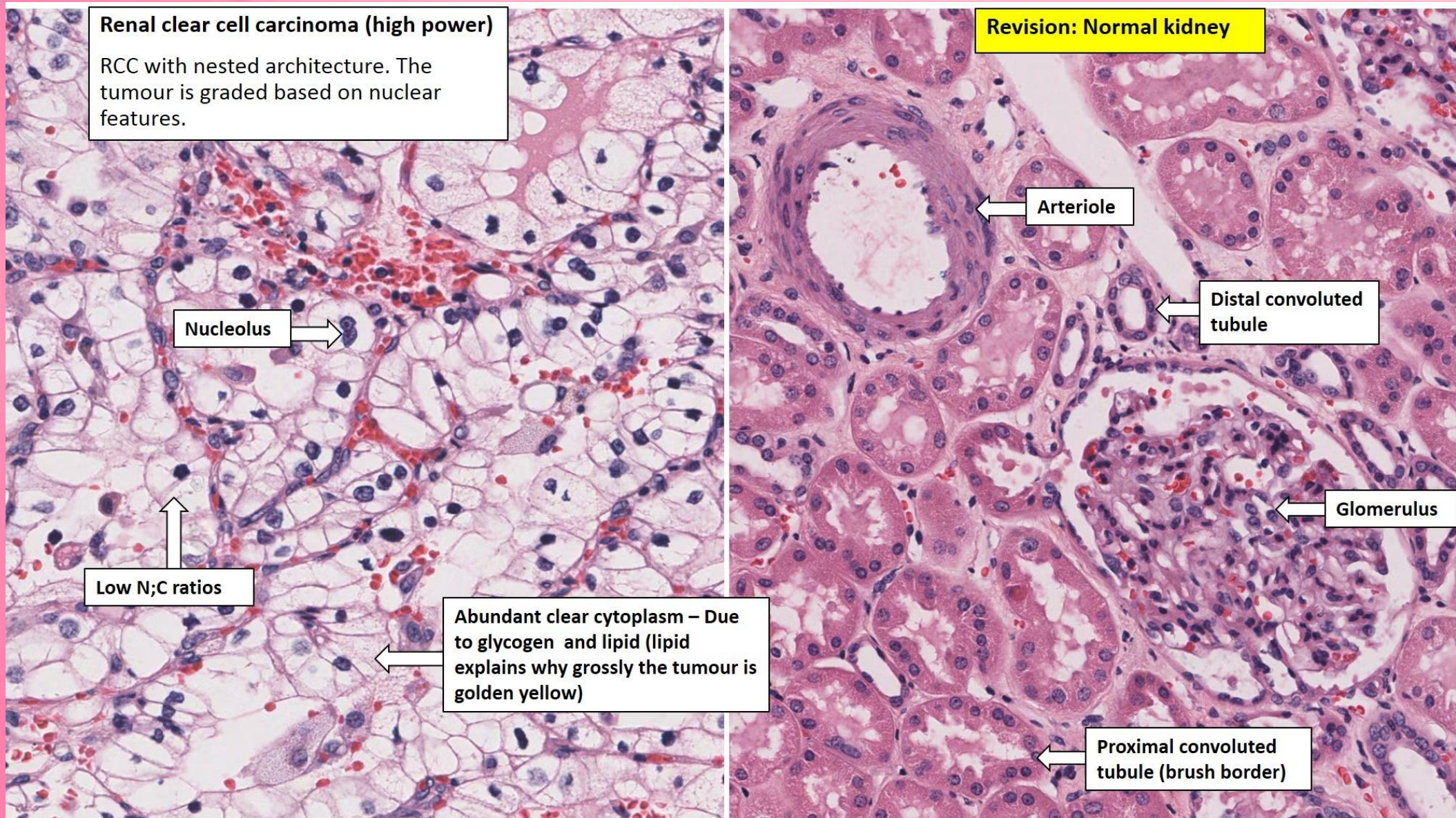


Large nests of tumor cells

Delicate septa containing thin walled blood vessels

Polygonal cells with clear cytoplasm and centrally placed small nuclei.

Mikroskopik



Selain Clear cell renal cell carcinom, terdapat juga tipe yang sering :

1. Papillary Renal Cell Carcinoma (pRCC)

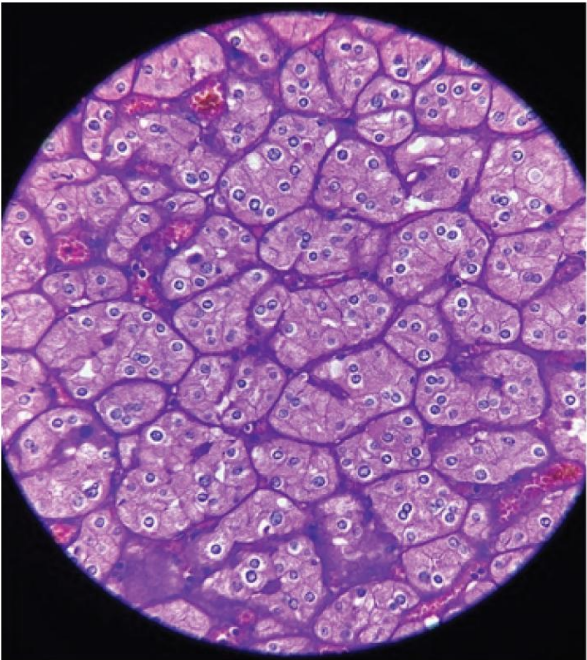
- Mencakup sekitar 15% dari kanker ginjal
- Memiliki pola pertumbuhan papiler yang khas

2. Chromophobe RCC & Oncocytoma

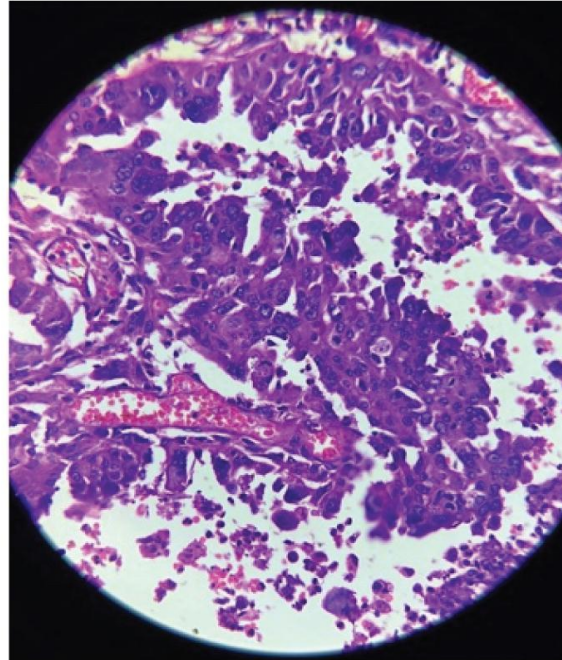
- Chromophobe RCC (4–8%): Memiliki struktur sarang (alveolar/nested pattern), batas sel yang tebal, inti sel ireguler, dan halo perinuklear
- Memiliki prognosis yang paling baik di antara RCC lainnya
- Oncocytoma: Merupakan tumor epitel renal yang bersifat jinak namun memiliki kemiripan morfologi yang dekat dengan chromophobe RCC

Microscopy

Chromophobe RCC

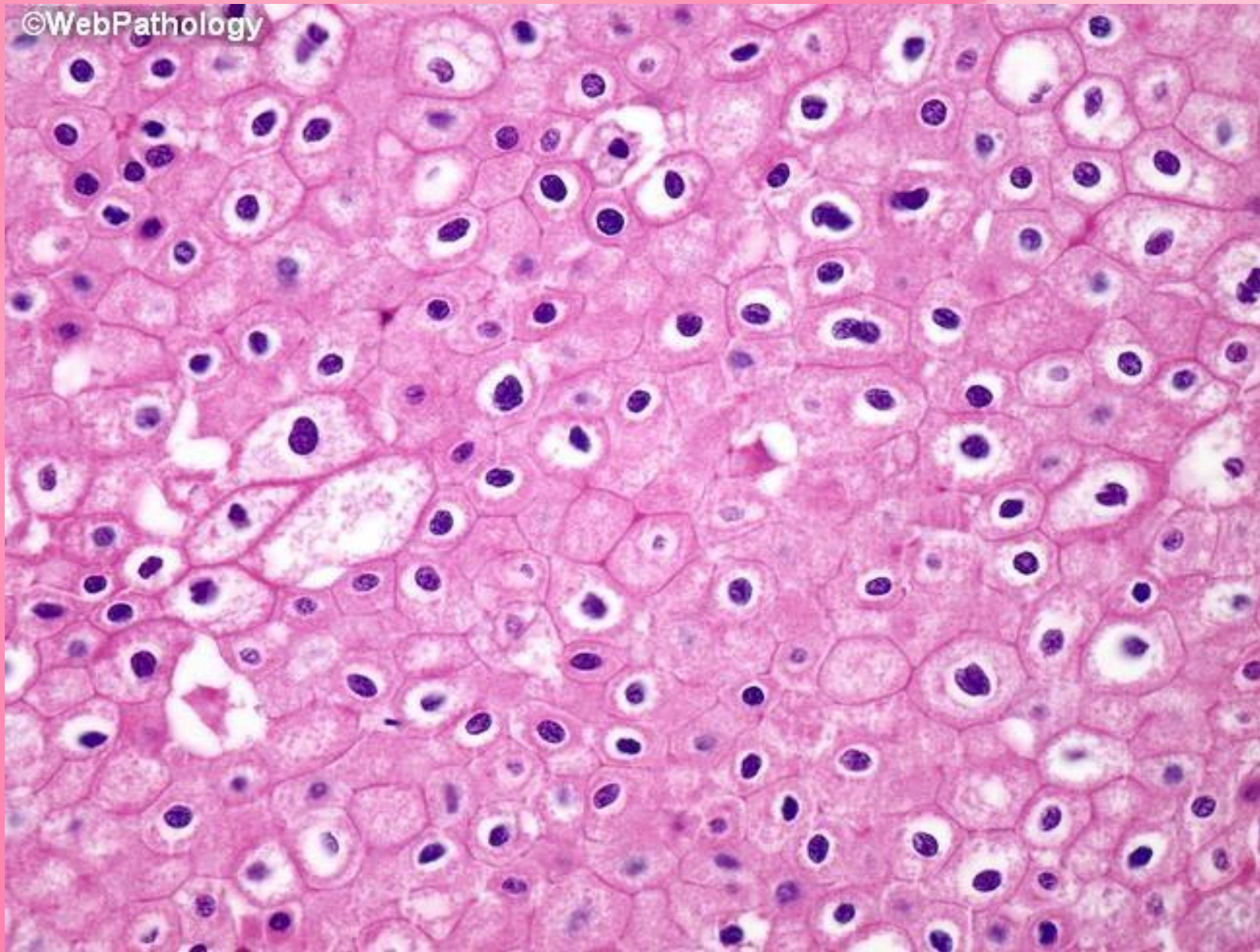


Papillary RCC

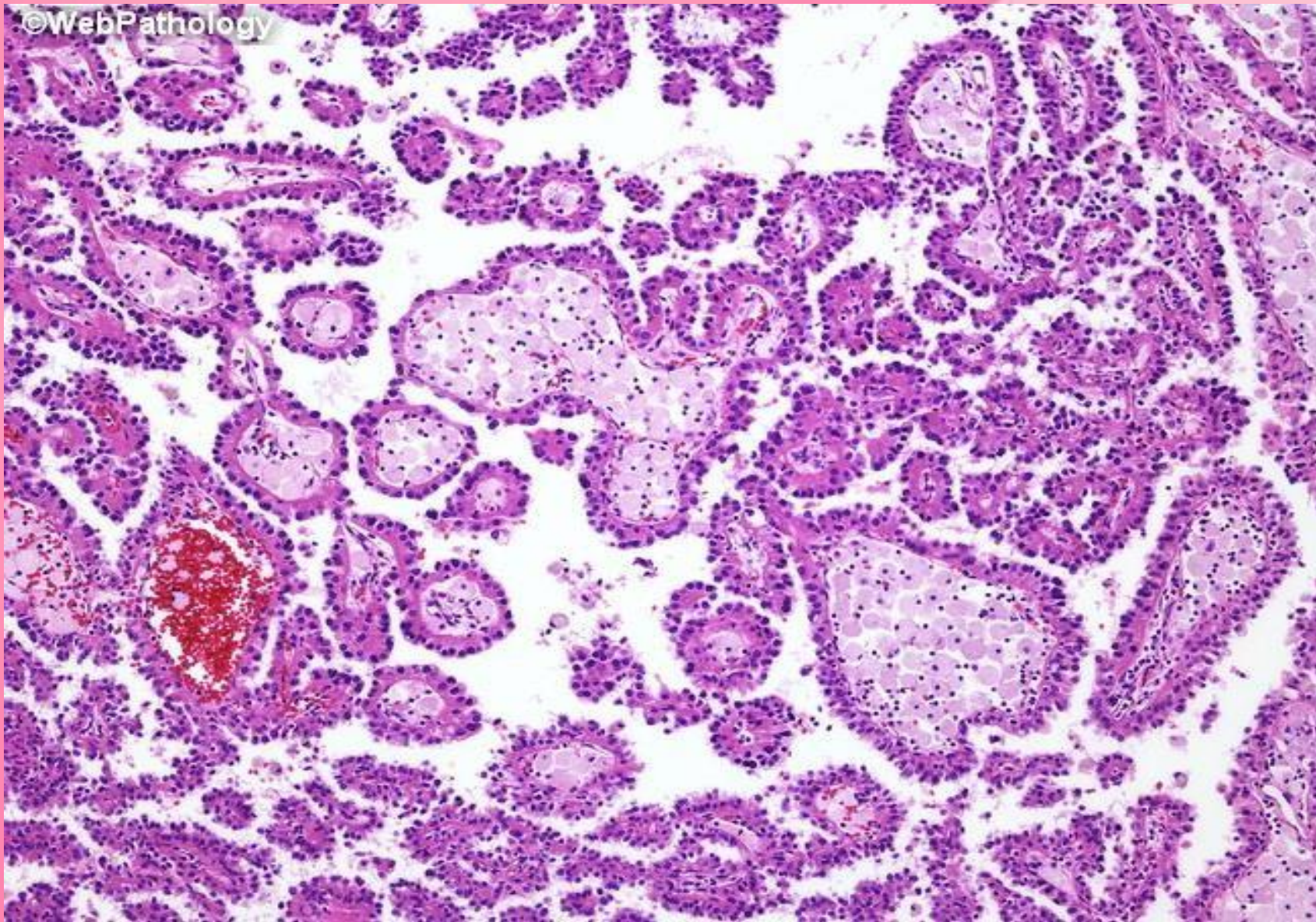


Dua subtype Karsinoma Sel Ginjal (Renal Cell Carcinoma / RCC):

- Chromophobe RCC (Kiri):
Menampilkan sel-sel berukuran besar dengan batas membran yang sangat tegas (sering disebut mirip dinding sel tumbuhan). Sitoplasmanya berwarna merah muda pucat/jernih dengan inti sel yang sering memiliki lingkaran bening di sekitarnya (perinuclear halo), tersusun rapat membentuk kelompok padat.
- Papillary RCC (Kanan):
Menampilkan sel-sel kanker yang tumbuh membentuk struktur tonjolan seperti jari-jari (papila). Di dalam tonjolan tersebut terdapat sumbu jaringan pembuluh darah (fibrovascular core), sehingga tampilannya tampak bercabang dan memiliki banyak rongga terbuka di antara susunan selnya.



CHROMOPHOBE RENAL CELL CARCINOMA.

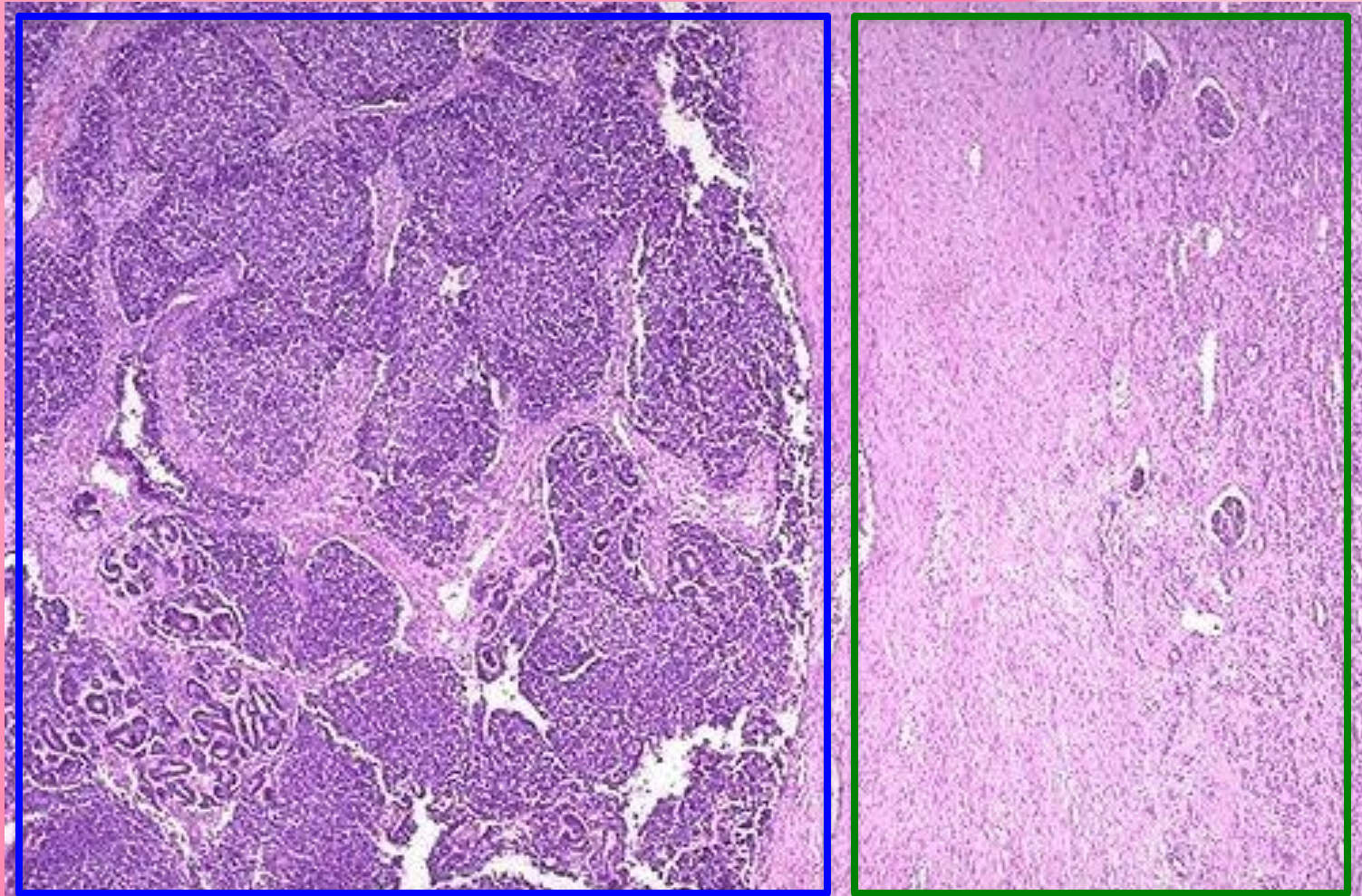


PAPILLARY RENAL CELL CARCINOMA

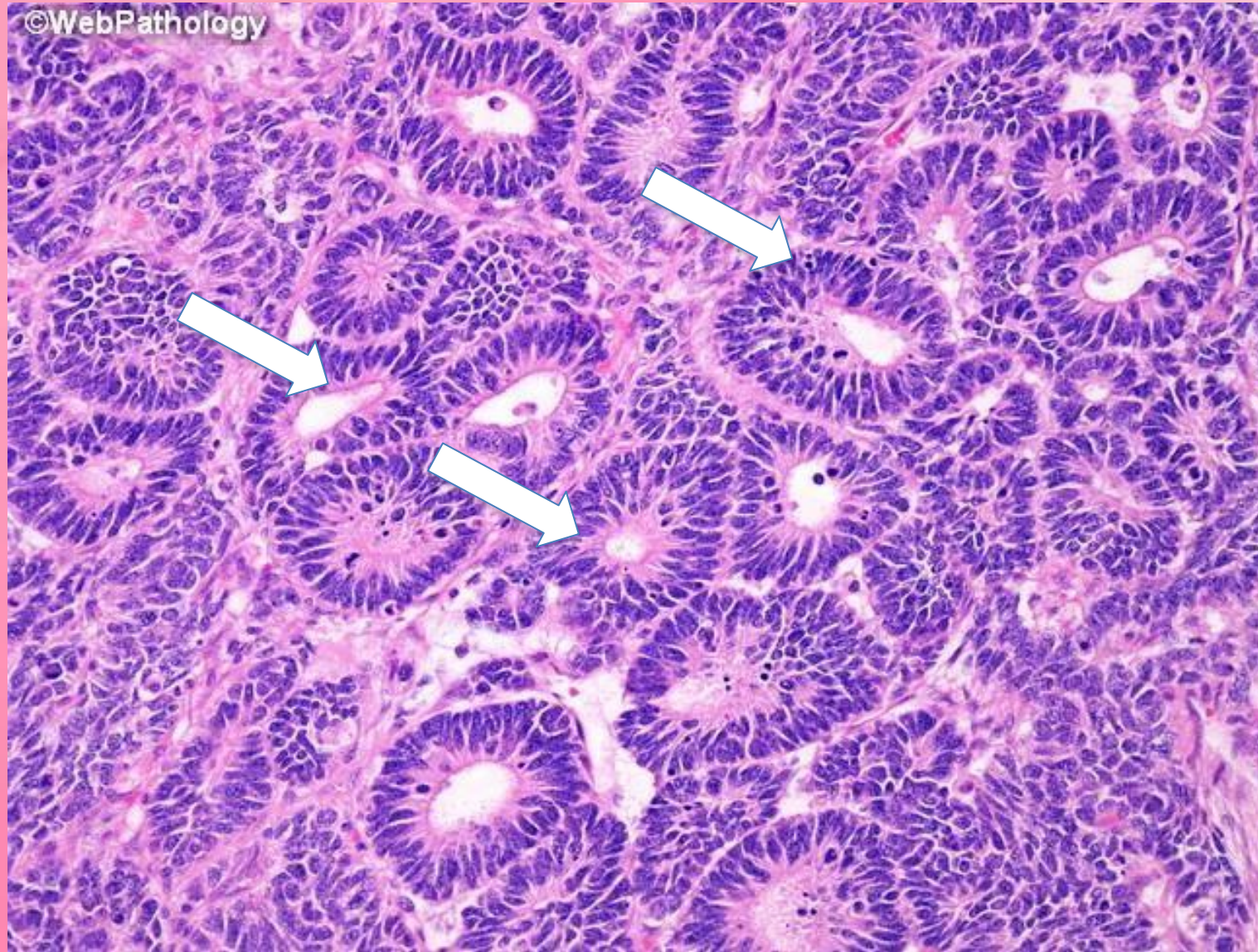
WILMS TUMOR OF THE KIDNEY



This small kidney from a 4 year old child contains a lobulated tan-white mass. This is **WILMS TUMOR OF THE KIDNEY**. Many are now known to be associated with genetic defects (WT1 gene) on chromosome 11. The children with Wilms tumor usually present with abdominal enlargement from the mass effect. Nowadays, treatment gives a better than 90% 5 year survival.



This is a **WILMS TUMOR** that is composed microscopically of **nests and sheets of dark blue cells** at the left with **compressed normal renal parenchyma** at the right. The WT1 gene at chromosomal band 11p13 encodes a transcription factor needed for renal development. There can be familial germline mutations (WAGR syndrome; Denys-Drash syndrome), but most Wilms tumors arise from acquired mutations.

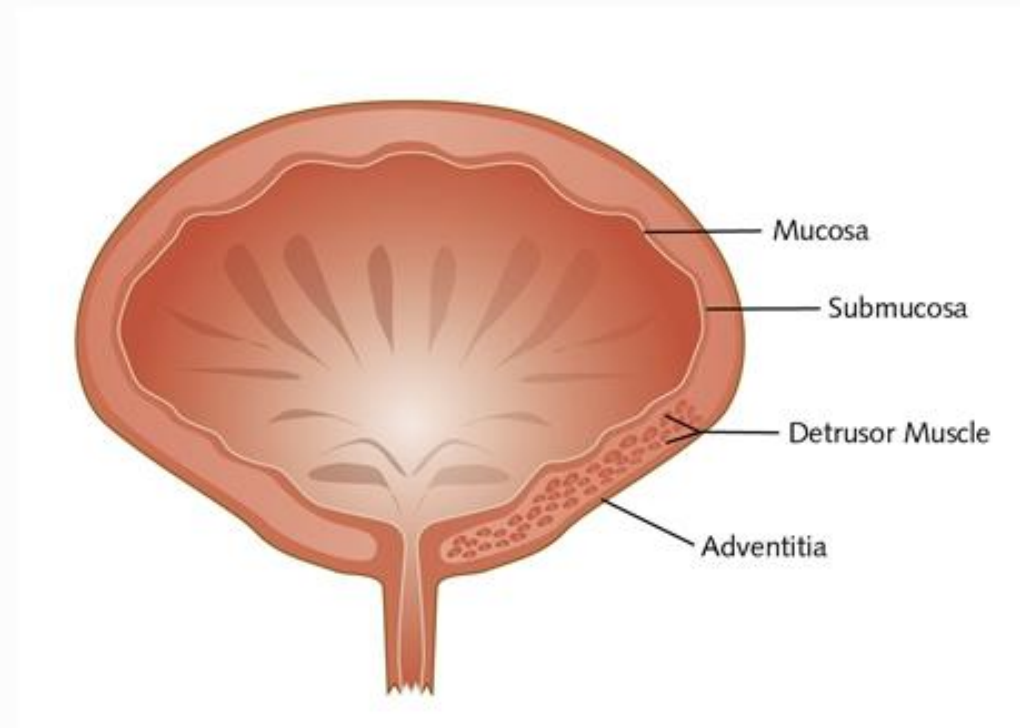


This example shows abundance of epithelial component (tubules) and nodules of blastema with only a scant amount of fibroblastic stroma.

2. Lesi Neoplastik pada kandung kemih (Urothelial carcinoma)

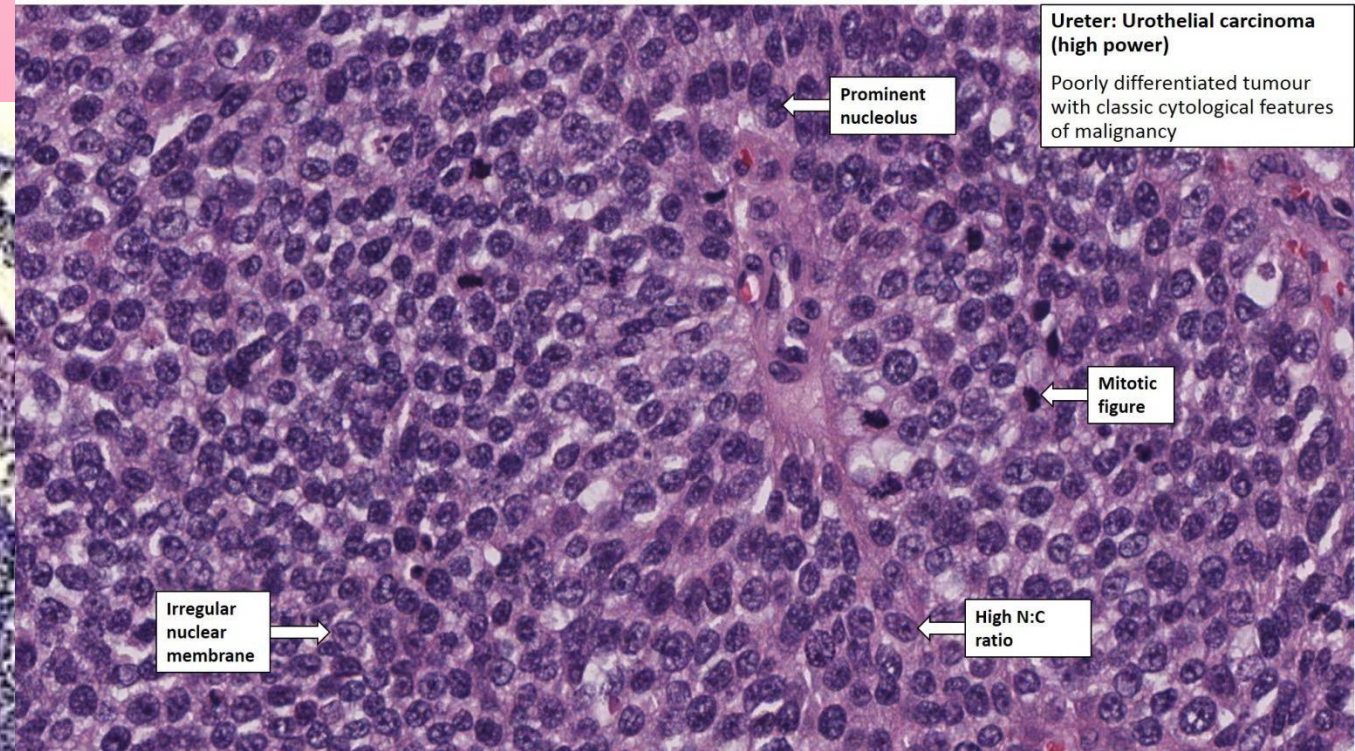
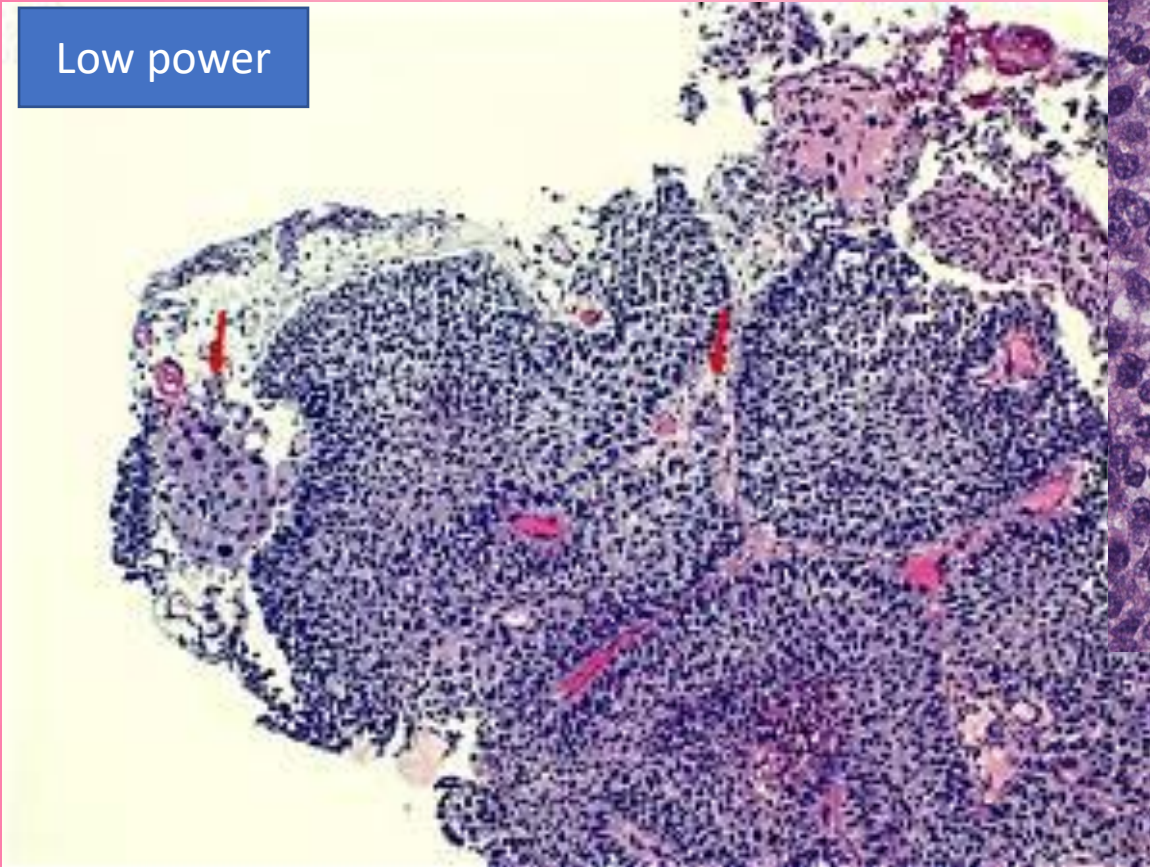
- Karsinoma urotelial (dahulu disebut Transitional Cell Carcinoma / TCC) merupakan keganasan saluran kemih yang paling sering ditemukan, mencakup 90% dari kanker kandung kemih, serta dapat tumbuh di pelvis renalis, ureter, atau uretra.
- Variant Histopatologi Karsinoma Urotelial yang Agresif :
 1. Varian Micropapillary: Sel tumor tersusun dalam kelompok-kelompok kecil tanpa inti fibrovaskular, dikelilingi celah retraksi kosong (retraction spaces) yang menyerupai pembuluh limfatik
 - Sebagian besar sudah menginvasi otot saat didiagnosis
 2. Varian Plasmacytoid: Sel tumor berbentuk bulat monomorfik yang saling terpisah (discohesive) mirip sel plasma, dengan sitoplasma eosinofilik homogen dan inti eksentrik
 - Sangat agresif, menyebar luas di sepanjang permukaan serosa, dan sering dikaitkan dengan mutasi hilangnya fungsi gen CDH1 (E-cadherin)
 3. Varian Sarcomatoid: Kanker derajat tinggi yang memperlihatkan koeksistensi antara komponen epitelial urotelial ganas dengan area sel spindle pleomorfik menyerupai sarkoma
 - Dapat disertai pembentukan osteoid metaplastik
 4. Varian Nested: Tampak sebagai sarang-sarang sel urotelial yang terlihat sangat tenang (deceptively bland) di jaringan suburotelial, sehingga harus dibedakan secara cermat dari sarang von Brunn (von Brunn nests) yang bersifat jinak
 - Sering dikaitkan dengan mutasi promotor TERT
 5. Divergent Differentiation: Menunjukkan diferensiasi skuamosa (paling sering, ~24%), glandular (~27%), atau trofoblastik (bizarre syncytiotrophoblast-like cells yang mengekspresikan beta-HCG, ~2.7%)

Anatomi Vesica urinaria



Mikroskopik Urothelial Carcinoma

Low power



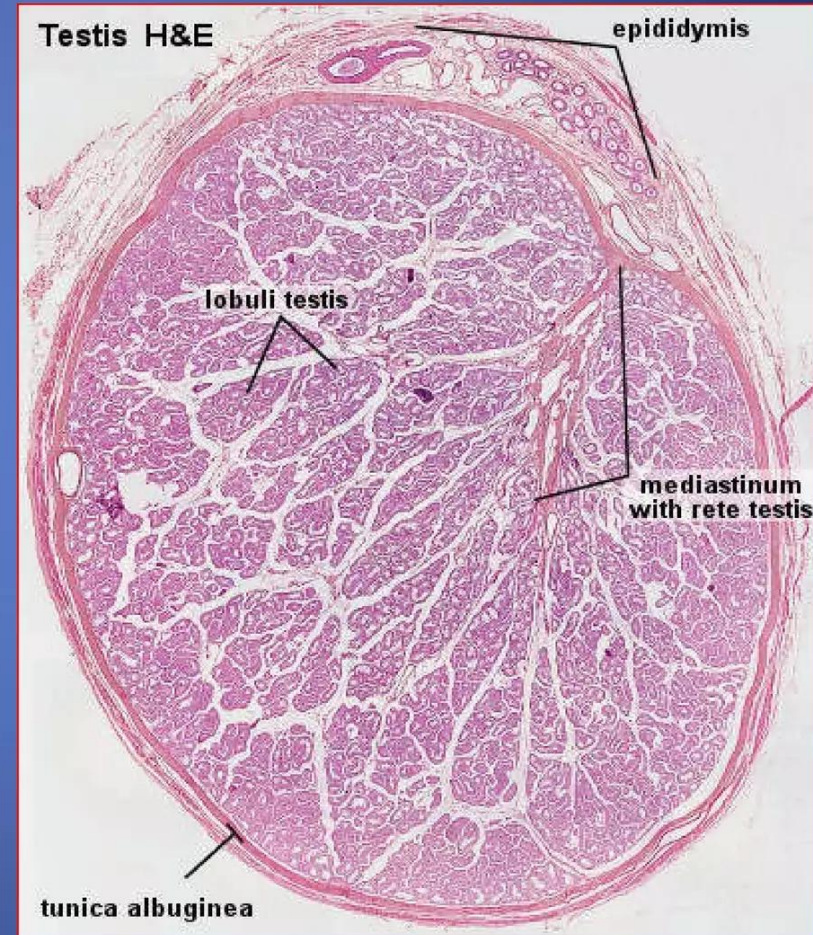
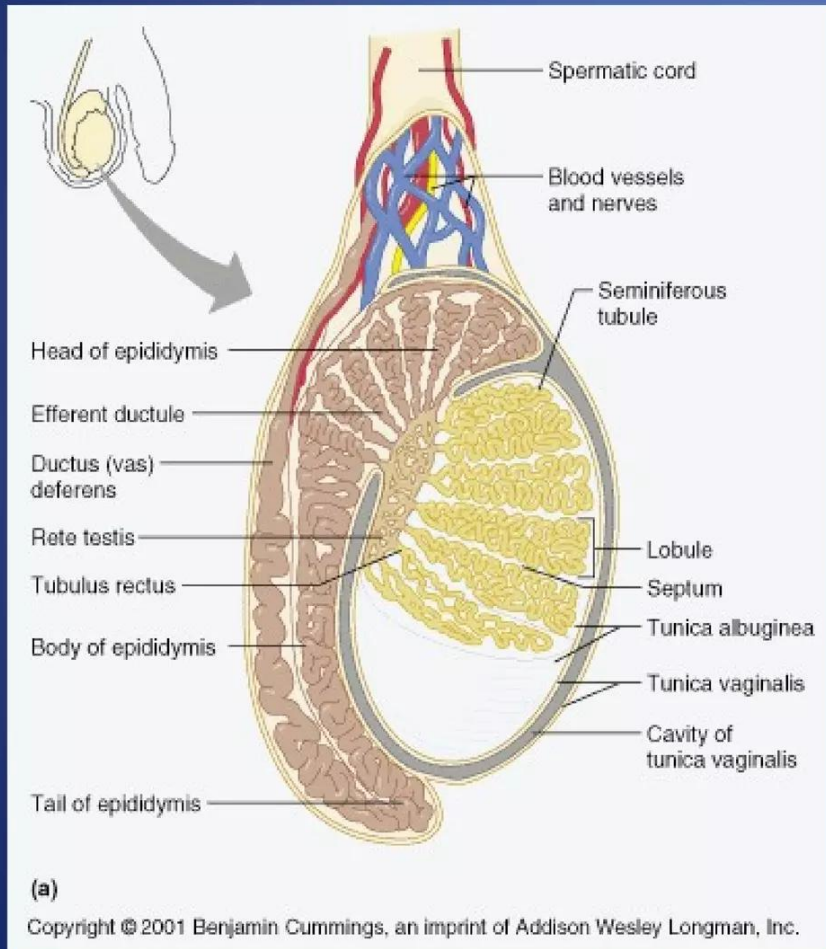
3. Lesi Neoplastik Pada Testis (Testicular Tumors)

- Kanker testis relatif jarang dijumpai (<1% dari keganasan pria), namun merupakan kanker utama pada laki-laki usia muda (15–45 tahun).
- Lebih dari 90% tumor testis primer berasal dari sel nutfah (Germ Cell Tumors / GCT) → paling sering **seminoma**

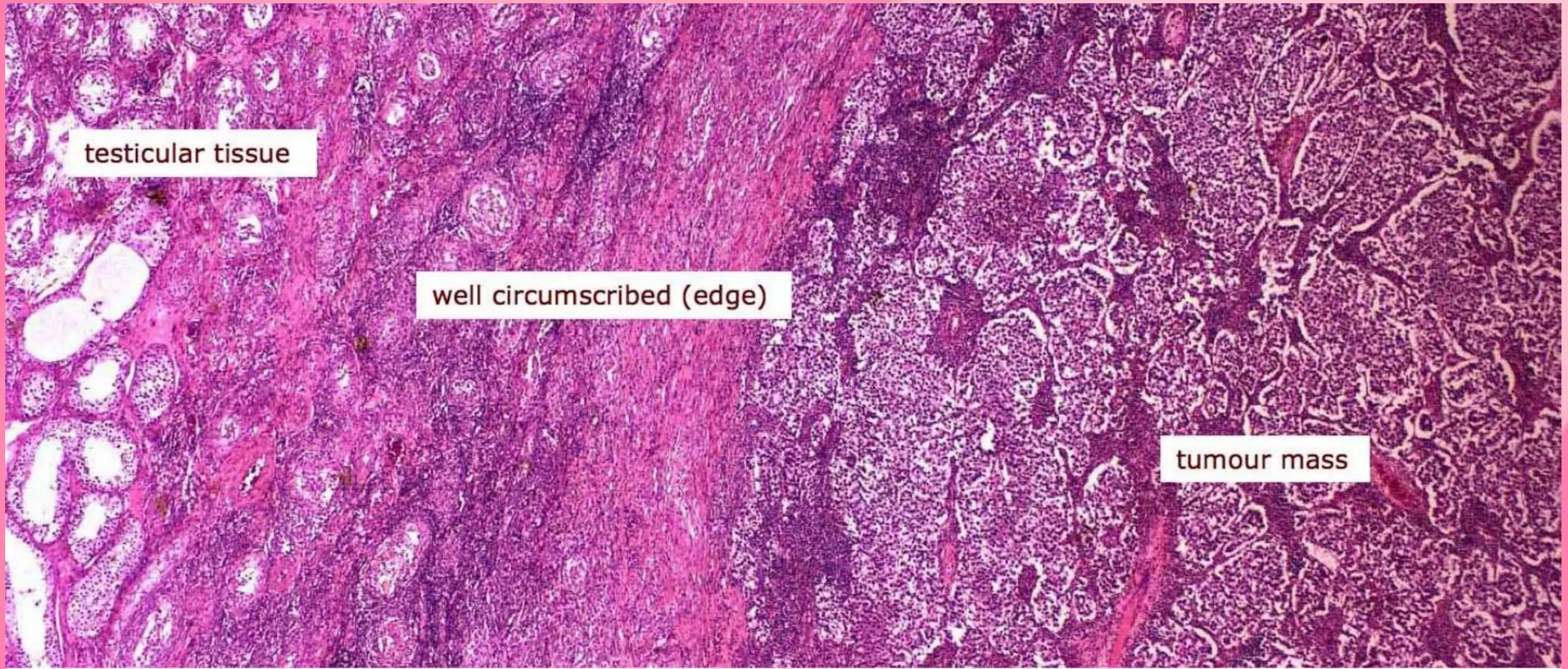
SEMINOMA: Mencakup 30%–40% dari seluruh tumor testis

- Di bawah mikroskop, sel seminoma tampak seragam, berinti bulat besar dengan nukleolus jelas, sitoplasma jernih, dan dipisahkan oleh sekat/septum fibrosa tipis yang mengandung infiltrasi sel limfosit.
- Seminoma Anaplastik: Varian seminoma yang lebih agresif, ditandai dengan ditemukannya 3 atau lebih gambaran mitosis per lapangan pandang besar (HPF) disertai pleomorfisme inti sel yang nyata dan area nekrosis yang luas.

ANATOMI

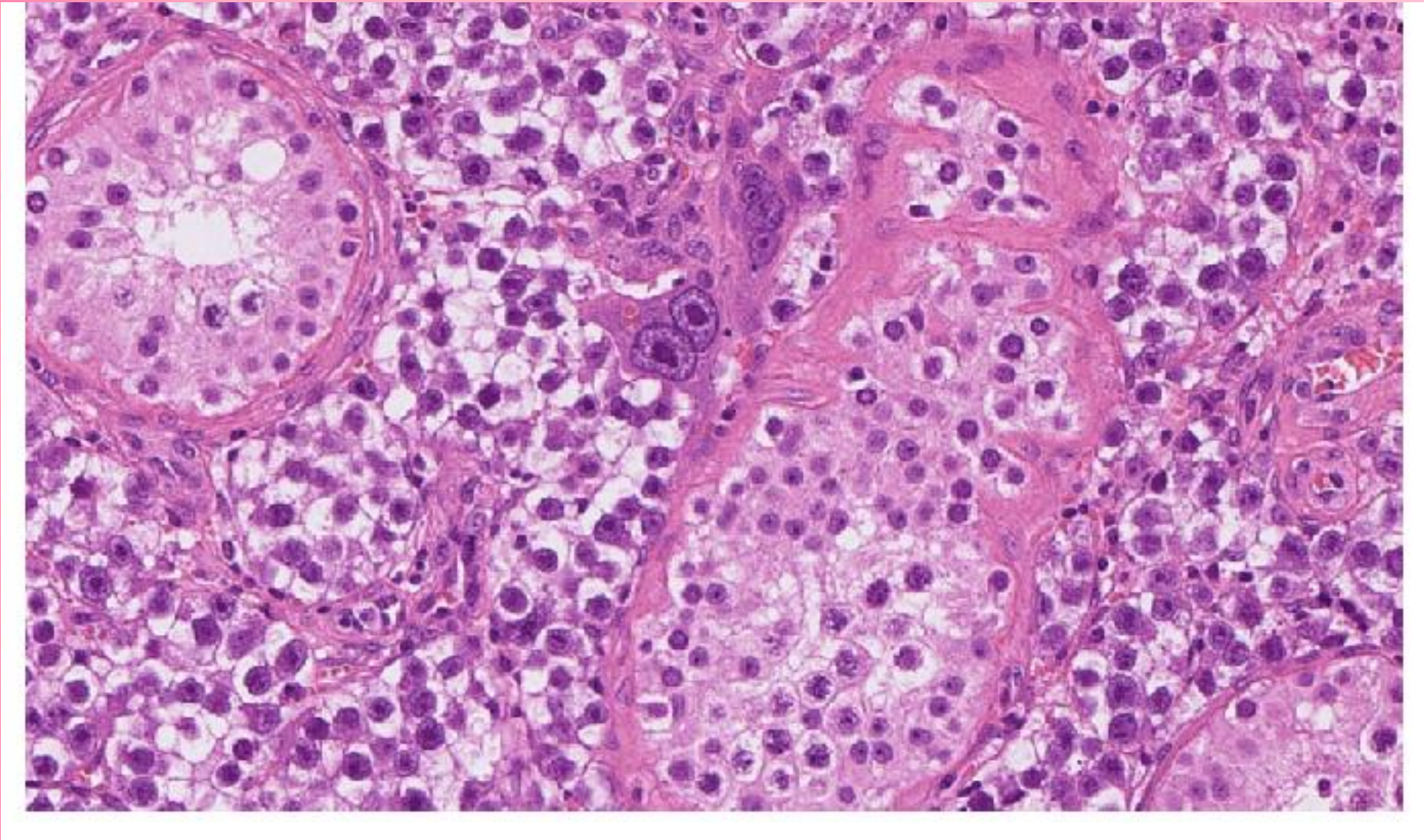


Mikroskopik Seminoma



Histology - Seminoma

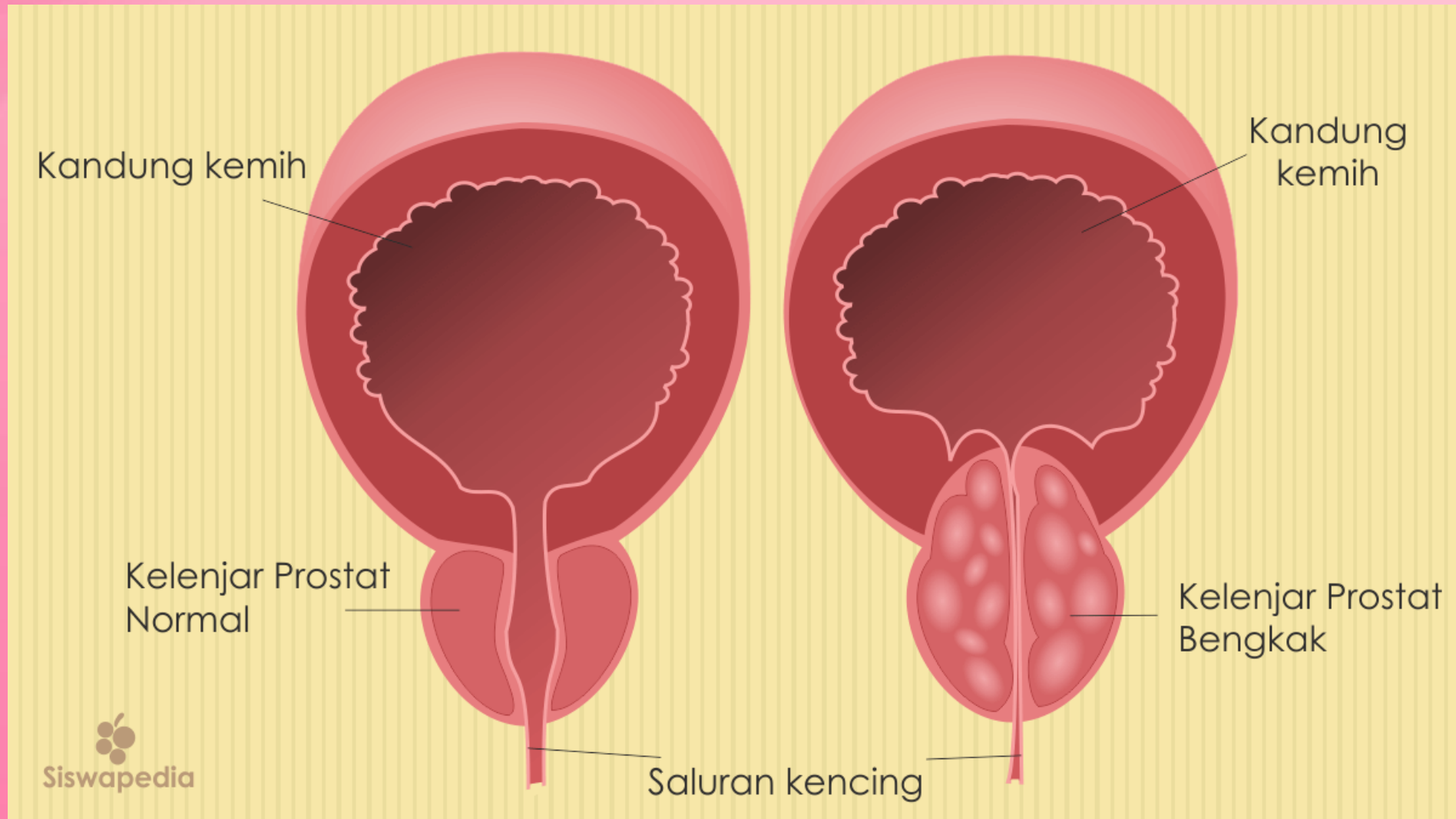
Mikroskopik Seminoma – high power

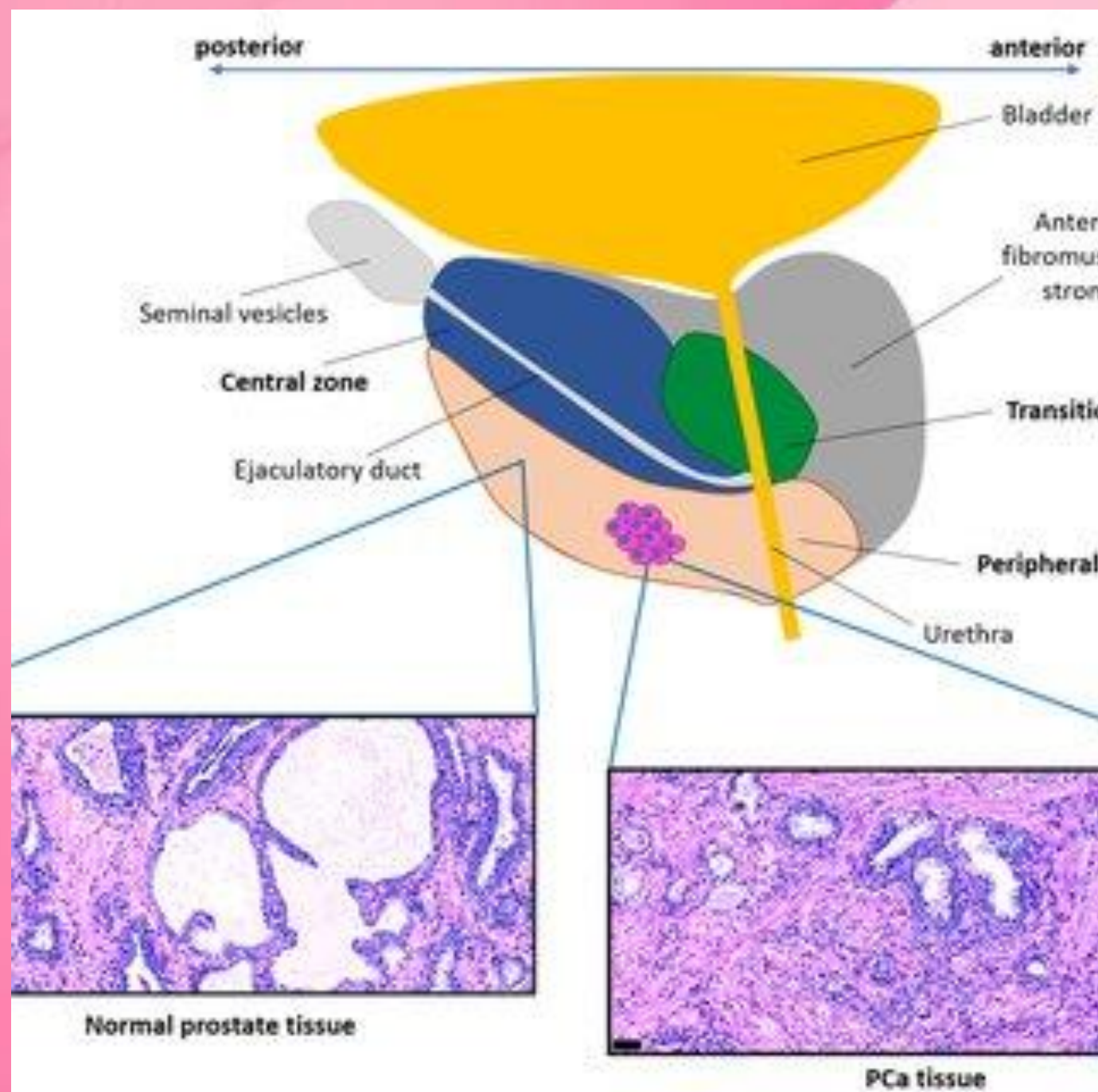


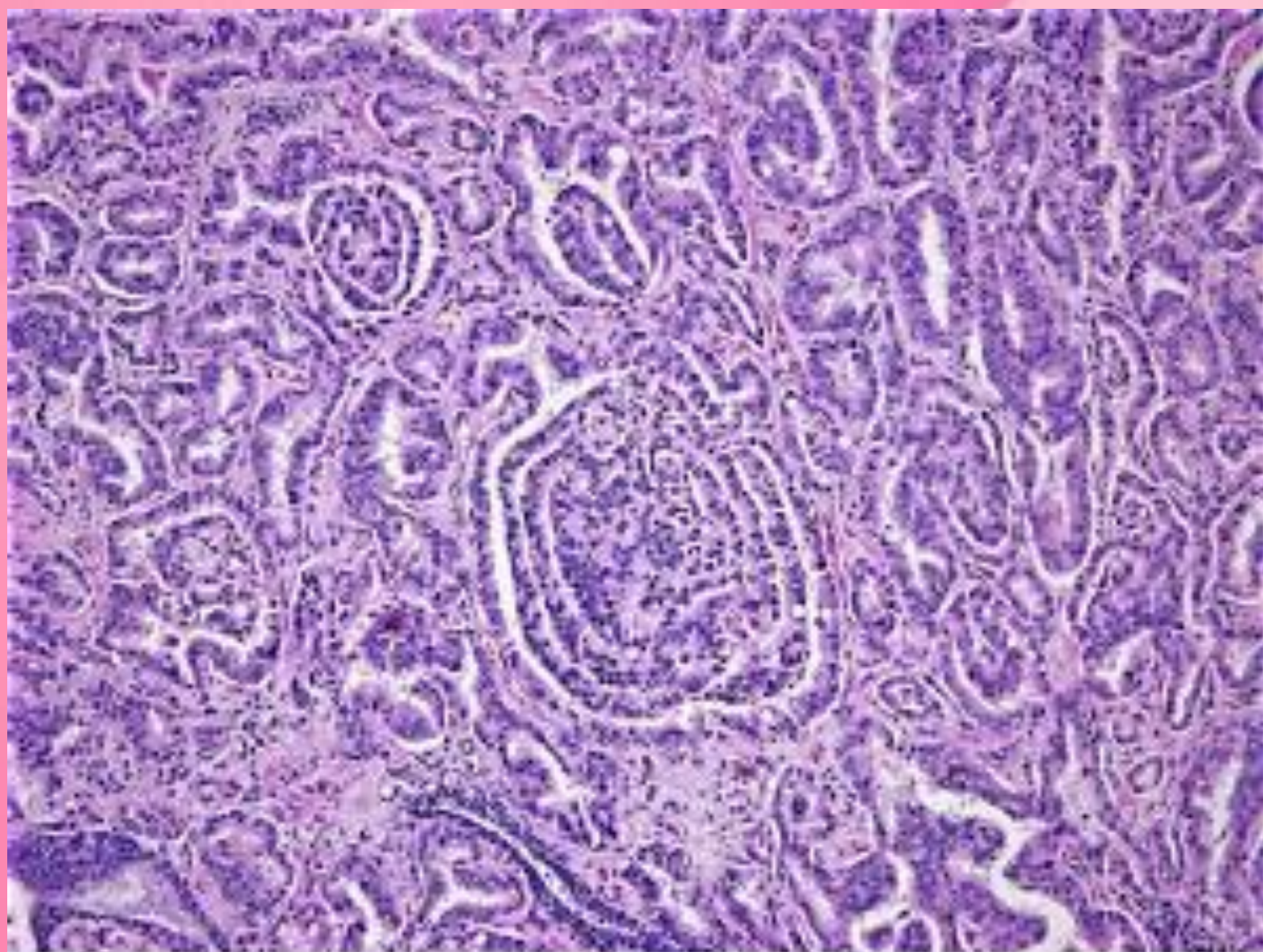
4. Lesi Neoplastik Pada Prostat (Adenokarsinoma Prostat)

- Kanker prostat merupakan keganasan non-kutaneus yang paling sering didiagnosis pada pria di seluruh dunia.
- Penyakit ini harus dibedakan secara patologis dari kondisi non-neoplastik seperti Benign Prostatic Hyperplasia (BPH) dan prostatitis.
- Diagnosis ditegakkan terutama melalui pemeriksaan spesimen core needle biopsy atau kerokan jaringan dari tindakan TURP (Transurethral Resection of the Prostate).
- Secara mikroskopis, adenokarsinoma ditandai dengan adanya proliferasi kelenjar-kelenjar kecil abnormal yang menginvasi stroma prostat, hilangnya lapisan sel basal kelenjar, serta adanya pembesaran inti sel dengan anak inti (nukleolus) yang sangat menonjol.

Anatomi Prostate



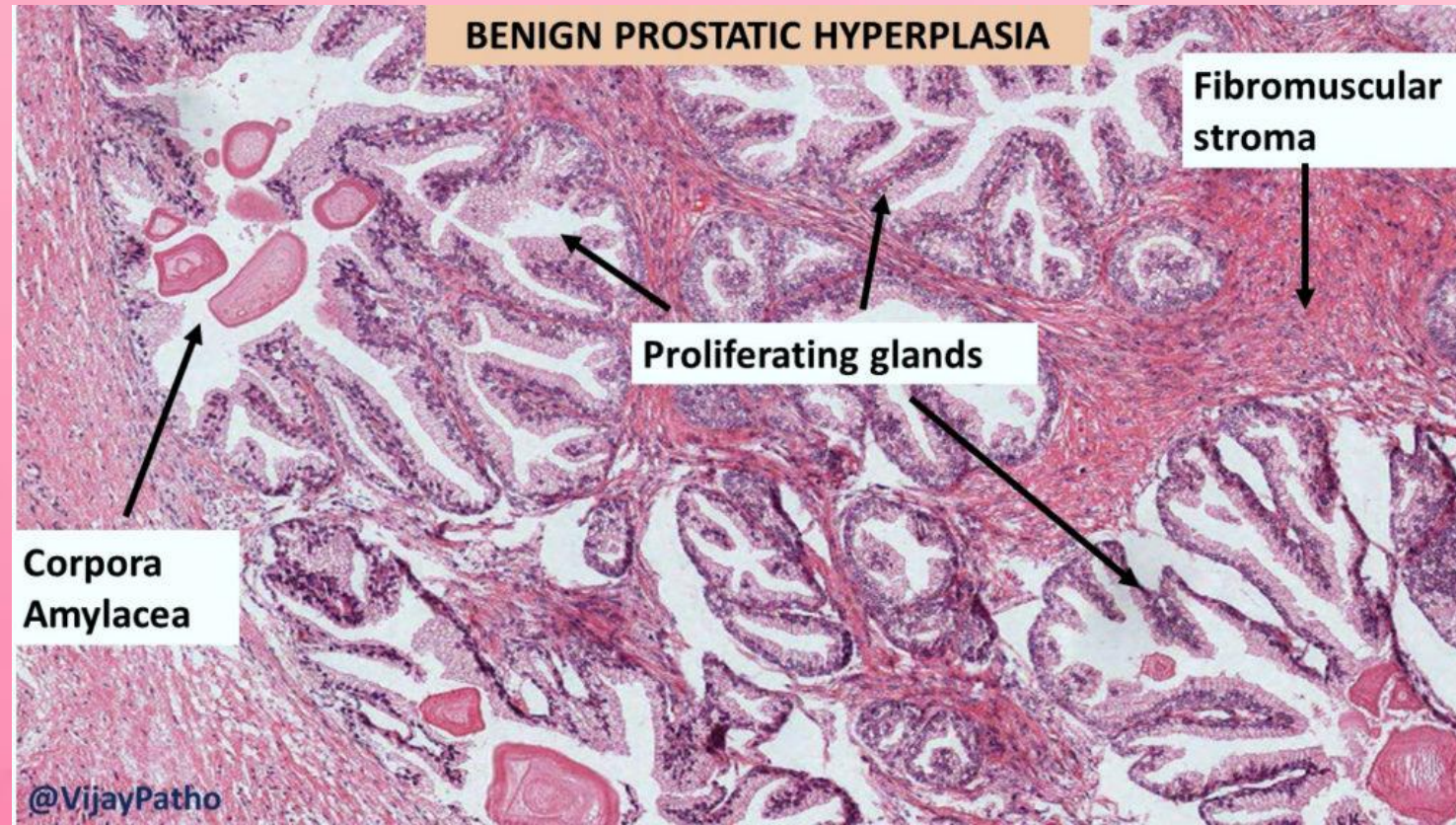




Characteristic morphologic features of
prostatic adenocarcinoma (10x)

Bedakan dengan benign prostat hiperplasia (BPH)

- Pembesaran prostat jinak atau benign prostatic hyperplasia (BPH) adalah kondisi ketika kelenjar prostat membesar. Akibatnya, aliran urine menjadi tidak lancar dan buang air kecil terasa tidak tuntas.
- Bukan merupakan lesi neoplastik pada prostat



thank
you!

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